Clean Fuel – Cleaner Skies
Ethanol to Jet Fuel

Senior Design Final Report

May 6th, 2025

Group 2

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Executive Summary

The aviation sector faces increasing pressure to reduce greenhouse gas (GHG) emissions while maintaining reliable, high-energy-density fuel for global transportation. Aviation is one of the most difficult sectors to decarbonize due to the energy density requirements of aircraft and the long operational lifetimes of aviation equipment. As a result, Sustainable Aviation Fuel (SAF) has emerged as the most practical near-term solution to reduce aviation emissions while maintaining compatibility with existing aircraft engines and fueling infrastructure. Government initiatives such as the Sustainable Aviation Fuel Grand Challenge aim to replace conventional Jet A fuel with SAF by 2050, ensuring long-term demand for low-carbon aviation fuels.

This project presents the design and economic evaluation of a commercial-scale ethanol-to-jet (ETJ) fuel facility producing 10,000 barrels per day of ASTM-certified sustainable aviation fuel using low-carbon ethanol feedstock. The facility is designed to support national decarbonization goals while maintaining strong technical and economic performance. The selected plant location in Wilkinson County, Georgia provides strategic advantages, including access to transportation infrastructure, proximity to major airports, and availability of ethanol feedstock supply chains. These factors contribute to reduced logistics costs and improved operational flexibility.

The ETJ pathway converts ethanol into jet-range hydrocarbons through three primary processing sections: dehydration, oligomerization, and hydroprocessing. In the dehydration section, ethanol is converted into ethylene through catalytic dehydration. The ethylene is then converted into longer hydrocarbon chains through oligomerization reactions, forming intermediate molecules suitable for jet fuel production. The final hydroprocessing section includes hydrogenation, hydrocracking, and isomerization reactions that refine the product into aviation-grade fuel meeting ASTM fuel specifications. This integrated reaction sequence produces high yields of jet fuel range hydrocarbons while minimizing unwanted byproducts and ensuring product quality.

The selected ethanol-to-jet pathway was evaluated against several competing sustainable aviation fuel technologies, including Hydroprocessed Esters and Fatty Acids (HEFA), gasification with Fischer-Tropsch synthesis, pyrolysis-based upgrading, and hydrothermal liquefaction. While HEFA technology currently dominates commercial SAF production, its long-term growth is constrained by limited feedstock availability, particularly fats, oils, and greases. Gasification and pyrolysis technologies offer broader feedstock flexibility but require significantly higher capital investment and operational complexity. In contrast, the ethanol-to-jet pathway provides a balanced combination of technical maturity, scalability, and operational reliability, making it a strong candidate for long-term commercial deployment.

From a safety and environmental perspective, the ethanol-to-jet process offers several advantages compared to alternative fuel production pathways. The process uses stable liquid ethanol feedstock, reducing storage and handling risks compared to unstable

intermediates such as bio-oil. Additionally, the process does not generate tar or slag byproducts commonly associated with gasification-based systems, simplifying waste handling and reducing environmental risk. Standard industrial safety systems, including pressure relief devices, emission controls, and wastewater treatment systems, are incorporated throughout the design to ensure safe and responsible plant operation.

A detailed economic analysis was conducted to evaluate the financial feasibility of the proposed facility. The total capital investment required for the plant is estimated at approximately $626 million, representing the cost of major process equipment, installation, utilities, and supporting infrastructure. Operating costs are primarily driven by ethanol feedstock, hydrogen consumption, utilities, and catalyst replacement. Despite these expenses, the plant demonstrates strong financial performance under current market conditions.

The facility generates an estimated annual revenue of approximately $1.565 billion, with the majority of this revenue derived from renewable fuel incentives and credits rather than direct fuel sales. These incentives include Renewable Identification Number (RIN) credits and federal clean fuel tax credits, which significantly increase the economic value of low-carbon fuel production. The strong contribution of credits to total revenue reflects the current economic structure of the sustainable aviation fuel industry, where government policy plays a critical role in enabling commercial-scale SAF deployment.

The project achieves an estimated net profit of approximately $714 million per year and a Net Present Value (NPV) of approximately $3.12 billion over a 20-year operating period, demonstrating strong long-term economic potential. Sensitivity analysis indicates that the economic viability of the facility is most strongly influenced by fuel prices and renewable fuel incentives, confirming that continued access to credits is a key factor in maintaining profitability.

Overall, the results of this project demonstrate that a commercial-scale ethanol-to-jet fuel facility can be both technically feasible and economically profitable while contributing to significant reductions in greenhouse gas emissions. The proposed design supports national sustainability goals, meets aviation fuel performance standards, and provides a scalable pathway for increasing sustainable aviation fuel production. With continued technological development and policy support, ethanol-to-jet fuel production represents a promising long-term solution for reducing aviation emissions while maintaining reliable fuel supply for the global aviation industry.

Project Charter

Ethanol to Jet Gantt Chart Senior Design

Introduction

The Global Warming Crisis

Global warming- the progressive increase in average temperatures around the globe due to the increased levels of green-house-gas emissions in the atmosphere- has been one of the longest ongoing struggles that our planet has faced in the post-industrial era. A collection of global warming data collected from NASA [1] is visualized below:

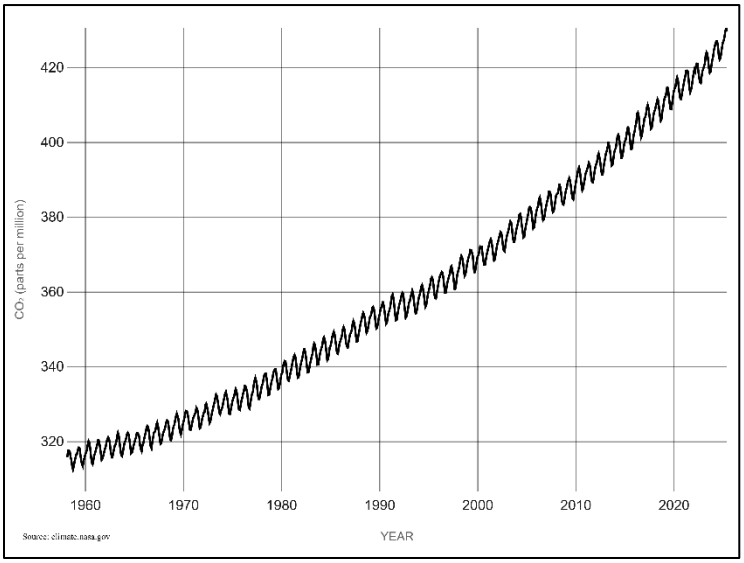


Figure 1- Global CO_2 Concentration

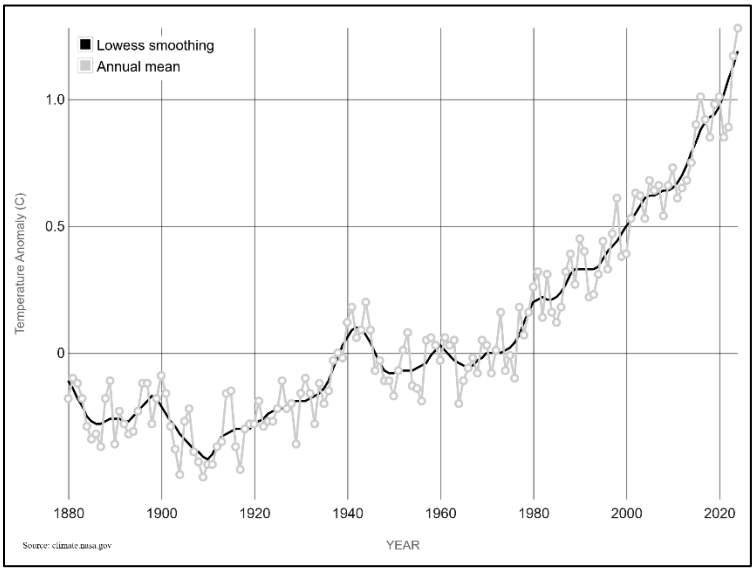


Figure 2 - Global Average Temperature Anomaly

The story confirmed within these figures is that greenhouse gas emissions have a direct impact on global warming- a well-known fact. Habitats, animals, and foods have all gone extinct each year and more parts of the planet will be lost if this warming crisis continues. For decades, efforts across industries have been underway to reduce emissions. This work

accelerated when President Joe Biden signed an executive order to catalyze American clean energy production through The Federal Sustainability Plan, which set a net-zero emissions goal by 2050, accompanied by government incentives to achieve this goal and other goals by 2030 [2]. However, since President Donald Trump took office, these goals have slowed and been effectively halted.

Transportation accounts for 29% of all greenhouse gas emissions globally, 9% of which is due to aircraft [3]. While the percentage seems small, with a CO_2 emission equivalent of 6,500 trillion metric tons in 2023, aviation contributes ~165 trillion metric tons to global warming. Such an amount is not to be underestimated, especially on a yearly basis. Furthermore, aviation is a particularly hard sector to decarbonize. Planes require incredibly high-energy-density fuel to mobilize large weights at high speeds. Thus, innovations like electric or hydrogen-powered cars are not yet feasible for aircraft. Additionally, many aircraft manufacturers have sold out through 2027 [4], and these planes can last 20-30 years. Thus, fossil fuel-powered planes will not be going anywhere for a long while. With such a large carbon footprint and few avenues for innovation in the physical plane body, the only reasonable solution to decarbonizing aviation is to replace traditional Jet Fuel A with Sustainable Aviation Fuel (SAF). This fuel is almost identical to its fossil-fuel counterpart, produced from a greener feedstock and process. Additionally, the U.S. government has released a Memorandum of Understanding (MOU) to replace all domestic Jet Fuel A with SAF by 2050. This initiative is known as the SAF Grand Challenge. [5] Many government incentives exist for companies to produce SAF, making it an economically feasible business and advancing net-zero goals. On top of this, manufacturing SAF can reduce greenhouse gas emissions by up to 98% compared to Jet Fuel A, making it extremely valuable for tackling global warming [6].

Our team aims to accelerate aviation decarbonization by producing SAF. Our innovative ethanol-to-jet fuel pathway aims to achieve record GHG emission-reduction targets, produce SAF with high selectivity, and become the world's largest SAF producer. By utilizing low-carbon ethanol derived from biomass, waste gas and more, we plan to design a plant to produce 10,000 barrels of SAF per day, helping the U.S. reach its goal of producing 35 billion gallons of SAF per year by 2050. We are eager to bring the world clean fuel and even cleaner skies through our economically robust ethanol-to-jet pathway.

Market Analysis Report

Market Size and Growth Outlook

The global sustainable aviation fuel (SAF) market is expanding because of regulatory pressure and aviation sector sustainability goals. However, SAF is not creating a new fuel market; rather, it is supplying a portion of the existing jet fuel demand by displacing conventional petroleum-based Jet A fuel. Global aviation fuel demand already exists and is expected to remain substantial as air travel continues to recover and grow. As a result, the opportunity for SAF producers is primarily driven by the need to replace a fraction of conventional jet fuel consumption with lower-carbon alternatives rather than generating entirely new demand.

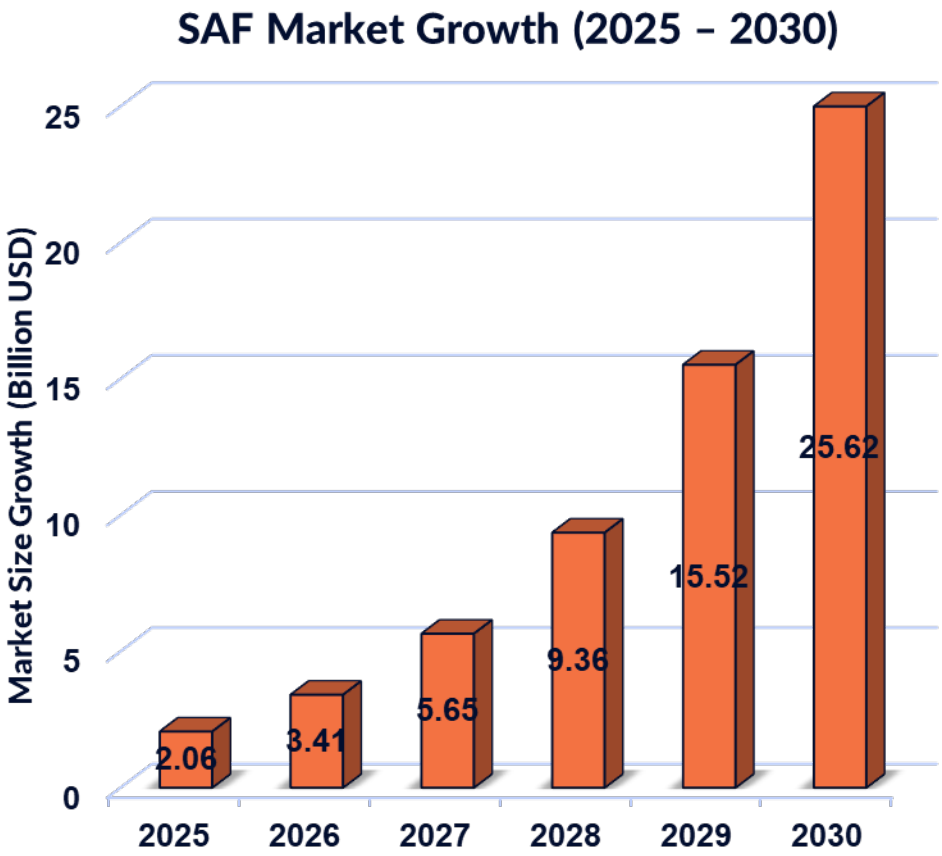
The SAF market is projected to grow significantly over the next decade as airlines, governments, and fuel suppliers work to meet decarbonization targets. Increasing regulatory mandates, airline sustainability commitments, and carbon reduction incentives are expected to drive continued expansion in SAF production capacity. The global SAF market is estimated to reach approximately $25.62 billion by 2030, reflecting strong investment in production infrastructure and supply chain development [7].

Feedstock availability and cost remain critical factors influencing SAF production economics. Ethanol feedstock for the ethanol-to-jet (ETJ) pathway can be produced from a wide range of sources, including corn, sugarcane, agricultural residues, and municipal or industrial waste streams. The flexibility in feedstock sourcing provides supply security and allows producers to adapt to regional availability and pricing conditions. However, feedstock cost typically represents the largest portion of operating expenses in SAF production and can significantly affect project profitability.

The primary customers for SAF include commercial airlines, cargo carriers, and airport fuel suppliers seeking to reduce lifecycle greenhouse gas emissions while maintaining compatibility with existing aircraft engines and fueling infrastructure. Airlines are motivated to purchase SAF to comply with regulatory requirements, meet corporate sustainability goals, and reduce exposure to carbon pricing mechanisms.

Competition in the SAF industry includes several established technology pathways, such as Hydroprocessed Esters and Fatty Acids (HEFA), Fischer–Tropsch synthesis from gasification, and alcohol-to-jet processes like the one proposed in this design. Among these technologies, HEFA currently dominates commercial SAF production due to its technical maturity, while ethanol-to-jet pathways are gaining attention because of their scalability and feedstock flexibility.

Government incentives and credits play a major role in enabling SAF deployment. Programs such as Renewable Identification Number (RIN) credits, Low Carbon Fuel Standard (LCFS) credits, and federal clean fuel tax incentives significantly improve project economics and encourage investment in new SAF production facilities. These policy mechanisms help bridge the cost gap between SAF and conventional jet fuel and



are expected to remain important drivers of industry growth.

Figure 3- SAF Market Growth (2025–2030)

The projected increase in sustainable aviation fuel production reflects a growing effort by airlines, governments, and fuel producers to reduce greenhouse gas emissions and transition toward lower-carbon energy sources. The expansion of SAF supply aligns with global decarbonization strategies and increasing regulatory pressure to reduce lifecycle emissions from the aviation sector.

Aviation accounts for approximately 9% of transportation-related greenhouse gas emissions, and the sector remains difficult to decarbonize because aircraft require high-energy-density liquid fuels for safe and efficient operation [3]. As a result, sustainable

aviation fuel is viewed as a practical near-term solution for reducing aviation emissions while maintaining compatibility with existing aircraft and fueling infrastructure.

At the same time, government programs are accelerating SAF deployment through long-term policy support and production targets. In the United States, the Sustainable Aviation Fuel Grand Challenge, led by the U.S. Department of Energy (DOE), Department of Transportation (DOT), and Department of Agriculture (USDA), aims to significantly increase domestic SAF production and replace a substantial portion of conventional Jet A fuel by 2050. These coordinated initiatives help create a stable, long-term demand for SAF and provide regulatory and financial support that encourages investment in new production capacity.

Overall, these policies and industry commitments demonstrate that the SAF market represents a sustained transition within the existing aviation fuel supply system rather than a short-term or emerging niche market.

SAF vs. Jet A: Changing the Future of Aviation Fuel

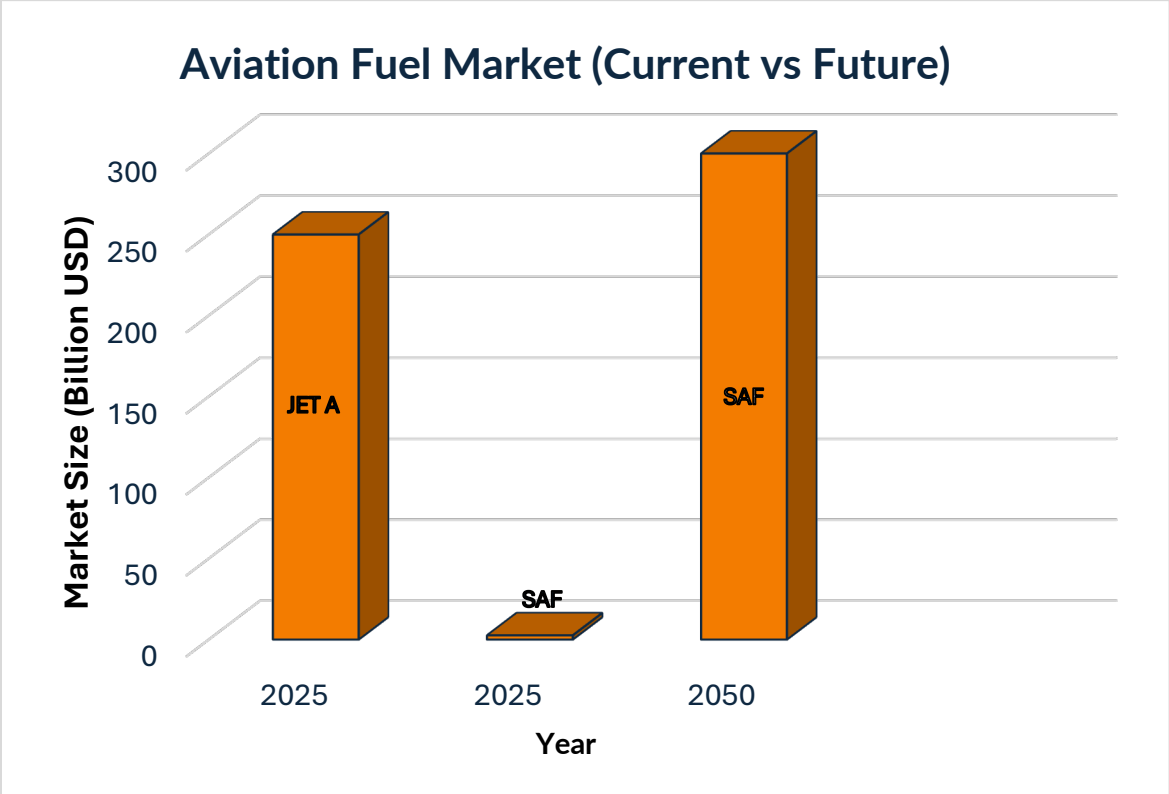


Figure 4- Jet A vs. SAF (2025–2050)

Right now, Jet A remains the main fuel that is used in aviation, but it is expected to change over the next few decades as the industry shifts toward more sustainable options. Demand for SAF is projected to grow rapidly and has the potential to become a significant portion of the aviation fuel supply in the future [5]. However, the key issue is not a lack of demand for aviation fuel, but rather the availability of sufficient SAF supply to replace conventional petroleum-based Jet A fuel. Air travel demand continues to increase, and governments are implementing stricter environmental regulations that require the aviation industry to reduce lifecycle carbon emissions. As a result, SAF is being introduced into the existing fuel market as a lower-carbon substitute for conventional jet fuel rather than creating an entirely new demand category.

Another major factor influencing this transition is the growing number of mandates that require airlines to blend SAF with Jet A in specified percentages. These blending requirements ensure that SAF becomes part of the overall aviation fuel supply. Many of these policies include annual escalation targets, meaning the required SAF blending percentage increases over time. For example, programs such as the U.S. Renewable Volume Obligation (RVO) gradually increase required renewable fuel volumes each year, creating a predictable long-term demand signal for SAF producers. As blending requirements become more common, airlines and fuel suppliers have stronger incentives to secure long-term SAF supply agreements to meet regulatory compliance and sustainability goals.

When it comes to production, SAF manufacturers have strong incentives to scale up capacity because government and industry incentive programs improve project economics and reduce financial risk. These incentives do not directly reduce the physical production cost of SAF; instead, they provide additional revenue streams that help offset the higher cost of producing SAF compared to conventional jet fuel. Programs such as D3 Renewable Identification Number (RIN) credits and the 45Z clean fuel production tax credit make SAF projects financially viable by increasing the value of low-carbon fuel production.

Over time, the actual production cost of SAF is expected to decline as operational knowledge improves and plant sizes increase. This effect is commonly described as the technology learning curve and economies of scale, where larger facilities operate more efficiently and capital costs are distributed over higher production volumes. As more SAF plants are constructed and the industry gains operating experience, improvements in process efficiency, supply chain logistics, and equipment design are expected to reduce unit production costs.

These improved economics make it easier for companies to invest in new facilities or upgrade existing ones. As additional production capacity is built, total SAF supply

increases to meet the existing demand for aviation fuel. Increasing supply also promotes market competition among producers, which can contribute to gradual price reductions over time. In this way, government incentives and industry investment play a critical role in accelerating SAF deployment and supporting the transition from conventional jet fuel to lower-carbon alternatives.

FEEDSTOCKS AND THE COST OF MAKING SAF

Feedstock availability and pricing play a major role in determining the amount of SAF that can be produced and supplied to the market. One of the biggest advantages of the ethanol-to-jet (ETJ) pathway is its ability to use ethanol derived from a wide range of feedstocks [6], including corn, sugarcane, and different types of waste materials such as industrial off-gases and agricultural residues (LanzaTech technology overview; [8]).

The availability of multiple potential feedstocks allows producers to design larger and more economically efficient plants by ensuring sufficient supply of ethanol to support continuous operation at scale. Larger plant sizes typically reduce the unit production cost of SAF because capital and operating costs are distributed over greater production volumes. However, in practice, feedstock selection may be influenced by regulatory requirements and incentive programs. For example, ethanol produced from certain waste-based or cellulosic feedstocks is more likely to qualify for D3 Renewable Identification Number (RIN) credits, which provide higher-value incentives compared to conventional renewable fuels. As a result, producers may prioritize feedstocks that meet these qualification criteria rather than using all technically feasible options. The environmental benefit of using different feedstocks is illustrated in Figure 5 , which shows the percentage reduction in greenhouse gas (GHG) emissions for several LanzaTech ethanol feedstock sources. Feedstocks such as forest residues and corn stover can achieve significant lifecycle emission reductions compared to conventional fossil fuels, making them more likely to meet regulatory thresholds required for low-carbon fuel incentives. These GHG reduction levels are important because eligibility for programs such as D3 RIN credits and other clean fuel incentives depends on demonstrating substantial lifecycle emission reductions relative to petroleum-based fuels

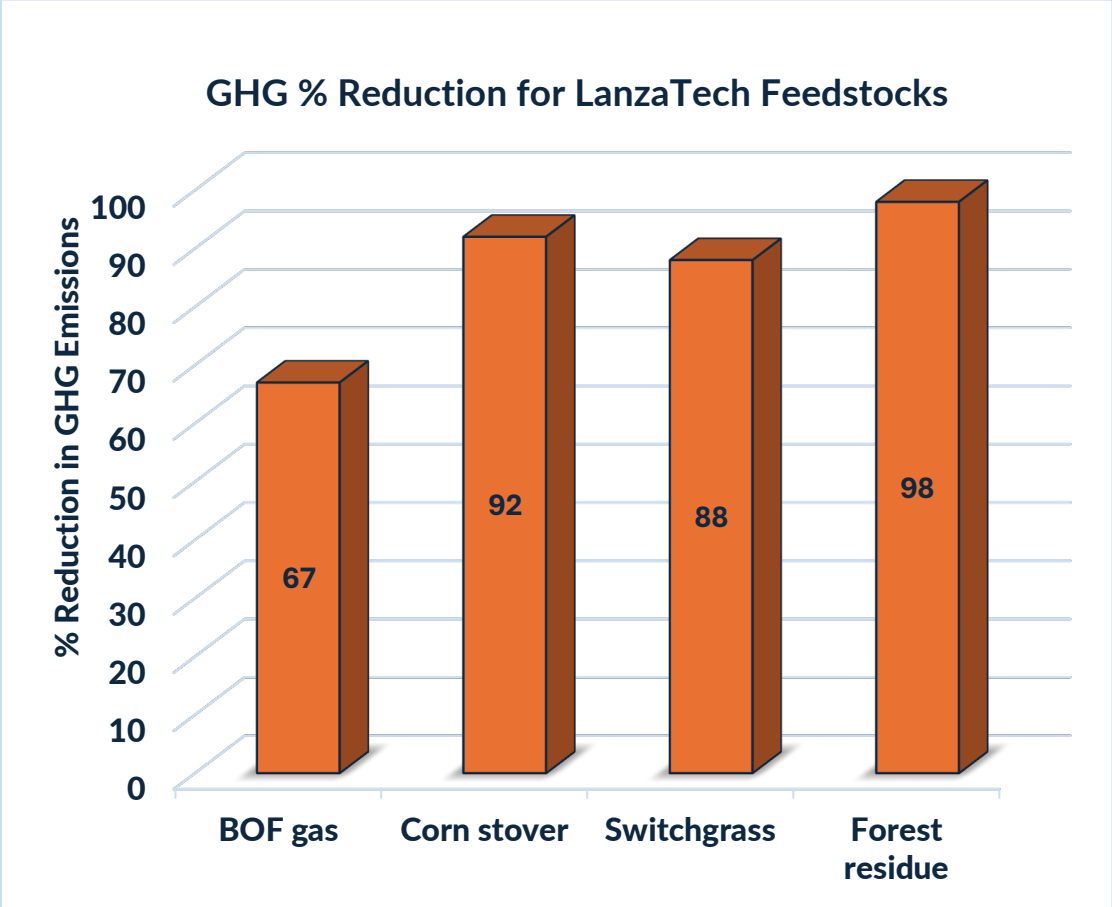


Figure 5- GHG Reduction for Ethanol from LanzaTech Feedstocks

Another factor that affects the economics of SAF through the ETJ pathway is the fact that producers can acquire credits and depending on what kind of credits they get will determine how profitable they will be as long as they are using low carbon ethanol they will be able to qualify for those credits and the lower the carbon intensity the higher credit in our case for this project we qualify for the highest credits.

Since SAF is still more expensive to make than traditional fossil fuels, these incentives will play an important role in lowering the overall costs.

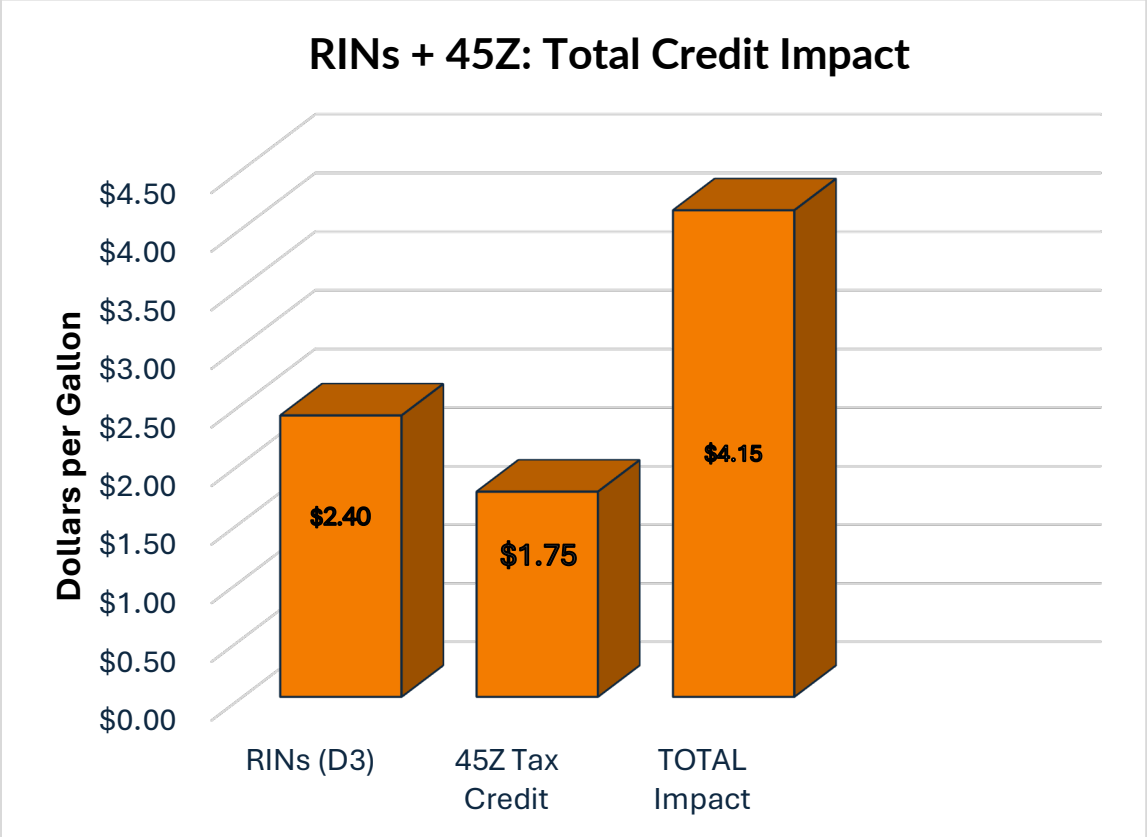


Figure 6- Combined Financial Impact of RINs and 45Z Credit

When refiners produce fuel from renewable feedstocks with low carbon intensity, they can earn Renewable Identification Number (RIN) credits under the Renewable Fuel Standard (RFS) program. RINs are categorized into different types based on feedstock and fuel pathway, such as D3 RINs, which are generated from cellulosic biofuels and typically have the highest market value. Under the RFS, refiners and fuel blenders are required to meet an annual Renewable Volume Obligation (RVO), which specifies the amount of renewable fuel that must be blended into the transportation fuel supply. If a refinery does not meet its required renewable fuel blending target, it must purchase RIN credits on the open market to remain compliant. Conversely, producers of qualifying renewable fuels earn one RIN for each gallon produced and can sell these credits to obligated parties, creating an additional revenue stream for renewable fuel producers.

RIN prices vary depending on fuel category and market conditions, and D3 RINs have historically commanded higher prices because of the limited supply of qualifying cellulosic fuels. In recent years, D3 RIN values have been approximately $2.00–$2.50 per gallon, making them a significant contributor to SAF project revenue. In addition to RIN credits, producers can also benefit from federal tax incentives such as the 45Z Clean Fuel

Production Credit, which provides payments based on lifecycle carbon intensity reductions relative to conventional fuels.

Together, these incentive mechanisms improve the financial viability of SAF production by increasing revenue rather than directly reducing the physical production cost of fuel. As more renewable fuel producers generate RIN credits and expand capacity, the SAF industry can scale production to meet regulatory requirements and increasing demand for lower-carbon aviation fuels. Historical Renewable Volume Obligations (RVOs) and RIN price trends are shown in Figure 7, illustrating the increasing regulatory demand for renewable fuels and the economic value of RIN credits over time.

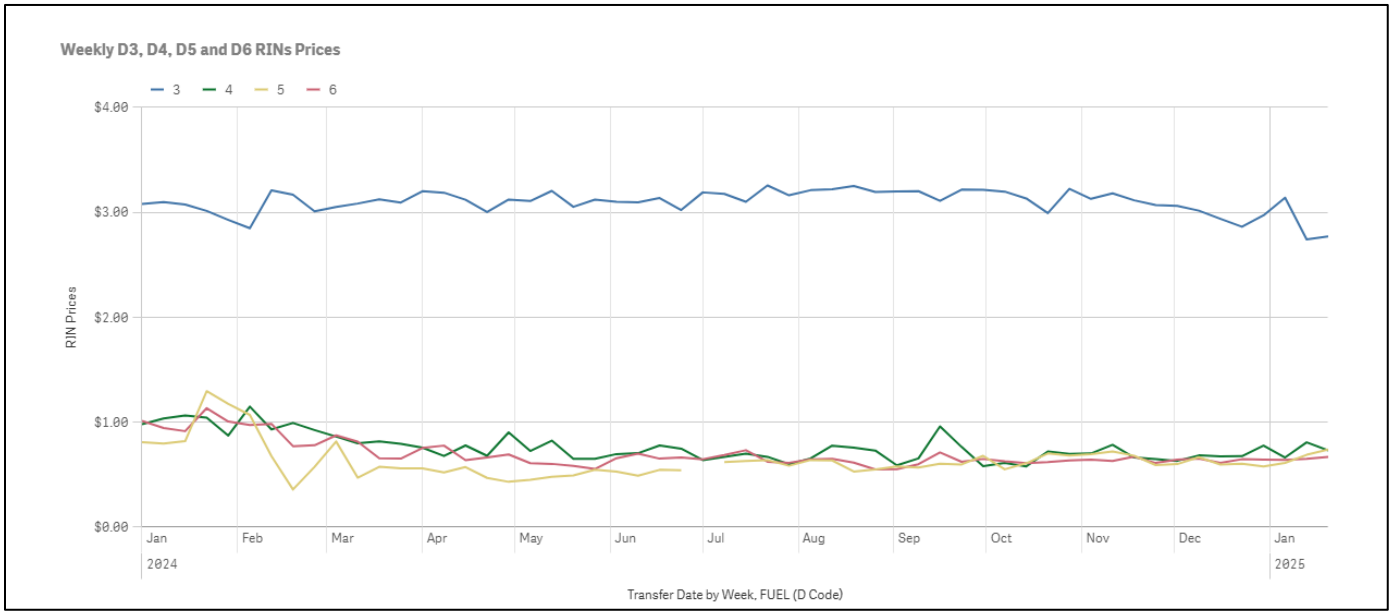


Figure 7- RIN price trends

Competing Processes

HEFA (Hydro processed Esters and Fatty Acids)

HEFA ((Hydro processed Esters and Fatty Acids) was the first process to achieve commercial scale SAF production and remains the most implemented process today. It converts fats, oils, and grease (FOG) into renewable fuels such as SAF and renewable diesel. This route is the dominant source of SAF currently in the market.

The process requires feedstock pretreatment as shown in Figure 8 [9].

The pretreated feedstock enters the hydroprocessing reactor where oxygen is removed by hydrodeoxygenation (HDO) reactions shown in Figure 9 by hydrogen gas in the presence

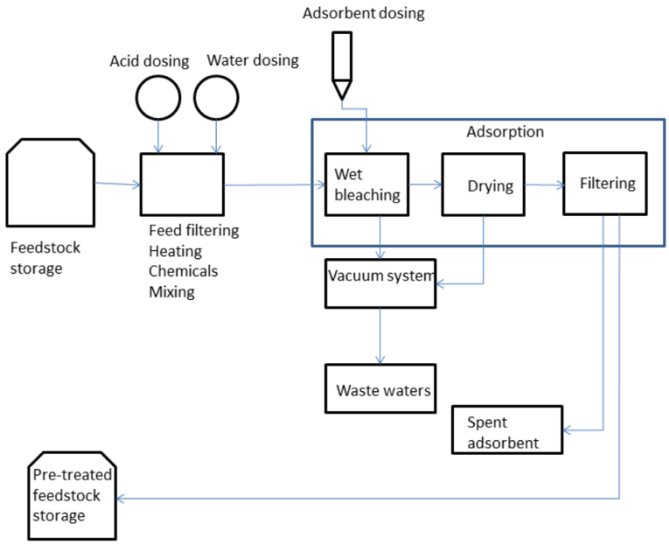


Figure 8: HEFA feedstock pretreatment

of fixed-bed catalysts (typically nickel-molybdenum supported on alumina substrate). The HDO process saturates the olefinic bonds in the triglycerides and systematically removes carboxyl and hydroxyl groups by hydrogenolysis of the C-O bonds, transforming triglyceride molecules into linear paraffinic hydrocarbons. In the second stage reactor, isomerization and cracking over Pt on chlorided alumina catalyst completes the conversion to jet-range hydrocarbons that meet the freeze point of < -40 ° and other

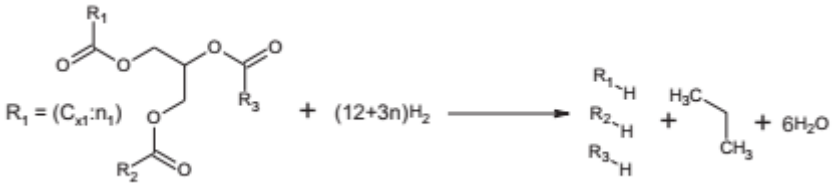


Figure 9:Triglyceride hydrotreating reaction

jet fuel requirements. The final distillation then separates the product into gasoline, kerosene/jet fuel and diesel giving suppliers a flexible product slate to meet market demand

Despite its strong commercial position, the HEFA pathway has limitations. HEFA plants depend on the limited supply of fats, oils, and greases (FOGs) such as used cooking oil and waste animal fats, which are also in high demand from biodiesel producers. As new HEFA capacity comes on board, competition for these low-carbon feedstocks is intensifying, and several studies suggest that sustainable HEFA feedstock supply could cap global bio-SAF production at only a small fraction of long-term jet fuel demand. For this reason, HEFA is viewed as a critical near-term solution rather than the sole, long-term answer for aviation decarbonization.

This is the block flow diagram of the HEFA pathway. **Error! Reference source not found.**

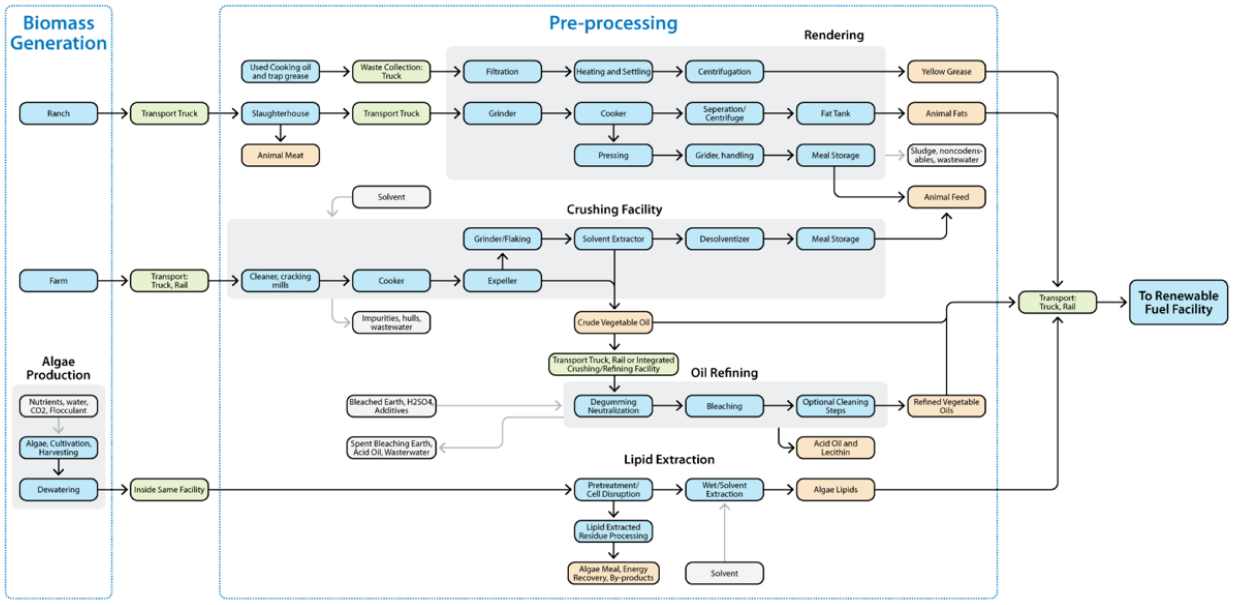


Figure 10: HEFA block flow diagram part 1

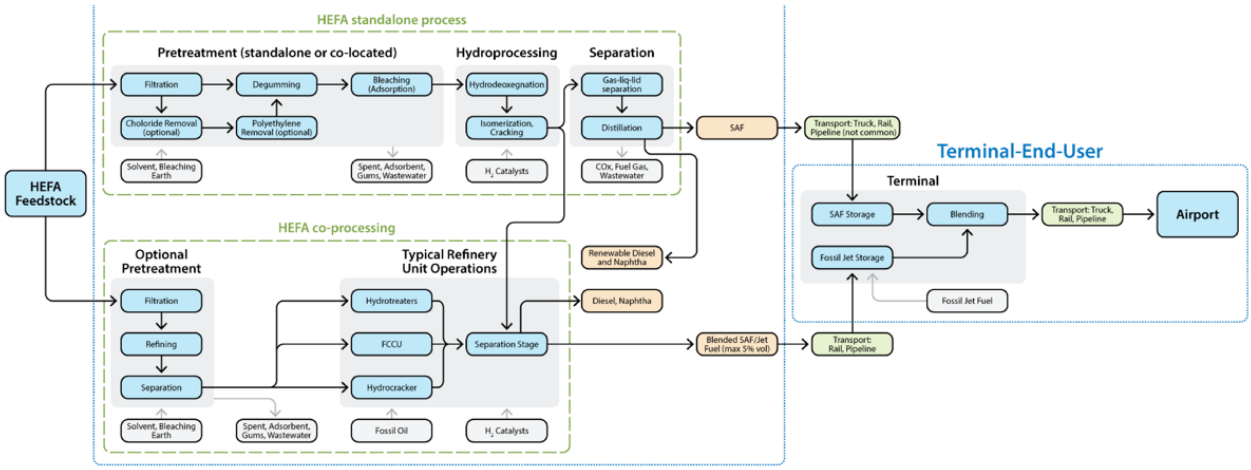


Figure 11: HEFA BFD part 2

GASIFICATION /FISCHER-TROPSCH

The Fischer-Tropsch (FT) synthesis converts syngas (a mixture of carbon monoxide and hydrogen) to liquid hydrocarbons via catalytic oligomerization reactions. The FT route to SAF starts with the gasification of biomass, where solid feedstock (wood chips, agricultural residues, energy crops, or municipal solid waste) undergoes a high-temperature (700-900°C or higher), decomposition reaction through controlled introduction of air or oxygen to partially oxidize sufficient feedstock to heat the feedstock to gasification temperature and produce raw syngas primarily consisting of CO_2, H_2O, CH_4, CO, and H_2. After gasification, strict syngas cleaning takes place, where the slag, particulates, condensable tars, nitrogen and sulfur compounds are removed to levels that are compatible with downstream FT catalysts (usually <1 ppb sulfur). The clean syngas is then sent to the FT synthesis reactor where the catalytic reactions over iron or cobalt catalysts build of the carbon chain through the Fischer-Tropsch mechanism:

$$nCO + (2n+1)H_2 \rightarrow C_nH_{2n+2} + nH_2O.$$

This process is exothermic and generates a wide range of hydrocarbons from C_1 (methane) to C_{20}+ (heavy waxes), and the distribution of products is determined by the reactor design, temperature, pressure and catalyst. Low-temperature Fischer-Tropsch (LTFT) reactors, working at 180-250°C, deliver products with narrow boiling point ranges more advantageous for SAF (Sustainable Aviation Fuel) production compared to their high-temperature counterparts. The FT products then undergo hydrocracking where high-molecular-weight waxes are split into jet-range hydrocarbons, and the linear paraffins are transformed into branched ones thereby lowering the freeze point of jet fuel to meet the Jet A specification of -40°C. Finally, the product fractions are distilled into

gasoline, kerosene (jet fuel), and diesel fractions.**Error! Reference source not found.Error! Reference source not found.**

One of the key advantages of the gFT pathway is its feedstock flexibility**.** In contrast to HEFA, which relies on limited supplies of waste oils and fats, gFT can use a broad range of low-value resources, such as forestry residues, agricultural wastes, and other biomass streams that are expected to be more scalable in the long term.

However, gFT also faces several important challenges. First, the process has high capital cost, because it requires large, high-temperature gasifiers, extensive syngas cleaning, and Fischer–Tropsch reactors that operate at elevated temperatures and pressures, making the pathway less attractive economically than some alternatives. Second, inorganic species in the biomass (mainly alkali and alkaline-earth metals such as Na, K, Ca, and Mg) can form low-melting slags that accumulate in the gasifier, causing fouling, corrosion, and downtime. Finally, biomass gasification tends to generate tars and other condensable organics that are difficult to remove and require additional scrubbing or reforming steps, increasing operating costs and process complexity. Besides that, the formation of tar leads to difficult waste streams. Solid waste contaminated with tar is usually referred to as special waste, which makes its disposal complicated and costly, while wastewater containing tars becomes a two-phase mixture that cannot be released before undergoing several treatment steps. A great number of water-soluble tar compounds are hardly biodegradable, so typical biological wastewater treatment is usually ineffective in such cases, which additionally raises operational costs and the need for environmental management. Here is an overall block flow diagram of gFt process [10].

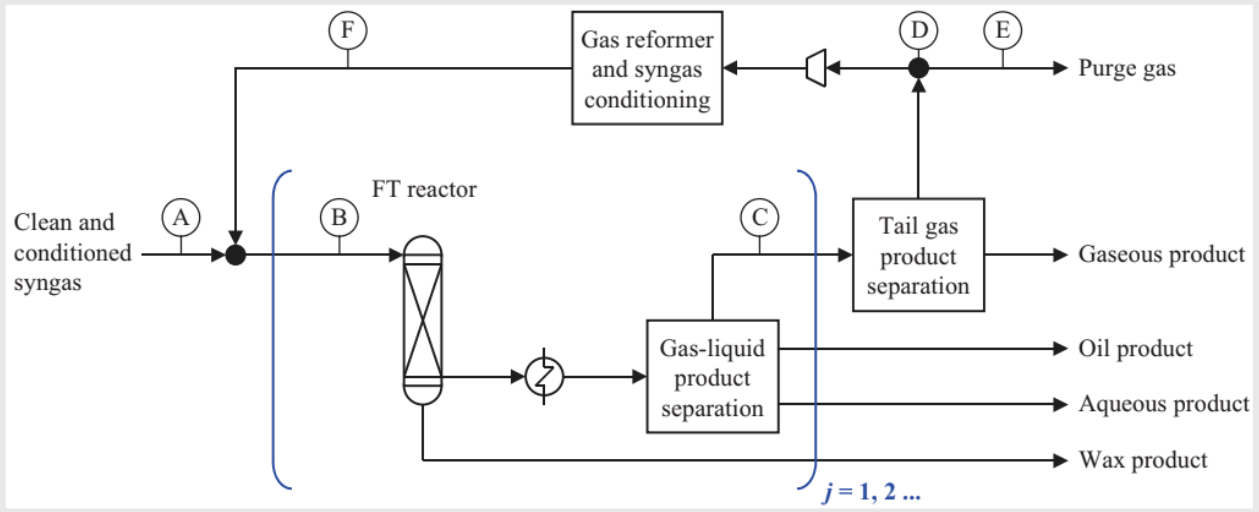


Figure 12: gFT block flow diagram

PYROLYSIS/HYDROTREATING (PTJ/PYC3) PROCESS

Pyrolysis is a decomposition pathway that involves rapid heating of solid biomass in an oxygen-free environment to break complex polymeric structures (cellulose, hemicellulose, lignin) into lower-molecular-weight liquid and vapor products that can be further upgraded. In fast-pyrolysis processes, the biomass feedstock, which is finely ground (usually below a few millimeters), is rapidly heated to about 450–600 °C in the absence of oxygen. The biomass polymers fragment to smaller molecules, and a liquid intermediate (bio-oil), non-condensable gases (mainly H_2O, H_2, CO, CH_4, CO_2), and solid char are produced. The bio-oil obtained contains 25–40 wt% oxygen which is distributed among hydroxyl, carbonyl, and carboxylic groups. This results in high polarity, chemical instability, low pH, and makes storage and logistics difficult. Char and product gas are generally combusted to dry the incoming biomass and to reheat the circulating sand that provides the endothermic heat of reaction for pyrolysis.

The condensed bio-oil is sent to two hydrotreating stages where it combines with hydrogen and reacts over a fixed-bed, sulfided Ni-Co-Mo on titania catalyst at approximately 180-350°C and about 800-2000 psig for oxygen removal and liquid stabilization. The first stage is at relatively mild temperature, 180 °C, and is aimed at reducing acidity and heteroatoms, while the second stage is at higher temperature and further deoxygenates the material. There is only a small amount of hydrocracking that is needed because most of the upgraded product is already in the gasoline boiling range after hydrotreating; a final fractionation step thus separates gasoline, kerosene/jet, and diesel cuts that are of refinery blend stock specification Here is a simplified process flow diagram of the fast pyrolysis process [11]

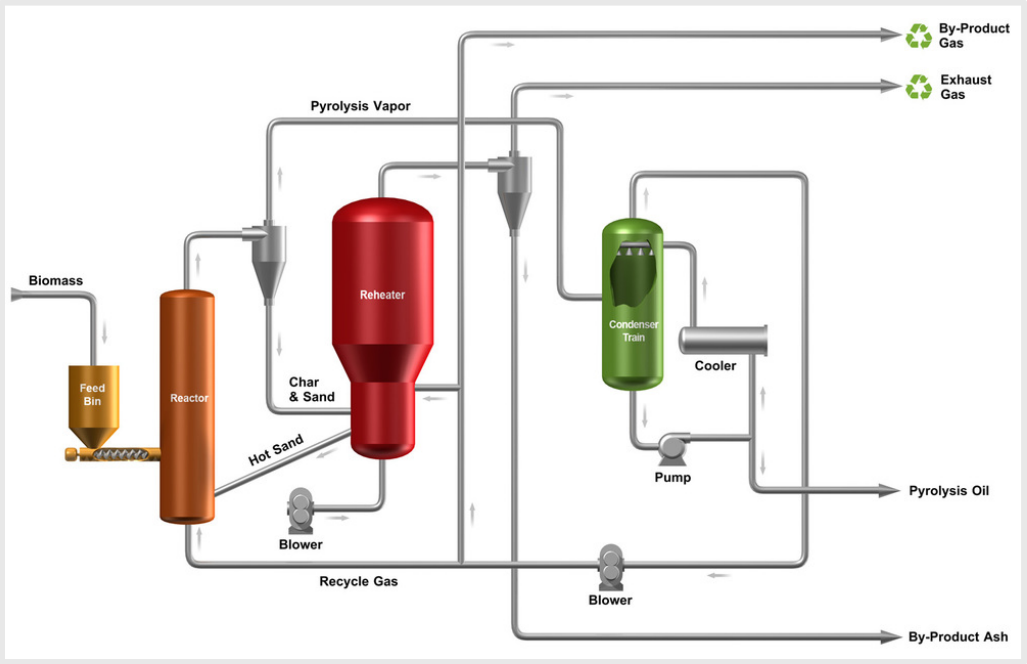


Figure 13:Pyrolysis Block flow diagram

Pyrolysis-hydrotreating has advantages over gasification-Fischer-Tropsch, but on the other hand, it also has disadvantages that make the overall performance of ATJ comparatively better. In pyrolysis, many problems related to gasification such as slag and tar formation are avoided, lower temperatures are used, and generally less CAPEX is required as compared to full gasification and FT synthesis. Also, the hydrodeoxygenation (HDO) reaction in the case of PHDO-SAF is simpler and less expensive than an FT loop with large reactors, recycle compression, and extensive product work-up. On the other hand, PHDO-SAF disadvantages are many. Firstly, bio-oil is unstable, the oxygen-rich liquid is acidic and chemically very reactive, it polymerizes upon heating, in storage, and during handling since it reacts with itself thus increasing viscosity and leading to phase separation over weeks to months. Therefore, stainless steel equipment, cooled storage, and stabilizer addition are necessary, which makes the logistics more complicated than those of ATJ where ethanol feed is very stable. Moreover, the reactor plugging and low catalyst life in HDO units is still not fully understood, thus on-stream time is limited, and a significant amount of R&D work is required before a reliable commercial operation can be achieved, whereas ATJ uses dehydration and oligomerization catalysts that are well-known and widely used in the petrochemical industry. Capital intensity is quite high because a commercially viable PHDO plant requires several fast-pyrolysis units (e.g., RTP) to feed a two-stage HDO section, thus the total CAPEX can be higher than that of an ATJ plant of the same size in terms of SAF output. Lastly, although PHDO might require less hydrogen per carbon than some other routes, it still requires roughly one mole of hydrogen for every atom of oxygen removed from the bio-oil, while ATJ typically requires less than one mole of hydrogen per molecule of oligomer formed, which makes for lower operating costs when low-carbon hydrogen is expensive.

ALDER PYROLYSIS OIL SOLVENT FRACTIONATION + HYDROTREATING PROCESS

Alder Fractionation is a bio-oil fractionation process that uses selective solvents to separate bio-oil into a concentrated organic phase, which contains the jet fuel precursors, and an aqueous phase, which contains the pyrolysis sugars. The Alder Renewables technology does this by mixing bio-oil with a solvent and an anti-solvent, separating the phases, and then recontacting the aqueous phase with a solvent to maximize organic recovery. The solvent and anti-solvent are recovered by distillation and recycled, the organic product is sent to the HDO unit, and the aqueous product is de-watered to make a liquid that can be used as a renewable fuel oil. Alder Renewable Crude (ARC), which is the organic stream, is hydrotreated as in the pyrolysis/HDO routes, to remove the oxygenates and hydrocracked to adjust the molecular weight to jet boiling range. The product can be blended with the jet fuel produced by the HEFA to form a 100% renewable jet fuel. The Alder product currently must be blended with conventional HEFA products to reach ASTM specifications. [12]

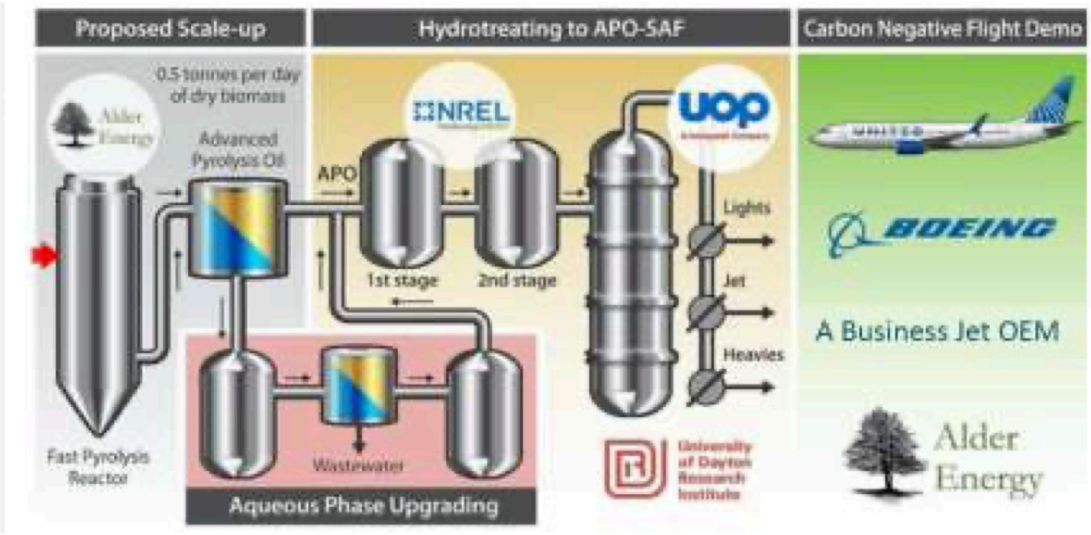


Figure 14:Block flow diagram of Alder Fractionation+UOP hydrotreating

UOP TWO STAGE PYROLYSIS OIL UPGRADING PROCESS (SLURRY REACTOR THEN FIXED BED)

The first stage reactor is a variation of the slurry hydrocracking reactor of UOP and the second stage is a traditional, fixed-bed reactor. Liquid product recycle is used in both

stages to control the exothermic heat of reaction. The first stage reactor of pyrolysis partially deoxygenates the oil and thereby stabilizes it to avoid polymerization/plugging in the second stage reactor. In the first stage, deoxygenation is accomplished with suspended catalyst particles in the liquid medium of hydrocarbons at 220-250 °C and 25-35 bar pressure. The slurry reactor tolerates the solids formed from the polymerization of the unstable feed and the heat generation from the deoxygenation reaction. The slurry reactor effluent is sent to the second stage which is a fixed-bed reactor that operates at higher temperatures (350-380 °C) and hydrogen pressure (60-100 bar) with a solid catalyst (nickel-molybdenum on alumina) to convert the remaining oxygenates to hydrocarbons. A final distillation step separates the product into gasoline, jet fuel and diesel boiling fractions.

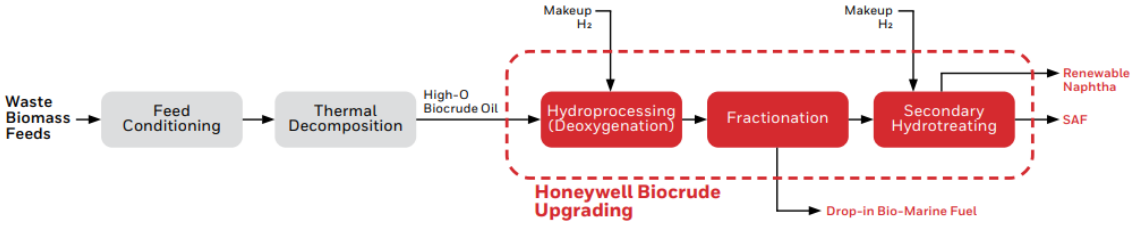


Figure 15- UOP Bio Crude Upragding BFD

Commonly used slurry-phase hydrotreating catalysts are molybdenum or nickel-molybdenum (and possibly nickel-cobalt-molybdenum) catalysts formulated on a titania or alumina support, in line with the general experience of slurry hydrocracking and heavy-oil upgrading technologies like Uniflex [13].

HYDROTHERMAL LIQUEFACTION (HTL/IH^2 - CRITERION/GTI) PROCESS

Hydrothermal liquefaction (HTL) is a thermochemical direct conversion route that converts wet biomass into a petroleum-like biocrude under pressurized, three-phase reaction conditions (solid biomass and catalyst, organic liquid product and water, and gases) at moderate temperatures of approximately 250-350 °C and high pressure, usually in the 10-25 MPa range. However, in contrast to pyrolysis or gasification, HTL uses liquid water at almost-critical temperatures thus water is used both as the reaction medium and an active reactant in the hydrolysis and reforming steps instead. Wet biomass is slurried to a concentration level of about 15-25 wt% dry solids and pumped into the HTL reactor. Lignin, cellulose, and hemicellulose, the macromolecular constituents of the feedstock, depolymerize in the HTL reactor, initially to smaller fragments and subsequently react by repolymerization, condensation, and rearrangement to form a biocrude phase. The average biocrude yields are around 30-45 wt percent on a dry-biomass basis, depending on conditions and feedstock. The biocrude produced has about 5-20 wt % oxygen and needs further catalytic hydrodeoxygenation to convert the product into finished hydrocarbon fuels [14].

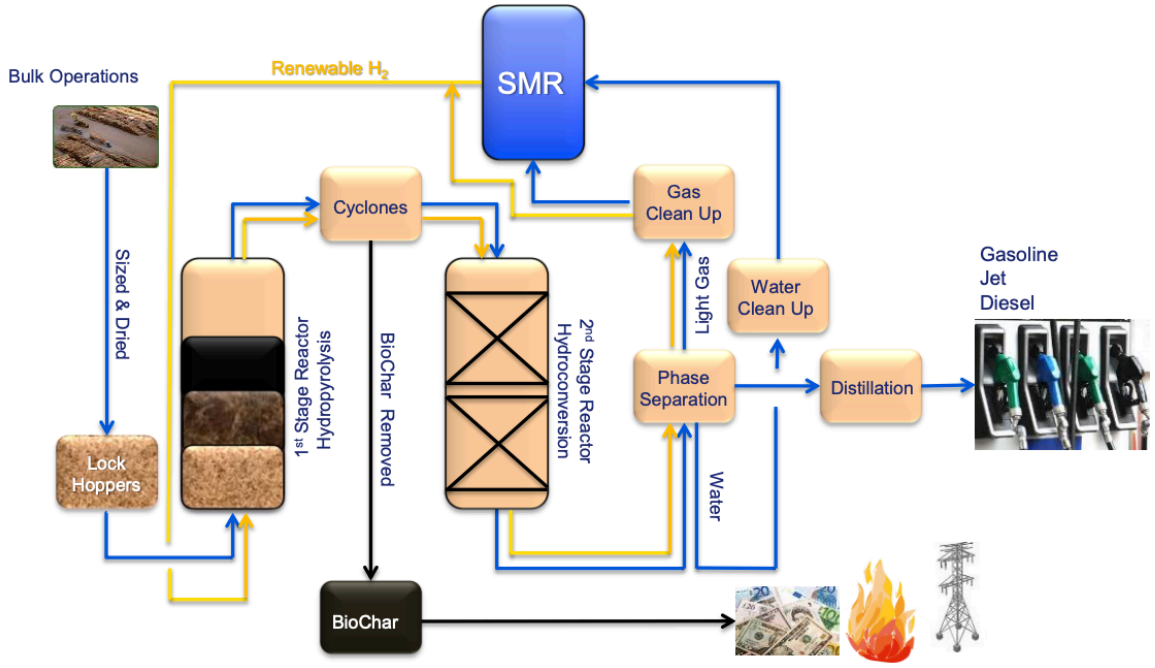


Figure 16:HTL BFD

GTI has remaining technical and economic barriers to commercializing their HTL SAF process. First, the Technology Readiness Level remains at 5-6 based on mostly pilot scale operation but also a demonstration-scale unit in Bangalore; no commercial-scale SAF production exists, giving significant execution and financing risk versus ATJ's proven commercial scalability demonstrated across multiple independent facilities [3][16][17]. Second, biocrude stability and upgrading complexity remain incompletely understood at the commercial scale, with proprietary data from limited pilot operations restricting third-party validation, vs ATJ's transparent technical disclosure and established chemistry proven through decades of ethanol dehydration research and development [3][10]. Third, the capital intensity of high-pressure reactor systems ($6-10 per gallon per year capacity) substantially exceeds ATJ's $0.50-1.00/gal-yr with commercial-scale cost reductions remaining speculative pending demonstrated operations [5][16][17].

Chosen Process

The selected process for producing sustainable aviation fuel is an ethanol to jet fuel pathway that transforms ethanol into jet range hydrocarbons through three major reaction blocks. These blocks are: catalytic dehydration of ethanol to ethylene, followed by oligomerization of light olefins into mid weight hydrocarbons, and ending with hydroprocessing, which involves hydrogenation, hydrocracking, and isomerization. Together they form an integrated system that converts material from low molecular weight alcohol into a drop-in, petroleum-product-compatible aviation fuel. Our process design is based on the configuration and information provided in the patents US 2024/02117895 A1 and US 2024/0218268 A1, which give significant details regarding the catalysts, reactor arrangements, and purification steps necessary for this ethanol to jet fuel conversion pathway. The overall flow scheme from the provided patents maximizes the yields in the jet range and ensures that each block supplies the necessary feedstock and product quality for the subsequent stage.

Process Description

DEHYDRATION SECTION

Referring to Figure 24, The ethanol feed enters the process at the azeotropic composition of ethanol and water, which is typical for fermentation-derived ethanol streams. The feed first flows into a surge tank, where it is temporarily stored to ensure a consistent and uniform composition and flow rate to downstream equipment. This surge capacity helps dampen fluctuations in feed supply and improves overall process stability and control.

From the surge tank, the ethanol stream is transferred using a centrifugal pump, which raises the pressure of the liquid to the required operating level for subsequent processing steps. The pressurized ethanol stream then enters a vaporizer, where heat is supplied to convert the liquid ethanol-rich stream into vapor prior to entering the dehydration reactor system. Vaporization of the feed is necessary to achieve the desired reaction conditions, improve heat transfer, and ensure efficient contact between the vapor-phase ethanol and the solid catalyst within the reactors.

The overhead stream D3 is joined with a recycled ethanol stream and then split into 2 ethanol streams before entering the dehydration reactor section. The first ethanol stream D7 is mixed with MP steam, preheated by exchanging heat with the R-102 effluent in a heat exchanger, and then heated to reaction temperature in fired heater F-100 before entering the first dehydration reactor R-100. The main reaction for the conversion of ethanol to ethylene by removing water as a byproduct: $C_2H_5OH \rightarrow C_2H_4 + H_2O$ dominates at our selected temperatures 390 to 420 degrees Celsius, but a secondary reaction: $2C_2H_5OH \rightarrow C_2H_5OC_2H_5 + H_2O$ can also occur, forming diethyl ether as a byproduct. However, at higher temperatures, diethyl ether cracks back through this reaction: $C_2H_5OC_2H_5 \rightarrow C_2H_4 + C_2H_5OH$, so ethylene formation is favored as long as the high temperature remains in the correct range [15].

Dehydration is an endothermic reaction so the first dehydration reactor R-100 cools as the reaction proceeds. The R-100 reactor effluent is mixed with the second split ethanol stream D13 mentioned above and enters charge heater F-101 and then the next reactor R-101. The temperature drops from the endothermic reaction again so the R-101 effluent is reheated in F-102 before entering the third and final reactor R-102. Three reactors are used because the temperature decrease in each reactor reduces the reaction rate and limits conversion, so reheating between reactors maintains the required reaction temperature. The final reactor is operated at the highest temperature to maximize ethylene selectivity and minimize the formation of dimethyl ether (DME). With this configuration of the reactors, the conversion of ethanol to ethylene will be maximized [16].

By the time the mixture exits R-101 in stream D16, the ethanol is largely converted, but the vapor phase still contains significant water, unconverted traces of ethanol, and a range of oxygenates species, including dimethyl ether. These compounds are removed as much as possible in R-102 before the oligomerization section because zeolite catalysts, especially those used in producing mid-range olefins, are severely deactivated by water

and oxygenates. The product stream D18 exchanges heat with both ethanol feed streams D8 and D12 to recover heat from the reactor effluent while avoiding fouling in the heat exchangers and minimizing pressure drop in the condenser system.

The cooled stream D23 enters the quench absorber T-101, which quickly reduces its temperature by direct contact with cooled, circulated liquid from E-104 and E-105. The cooling water to E-105 is supplied from the outlet of E-104 to prevent overheating of the cooling water and reduce the potential for scaling within the exchanger system. As stream D23 cools, both water and heavier oxygenates condense out of the ethylene product. These condensed materials form a liquid stream D25 that exits the bottom of T-101 and flow to the wastewater stripper T-104. The vapor stream D26 leaves the top of the quench absorber and heads into the compression train, but at this point it still contains some impurities that must be removed.

The compression system raises the wet ethylene gas stream to oligomerization pressure while allowing staged removal of water and oxygenates. In the first stage compressor K-100, the gas temperature increases due to the compression work. It is then cooled in an air cooler E-106 and a trim condenser E-107, which causes additional water to condense. That liquid is then removed in the knockout drum D-101. The second compression stage follows the same logic: K-101 raises the pressure again; the air cooler and trim condenser E-108 and E-109 cool the discharge and D-102 removes the condensed liquid. Without interstage cooling, the gas would reach high temperatures that could damage the compressor.

The compressed gas still contains dissolved oxygenates. Removing these oxygenates requires selective washing steps. The first step occurs in water wash tower T-102, where hydrophilic impurities dissolve in the wash water. Downstream of the wash towers is a knock-out drum to capture any entrained liquid before the stream enters the two-stage dryer system, DR-100 and DR-101, to lower the water content to sub-ppm levels, typically corresponding to a dew point of approximately −100°F, which is required to avoid ice formation in the refrigerated section and protect downstream equipment.

The two dryers operate in a lead–lag configuration, where one dryer is in service removing moisture while the second dryer is offline undergoing regeneration. During regeneration, the saturated adsorbent is heated and purged to remove accumulated water, restoring its adsorption capacity for continued operation. The extra drum downstream of the dryers serves as part of this regeneration and separation system, ensuring that any condensed liquid formed during regeneration or cooling is removed before the gas continues through the process.

Then, the ethylene-rich stream is sent to a final distillation column T-106 to purify it from the remaining oxygenates in the mixture. The ethylene stream D58 exits from the top of the column and is sent to a compressor to raise the gas to the required pressure range for oligomerization, approximately 1000 psig, while the bottoms stream D57 contains the remaining oxygenates and heavier impurities. In addition, a portion of the recovered ethanol from the wastewater stripper T-104 is recycled back to the process as stream D45

via drum D-103, where it is combined with the fresh feed to improve overall ethanol utilization and reduce raw material losses.

OLIGOMERIZATION

Referring to Figure 28, the oligomerization section is where the core carbon backbone for jet fuel molecules is built. Converting ethylene to oligomers is a highly exothermic reaction and without mitigation of this high heat, it could lead to a runaway reaction producing undesired products, deactivating catalyst and potentially causing damage to the reactor. One of the measures taken to manage this exotherm is splitting the first stage reactor into 4 catalyst beds with interstage steam generators to cool the product between reactors. A second measure is to provide a paraffin diluent from the hydroprocessing fractionator, to act as a heat sink and absorb a significant amount of this heat. The diluent is carried through the process unchanged as it is inert to the oligomerization and hydrogenation reactions. The ethylene stream D59 from the dehydration section mixes with paraffinic diluent stream H24 (coming from the final jet fuel product in the hydroprocessing section) forming stream O3A. This stream is then cooled in E-200A before mixing with the light olefin recycle stream O13 and the combined stream enters the first catalyst bed R-200A. Inside the first fixed catalyst bed, ethylene reacts over the alumina-bound MTT zeolite catalyst, through a carbocation mechanism [17]. Ethylene is protonated to form $C_2H_5^+$, which reacts with more ethylene to form $C_4H_9^+$. This propagation continues stepwise, producing species such as $C_6H_{13}^+$ and $C_{18}H_{17}^+$. In general, the growth follows $C_2H_4 \rightarrow C_{2(n+1)}H_{4(n+1)}^+$. Termination typically occurs through beta-hydride elimination, forming an alpha olefin according to $C_{2(n+1)}H_{4(n+1)}^+ \rightarrow C_{2n}H_{4n} + H^+$. The general chemical equation for oligomerization can be summarized with: $nC_2H_4 \rightarrow nC_{2n}H_{4n}$. The same reaction applies across the remaining 3 catalyst beds R-200B, R-200C, and R-200D. These four fixed beds enable growth to mid-range olefins while preventing runaway temperature. The MTT pore structure imposes a natural limit on chain length, usually centered around the C_8 to C_{12} range, which is suitable for jet fuel blendstock.

To form heavier oligomers, the effluent from the fourth bed, stream O4D, is cooled in steam generator E-201 before entering R-201 as stream O6. The catalyst in R-201 is a Nickel on amorphous silica-alumina. Its pore structure is more open than MTT, allowing larger molecules to diffuse in and out of the catalyst. In this reactor reactions such as

$$nC_2H_4 \rightarrow C_{12}H_{24},\ C_{14}H_{28},\ C_{16}H_{32}$$

and others occur, producing the heavier molecules needed to achieve the full jet fuel boiling range. The R-201 product stream O7 first enters the deethanizer column, where unreacted ethane and lighter components are removed from the heavier olefin stream to improve downstream separation efficiency. The resulting stream then proceeds to the olefin splitter T-200, where the reactor operating conditions were selected based on

literature and patent data consistent with LanzaTech and related oligomerization technologies [18] [19] [20].

The olefin splitter separates the olefins lighter than heptane from the heavier olefins. Light olefins are taken overhead as stream O11, after compression in K-200 and recycled as O13. Recycle loops are common in oligomerization because equilibrium limitations and selectivity considerations often prevent complete conversion in a single pass, so recycling unreacted light olefins improves overall conversion while maintaining optimal reaction conditions without requiring unnecessarily large reactor volumes. A portion of stream O11 is purged as stream O10 if needed to remove impurities such as carbon dioxide, carbon monoxide, nitrogen compounds, oxygenates, or other inert species that may accumulate in the recycle loop and negatively impact catalyst performance or system operation.

C7− olefins leave as a drag stream and are sent to the hydroprocessing section. The heavy oligomers leave the bottoms in O14. This stream is then cooled in E-202 by heat exchange with the R-201 effluent stream and is then sent to the hydroprocessing section.

HYDROPROCESSING

Referring to Figure 31, in the hydroprocessing section, three key transformations take place: hydrogenation, hydrocracking, and isomerization. The heavy oligomer feed stream H1 (O16) mixes with the C_7-drag stream and compressed hydrogen and is precooled in an air cooler to the reaction temperature for hydrogenation in R-300. The hydrogenation reaction saturates the double bonds through reactions such as $C_nH_{2n} + H_2 \rightarrow C_nH_{2n+2}$. The reaction utilizes a platinum on alumina catalyst to produce paraffins. The reactor effluent stream H6 moves through E-300, E-303, and a fired heater F-300 to raise its temperature to reaction conditions for R-301.

The first catalyst bed in R-301 uses a nickel tungsten catalyst supported on FAU and BEA zeolite. These catalysts have large pores that permit the diffusion of long-chain hydrocarbons.

Once protonation takes place, these long chain molecules undergo beta scission following general chemical reaction: $C_mH_{2m+2} \rightarrow C_xH_{2x+2} + C_{(m-x)}H_{2(m-x)+2}$. An example of one of the reactions that can take place in this hydrocracking reactor can be represented as: $C_{20}H_{42} \rightarrow C_{12}H_{26} + C_8H_{18}$. This cracking step is essential because the mixture coming from oligomerization is often heavier than the desired jet fuel range molecules. After the hydrocracking reactions, the mixture continues into the isomerization bed which contains a molecular sieve catalyst SAPO-34 with alumina binder. This catalyst promotes molecular rearrangements such as $nC_{12}H_{26} \rightarrow isoC_{12}H_{16}$ that improve cold flow properties that are ASTM required for jet fuel by generating branched paraffins.

The effluent H10 is cooled before entering the hot separator D-300 as stream H11. The overhead vapor H13 contains hydrogen and light hydrocarbon gases, while the bottoms H14 contains the heavier liquid hydrocarbons. After additional cooling, the stream H12

gas mixture enters a cold separator D-301 to separate any liquid where H17 the overhead is removed as hydrogen rich vapor and recycled to reactor R-300 via compressors K-300 and K-301. A small purge H26 is required to remove impurities such as CO, CO_2, ethane and methane to maintain recycle gas hydrogen purity in the process. The liquid from the bottoms of both separators pass through a heat exchanger and enter the hydrocracking fractionator T-300. C_7- Light Naphtha exits the overhead as H19, C_8-C_{17} jet fuel is withdrawn as H20, and C_{18}-C_{22} diesel range hydrocarbons exits the bottoms as H22. A portion of the paraffinic jet fuel H24 is sent to the oligomerization reactor as inert stream O2 to help manage the exothermic reaction. The figure below shows the relationship of jet fuel yield, temperature, and the combination of catalyst required to reach maximum jet fuel yield of about 78% at about 342 ℃ with a 5% Ni/W on FAU + 95% SAPO-34 hydrocracking catalyst blend [21].

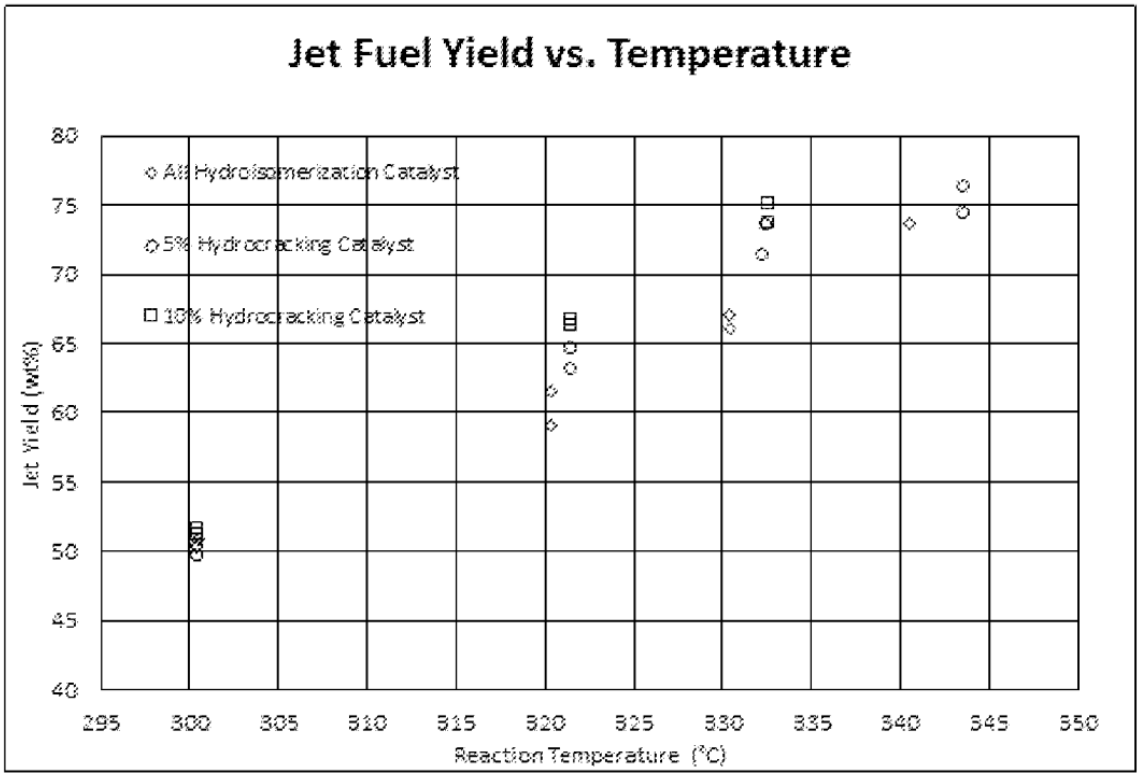


Figure 17- Jet Fuel Yield vs. Temperature [21]

Advantages & Disadvantages

The chosen process has meaningful advantages over the comparative processes. The combination of oligomerization and hydroprocessing produces high yields of hydrocarbons in the jet fuel range. The isomerization step tailors the freeze point, enabling the fuel to meet the ASTM specification. Green ethanol, the feedstock for this process, can be derived from many sources such as bio-fermentation, pyrolysis, or gas-fermentation from corn, sugarcane, waste, and other starch/sugar feedstocks. Another advantage is that there is no formation of char or tar, resulting in a clean process with a high carbon efficiency. Also, the addition of the recycling loops across the plant helps conserve hydrogen and olefins. An ETJ process has recently been commercialized by LanzaJet, and their product has achieved ASTM D7566 specification.

Along with these advantages, there remain a few disadvantages with the ETJ SAF pathway. The process requires a significant hydrogen supply for hydrogenation, hydrocracking, and saturation reactions; however, the overall hydrogen consumption should be evaluated relative to other refining and upgrading processes before characterizing it as unusually high. The catalysts are sensitive to impurities, and the process needs multiple reactors and requires careful thermal management. In addition, the dehydration and hydroprocessing reactions generate a substantial amount of byproduct water, which must be treated before discharge or reuse. The design includes wastewater treatment units such as stripping and separation systems to remove dissolved organics and oxygenates. While water treatment adds operational complexity, the resulting wastewater stream is primarily composed of water and dilute organics, making treatment technically feasible using conventional industrial wastewater processes. Despite having these challenges, ethanol to jet fuel remains one of the most technically promising options to produce sustainable aviation fuel.

Process Safety

The purpose of this section is to identify the main hazards, emissions, waste streams, and control measures associated with the ETJ process. The scope includes major material hazards, process-wide operational safety concerns, regulatory context, environmental controls, and a node-based HAZOP for the product fractionation column.

Process Inputs, Outputs, Emissions, and Waste Streams

The environmental assessment of the ethanol-to-jet (ETJ) process begins by identifying all material streams that enter and leave the plant. These outputs include final products, intermediate by-products, atmospheric emissions, aqueous discharges, and solid wastes. Not every stream leaving the process is considered waste; some streams are valuable products or recyclable materials, while others must be treated or controlled before discharge.

Classifying these streams into categories such as product streams, vent gases, aqueous effluents, and solid wastes provides a clear framework for evaluating environmental impacts. This classification also helps determine which regulatory standards and environmental controls apply to each stream.

Table 1- Major inputs, outputs, emissions, and waste streams for the ETJ facility

Category	Representative Stream	Why It Matters	Typical Handling
Primary product	Sustainable aviation fuel (C_8–C_{17} range)	Main economic output; final product quality controls determine freeze point, volatility, and marketability	Product storage, QA/QC testing, controlled transfer
Byproduct	Light naphtha overhead	Flammable light-end stream with higher vapor pressure than jet fuel; may be sold as blendstock	Condensation, reflux drum handling, product storage
Byproduct	Diesel-range bottoms fraction	Heavier hydrocarbon stream; combustible liquid with lower volatility but high fuel value	Bottoms withdrawal and storage
Air emission	Heater flue gas	CO2, NOx, SOx, and CO from combustion equipment; subject to air permitting	Low-NOx burners and air permit compliance
Fugitive emission	Hydrocarbon vent / purge streams	Potential VOC and flammable vapor release if uncontrolled	Closed vent system, flare, or recovery
Liquid waste	Process water from dehydration and condensate knockouts	May contain oxygenates and dissolved organics requiring treatment before discharge	Wastewater treatment
Liquid waste	Cooling tower / boiler blowdown	Contains salts and treatment chemicals from utilities	Industrial wastewater system

Category	Representative Stream	Why It Matters	Typical Handling
Liquid waste	Lubricants, seal oils, cleaning fluids	Maintenance fluids can leak	Spill kits
Solid waste	Spent catalysts and adsorbents	May contain metals or contaminated solids; disposal depends on composition	Regeneration, metal recovery, return-to-vendor, or licensed disposal

EPA air rules use technology-based standards to control hazardous air pollutant emissions from applicable industrial equipment, including requirements built around closed vent systems and control devices in many chemical-sector contexts [22]. For the ETJ process, the most realistic environmental strategy is therefore to avoid routine atmospheric venting of hydrocarbon service, route non-condensable hydrocarbon streams to a control device or recovery system, and design storage and transfer operations so that evaporative losses are minimized. On the water side, aqueous streams should be characterized before discharge because dehydration, quench, and scrubbing can move oxygenates into water. Spent catalysts and driers should be managed based on actual composition and vendor guidance rather than assumed to be nonhazardous.

Major Chemical Hazards

This section identifies the main chemicals in the ethanol-to-jet (ETJ) process that present the most significant safety hazards. Rather than listing every compound present in the process simulation, the focus is on materials that contribute most to operational risk. These include ethanol, ethylene, hydrogen, hydrocarbon product streams, and concentrated oxygenate streams. Utility streams such as water generally present lower hazards and are not discussed in detail unless they introduce concerns related to corrosion, operability, or pressure control.

Table 2- Major material hazards

Material	Physical State	Primary Hazard	Main ETJ Concern	PSM Relevance
Ethanol	Liquid	Flammable liquid	Pool fire, flash fire, ignition during handling, vapor release in hot service	Relevant if part of covered flammable process
Ethylene	Gas	Highly flammable gas	Explosion or flash fire risk; compression and purification service increase consequence	Strongly relevant
Hydrogen	Gas	Highly flammable gas	Very low ignition energy, leak detection challenge, high-pressure recycle gas service	Strongly relevant
Naphtha	Liquid/vapor	Flammable hydrocarbon	Higher volatility than jet/diesel; important in condenser, reflux drum, and storage	Relevant
Jet / diesel fractions	Liquid	Combustible / flammable liquid depending on cut	Hot liquid release, tank and pump fire exposure, product contamination	Relevant
Oxygenate-rich condensate	Liquid	Flammable / irritant mixture	Potential VOC release and wastewater loading	Case-specific

From a process safety perspective, hydrogen and ethylene are the highest-concern materials because they combine high flammability with pressurized service and, in

hydrogen's case, require reliable monitoring for leaks using LEL (Lower Explosive Limit) detectors with alarm systems, which are standard safety measures in industrial facilities. Ethanol is easier to handle than hydrogen gas but still presents a major fire risk when stored, pumped, or heated, particularly in the event of liquid spills that can form pool fires. To reduce this risk, the equipment is designed to be inherently safe, including the use of non-sparking tools, sealed instruments and electrical supplies, closed systems to minimize vapor release, and inert gas blanketing where appropriate.

Liquefied petroleum gases (LPG), particularly butane and butene, are also key hazard drivers in the fractionation system because they are the lightest hydrocarbons in the system and therefore the most likely to form a rapidly expanding vapor cloud if containment is lost. Upon release, these materials can undergo rapid cooling and flashing, producing a dense, highly flammable vapor cloud that can travel and ignite if an ignition source is present, making them the most critical materials to control from a vapor cloud explosion standpoint.

Exposure Limits and Hazard Communication

Not every important chemical hazard is toxicology-driven, but occupational exposure limits still matter because they define expectations for worker protection, ventilation, and PPE selection. OSHA's annotated Z-tables list ethanol with an OSHA PEL of 1000 ppm and sodium hydroxide with a ceiling limit of 2 mg/m^3 [23]. Ethylene is a flammability/asphyxiation concern not a toxic inhalation hazard in this process. For hazard communication inside the plant, NFPA 704 ratings are commonly used on labels, door markings, and response plans to communicate relative health, flammability, and reactivity hazards. The table below displays the NFPA ratings for the major chemicals only.

Table 3- Selected exposure and hazard-communication values

Chemical	OSHA limit / note	NFPA Health	NFPA Flammability	NFPA Reactivity
Ethanol	PEL 1000 ppm	2	3	0
Hydrogen	Flammability hazard more significant than toxicity	0	4	0
Ethylene	Simple asphyxiation / flammability concern	1	4	2
Hydrocarbon	Exposure controlled mainly by vapor/dermal management and fire prevention	1	2–3	0
Alumina (dehydration catalyst support)	OSHA PEL 15 mg/m^3 dust, 5 mg/m^3 respirable	1	0	1
MTT zeolite with alumina binder	Treat as catalyst/zeolite dust; control particulate exposure	1	0	0
Ni on silica-alumina catalyst	Metal-containing catalyst dust; avoid inhalation	2	1	0
Pt on alumina hydrogenation catalyst	Catalyst dust; irritation hazard	2	0	0

Chemical	OSHA limit / note	NFPA Health	NFPA Flammability	NFPA Reactivity
Ni-W on FAU zeolite hydrocracking catalyst	Metal catalyst dust; possible irritation. Toxic nickel carbonyl.	2	0	0
SAPO-34 with alumina binder	Molecular sieve dust exposure control	1	0	0

Regulatory and Management Framework

A commercial-scale ethanol-to-jet (ETJ) facility operates under several regulatory and safety management requirements. In particular, the presence of flammable gases and liquids such as hydrogen, ethylene, and hydrocarbon products means that the facility would likely fall under OSHA's Process Safety Management (PSM) standard [23]. The purpose of the PSM regulation is to prevent errors that can lead to injury or catastrophic releases of toxic, reactive, flammable, or explosive chemicals. Facilities that store or process large inventories of flammable gases or liquids are therefore be required to implement formal safety management systems.

Because the ETJ process involves significant quantities of hydrogen, ethylene, and other hydrocarbons, the plant design should assume the need for key safety management elements such as operating procedures, personnel training, mechanical integrity programs, management of change, pre-startup safety reviews, and emergency response planning.

In addition to process safety requirements, the facility also needs to comply with environmental regulations related to air emissions, wastewater discharge, and solid waste handling. While the specific permitting requirements depend on the final plant location and operating capacity, the design should incorporate measures to minimize environmental releases, such as vapor recovery, flaring for hydrocarbon vent streams, treatment of wastewater containing organic compounds, and proper handling of spent catalysts and other process wastes.

Process-Wide Operational Safety Review

This section reviews operational safety considerations across the entire ethanol-to-jet (ETJ) process rather than focusing only on the single unit selected for detailed HAZOP analysis. Large process facilities contain a range of plant-wide hazards, including fire and explosion risks, rotating equipment, high-temperature surfaces, confined spaces, and work performed at elevated platforms. These hazards must be considered in addition to the specific process deviations evaluated in the HAZOP study.

The HAZOP analysis provides a detailed examination of one selected unit, while this section addresses broader operational hazards that may occur throughout the facility. The table below summarizes major operational hazards associated with the ETJ process and the design features or management practices used to mitigate them.

Table 4- Operational Consequences & Safeguards

Hazard Type	Where It Appears	Potential Consequence	Primary Safeguards	Residual Concern
Fire / flash fire	Ethanol storage and transfer; naphtha system; hydrocarbon pumps	Personnel injury, product loss, equipment damage	Area classification, ignition source control, leak prevention, drainage, fire protection	Moderate
Explosion / VCE	Ethylene compression and purification; hydrogen recycle system	Catastrophic overpressure, major equipment damage	Gas detection, ventilation, ESD, pressure protection, leak testing	High
High temperature	Dehydration reactors, fired heaters, hot piping, reboilers	Burn hazard, thermal degradation, ignition of leaked material	Insulation, temperature control, guards, hot-work management	Moderate
High pressure	Compressors, hydroprocessing loop, fractionation pressure boundary	Rupture, jet release, escalation to fire	PSVs, pressure alarms, interlocks, design codes and inspection	Moderate to High
Exotherm control	Oligomerization and some hydroprocessing reactions	Runaway temperature, catalyst deactivation, off-spec product	Staged beds, heat removal, temperature alarms, feed cutback logic	Moderate
Rotating equipment	Pumps, compressors, fans	Seal leak, mechanical failure, caught-in hazards	Machine guarding, seal plans, vibration monitoring	Moderate
Confined space	Columns, drums, tanks, exchangers during maintenance	Asphyxiation, exposure, rescue complications	Permit-required entry, gas testing, isolation and LOTO	Moderate

Hazard Type	Where It Appears	Potential Consequence	Primary Safeguards	Residual Concern
Working at height	Tall columns, platforms, overhead lines	Falls during inspection or maintenance	Handrails, fall arrest, access control, scaffolding procedures	Moderate
Chemical exposure	Caustic wash systems, wastewater handling, sample points	Eye/skin burns, inhalation irritation	PPE, eyewash/shower, procedures, labeling	Low to Moderate
Loss of utilities	Cooling water, steam, instrument air, power, lubricants, seal oils	Column upset, loss of control, overpressure or off-spec product	Utility alarms, fail-safe valves, PRV's, shutdown procedures, backup power where justified	Moderate
High-voltage electrical hazard	Motors, MCCs, control panels, switchgear, heaters, VFDs, lighting, field wiring in process areas	Electric shock, arc flash, ignition of flammable vapor, equipment trip, fire	Area classification, grounding/bonding, arc-flash protection, LOTO, explosion-proof equipment, preventive maintenance	Moderate

The hazards summarized in the table above reflect common operational risks associated with large hydrocarbon processing facilities. Many of these hazards arise from plant operation and maintenance activities rather than from a single process deviation. For example, heater operation in the dehydration section introduces burn and fire hazards, oligomerization reactors require careful management of exothermic reactions, and the hydroprocessing section involves hydrogen recycle systems that present flammability risks. In addition, routine plant activities such as working at height, equipment maintenance, and handling pressurized systems introduce further operational hazards.

HAZOP Methodology and Risk Ranking

The selected HAZOP node is the product fractionation column that removes naphtha overhead while recovering jet and diesel-range hydrocarbons in the bottoms. This unit was chosen because it is strongly tied to the process control discussion and because it contains several interacting variables: pressure, reflux, reboiler duty, condenser duty, feed rate, and inventory levels that can quickly affect both safety and product quality. The HAZOP is organized by node: feed system, column body, overhead condenser and reflux drum, reboiler system, and bottoms withdrawal.

Table 5 - HAZOP Analysis of Jet Fuel Fractionator

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
T-700 Feed	FV-700	low flow	valve fails close, loss of level in D-700 or D-701, low pressure in drums	process upset, jet off spec	4	FI-700 will indicate low flow, naphtha spec allows some contamination from jet	2	4	FI-700	
T-700 Feed	FV-700	high flow	LV-700 or LV-701 fails open or liquid level lost in D-700 or D701	Hydroprocessing reactor section depressures into T-701, possibly causing loss of containment and fire	1	FI-700 will indicate high flow, column pressure indication	2	1	FI-700	provide PSV on column and overhead receiver, high pressure trip and low drum level trip to feed valves
T-700 Feed	TI-700	high temp	inadequate cooling in E-700 or too much heat transfer in E-603	process upset, naphtha off spec	2	TI-700 will indicate high temp	2	2	TI-700	Provide TI's on overhead vapor, reboiler vapor, reflux liquid, bottoms liquid, feed liquid, maybe feed tray provide low level alarms on D700 & D701
T-700 Feed	TI-700	low temp	too much cooling in E-700 or low outlet temperature from E-603	process upset, Jet off spec	3	TI-700 will indicate low temp, column temperatures will warn of upset conditions	2	3	TI-700	^
T-700 Feed	FI-700	loss of containment	flange leak, valve stem leak	venting hydrocarbon, possible fire or explosion	1	equipent design & maintenance	3	2	-	Install automatic ESD isolation on fractionator feed, improve valve/flange mechanical integrity with leak detection, LEL monitor

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
T-700 Feed	outside of section scope	variation in composition	temperature excursion in R-601, catalyst deactivation, or temperature/pressure variation in D-700 or D-701	process upset, products off spec	3	temperature, pressure and flow measurements in R-601, regular sampling analysis	4	4	-	
T-700 Overhead	PV-700	low pressure	valve fails open, ambient conditions such as rainfall cause too much cooling in E-701	process upset, naphtha off spec	4	PI-700 indicates pressure	3	4	PI-700	provide offgas flowmeter for material balance and troubleshooting
T-700 Overhead	PV-700	high pressure	valve fails close, ambient conditions cause inadequate cooling in E-701	process upset, jet off spec, pressure build up into loss of containment	1	PI-700 indicates pressure	3	2	PI-700	installation of PRV's and PSV's on column and accumulator, alarm
T-700 Overhead	TI-701	high temp	Ambient conditions cause inadequate cooling in E-701	process upset, naphtha off spec	4	TI-701 indicates temperature	4	4	TI-701	
T-700 Overhead	TI-701	low temp	Ambient conditions cause too much cooling in E-701	process upset, jet off spec	3	TI-701 indicates temperature	1	2	TI-701	provide TI-701 high alarm, consider louvers on E-701
T-700 Overhead	PI-700	loss of containment	flange leak, valve stem leak, external fire	venting non-condensables, possible environmental issue, fire, severe equipment damage	1	unit design & maintenance	4	3	-	provide local LEL alarms where leaks are possible
D-700 Overhead Receiver	D-700	low level	feed loss, cooling loss, high pressure, reboiler heat loss	process upset, jet & naphtha off spec	2	level, flow and pressure indicators	2	2	-	provide level alarms
D-700 Overhead Receiver	D-700	high level	drop in column pressure, TV-700 fails close, LI-702 error, P-700 trips	liquid into fuel gas system, possible loss of containment and fire	3	pressure, temperature, and level indication	4	4	-	provide level alarms
T-700 Liquid Product	TV-700	high flow	TV-700 fails open, column temperature reading incorrect	process upset, naphtha off spec	4	FI-701 will indicate high flow, maintenance	2	4	FI-701	
T-700 Liquid Product	TV-700	low flow	TV-700 fails close, P-700 trips	process upset, jet off spec	3	FI-701 will indicate low flow, maintenance	2	3	FI-701	

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
T-700 Liquid Product	TI-704	high temp	inadequate cooling in E-701	process upset, naphtha off spec	3	TI-702 indicates temperature, column temperature indication	2	3	TI-704	
T-700 Liquid Product	TI-704	low temp	too much cooling in E-701	process upset, jet off spec	3	TI-702 indicates temperature, column temperature indication	2	3	TI-704	
T-700 Liquid Product	FI-701	loss of containment	flange leak, valve stem leak	venting hydrocarbon, possible fire or explosion	1	unit design & maintenance	5	NH	-	provide LEL monitor near areas of concern
T-700 Liquid Product	outside of section scope	variation in composition	temperature variation in column, reflux valve failure	process upset, product off spec	2	column temperature indication, valve maintenance, product sampling and analysis	2	2	-	temperature & pressure alarms, flow indicators
T-700 Bottom Product	FV-705	high flow	FV-705 fails open, high sump level	possible lost sump level, process upset, Jet off spec	2	FI-705 indicates flow, LI-705 level, maintenance	3	3	FI-702, LI-705	installation of PRV's and PSV's downstream of control valve, alarms
T-700 Bottom Product	FV-705	low flow	FV-705 fails close, low sump level, P-704 trips	possible overfill sump covering reboiler vapor return causing tray damaage, process upset, jet off spec	2	FI-705 indicates flow, LI-705 level, maintenance	3	3	FI-702, LI-705	low level alarms
T-700 Bottom Product	TI-706	high temp	steam header upset or FV-704 malfunction - too much heating in reboiler	process upset, naphtha off spec	3	TI-704 indicates temperature, column temperature indication	2	3	TI-706	
T-700 Bottom Product	TI-706	low temp	steam header upset or FV-704 malfunction - inadequate heating in reboiler	process upset, jet off spec	2	TI-704 indicates temperature, column temperature indication	2	2	TI-706	
T-700 Bottom Product	FI-705	loss of containment	flange leak, valve stem leak, pump seal leak	venting hydrocarbon, possible fire or explosion	1	unit design & maintenance	5	NH	-	provide LEL monitor in area

Process Control

The ethanol-to-jet (ETJ) process consists of several major reaction and separation steps including ethanol dehydration, olefin oligomerization, hydroprocessing, and final product fractionation. Each of these units operates under different temperature, pressure, and flow conditions, which means that reliable process control is necessary to ensure both safe operation and consistent product quality. Without appropriate control strategies, disturbances in feed composition, utility supply, or recycle streams could propagate through the process and lead to unstable operation or off-spec products.

The main goal of the control system is to maintain the desired operating conditions throughout the plant while minimizing the effect of disturbances. In practice, this means regulating key process variables such as temperature, pressure, flowrate, and liquid level. Maintaining these variables within acceptable ranges helps protect equipment, stabilize the overall plant throughput, and ensure that the final hydrocarbon products meet the required jet-fuel specifications.

Control Strategy by Process Section

DEHYDRATION SECTION

The dehydration section converts ethanol to ethylene through catalytic dehydration reactors. Control of this unit is important because both the reactor performance and downstream separation stability depend on consistent ethylene production rate and purity. The primary objective of the control system in this section is to maintain stable reactor conditions while ensuring that downstream compressors and separation equipment receive a steady feed.

The ethanol feed flowrate is regulated using a flow control loop to maintain consistent throughput to the dehydration system. A steam injection stream is also controlled relative to the ethanol feed in order to maintain the desired steam-to-ethanol ratio. This ratio helps regulate reactor temperature and reduces coke formation on the catalyst.

Temperature control is critical in the dehydration reactors because conversion and catalyst stability depend strongly on reactor temperature. Reactor inlet temperatures are regulated using heat exchangers and heaters upstream of the reactors. Multiple temperature measurements are typically installed along the reactor beds to monitor temperature profiles and detect potential hot spots.

Pressure control is maintained across the dehydration system to ensure stable operation of the downstream ethylene compression stage. Drums and separators in the dehydration section use level control loops to maintain liquid inventory and prevent liquid carryover into gas-handling equipment. Pumps transferring recycled ethanol and condensate are controlled using flow and pressure measurements to ensure stable circulation throughout the section. The table below summarizes the controls for the dehydration section

Table 6- Dehydration Section Controls

Types of Equipment	Controls
Dehydration Reactors	Temperature, Pressure, Flow, Steam-Ethanol Ratio
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Driers	Regeneration Temp, Moisture Content
Drums	Pressure, Level

Types of Equipment	Controls
Columns	Temperature, Pressure, Flows, Level

OLIGOMERIZATION SECTION

The oligomerization section converts light olefins produced in the dehydration step into longer hydrocarbon chains suitable for further upgrading into jet fuel. The main control challenge in this section is managing the strongly exothermic reaction while maintaining stable reactor operation and recycle flow.

Olefin feed flow to the oligomerization reactors is controlled using flow control loops to maintain a constant reactor loading. A paraffin diluent stream is also introduced and controlled relative to the olefin feed using a ratio control strategy. The diluent helps moderate the exothermic reaction by absorbing heat and reducing reaction severity.

Temperature control is achieved using interstage heat exchangers between the reactor beds. These exchangers remove heat from the reaction mixture and help maintain the desired reactor temperature profile. Maintaining proper temperature is important for catalyst stability and product selectivity.

Pressure control is applied to the separation columns in this section, such as the deethanizer and olefin splitter, to stabilize vapor–liquid equilibrium and maintain proper separation of light and heavy hydrocarbons. Reflux flow and reboiler duty are adjusted to maintain the desired column temperature profiles and separation performance.

Compressors used in the recycle loop are controlled using pressure and flow control loops to maintain stable circulation of light olefins and prevent accumulation of inert gases. Level control loops in the associated drums and separators maintain liquid inventory and prevent flooding or pump cavitation. The table below summarizes the oligomerization section controls.

Table 7- Oligomerization Section Controls

Types of Equipment	Controls
Oligomerization Reactors	Temperature, Pressure, Flow, Paraffin-Olefin Ratio
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow

Types of Equipment	Controls
Compressors	Pressure, Flow
Columns	Temperature, Pressure, Flows, Level

HYDROPROCESSING SECTION

The hydroprocessing section upgrades oligomerized hydrocarbons through hydrogenation, hydrocracking, and isomerization reactions to produce jet-range hydrocarbons. These reactions require hydrogen at elevated pressures and temperatures, making careful control of operating conditions essential for both safety and catalyst performance.

The hydrocarbon feed and hydrogen feed streams are controlled using flow control loops, and a ratio control strategy is implemented to maintain the desired hydrogen-to-hydrocarbon ratio entering the reactors. Maintaining sufficient hydrogen partial pressure prevents coke formation and ensures proper conversion.

Reactor temperature is regulated using heat exchangers and fired heaters upstream of the reactor beds. Temperature measurements along the reactor help detect deviations from the desired temperature profile, which could indicate catalyst deactivation or abnormal reaction behavior.

Pressure control loops regulate the hot and cold separators in the hydroprocessing system. These separators remove hydrogen and light gases from the liquid hydrocarbon stream and maintain stable conditions for the hydrogen recycle loop. Compressors in the hydrogen recycle system maintain the required reactor pressure and recycle flowrate.

Liquid levels in the hot separator, cold separator, and associated drums are controlled to ensure proper phase separation and to prevent liquid carryover into compressors or gas carry-under into liquid handling equipment. The table below summarizes the controls for the hydroprocessing section.

Table 8- Hydroprocessing Section Controls

Types of Equipment	Controls
Hydrogenation & Hydrocracking Reactors	Temperature, Pressure, Flow, Hydrogen-Hydrocarbon Ratio
Heat Exchangers	Temperature, Flow

Types of Equipment	Controls
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Drums	Pressure, Level
Columns	Temperature, Pressure, Flows, Level

Detailed Control Loop Analysis on Hydrocracker Product Fractionator

The selected unit for detailed control analysis is the fractionation column T-700 (Gasoline fractionator) shown in the attached control loop schematic Figure 18. The system contains multiple interacting loops and directly affects final product quality. The column separates lighter (lower-boiling) material overhead from heavier jet and diesel fractions in the bottoms system. Stable operation requires simultaneous control of pressure, reflux, overhead withdrawal, reboiler duty, and sump inventory. These control loops maintain vapor–liquid equilibrium in the column and ensure that the final products meet the desired specifications.

Feed Flow Control

The fractionator receives feed from upstream separation units. Feed flow is regulated using flow control valves on the incoming feed streams. Controlling the feed flow establishes the load on the column and helps prevent sudden disturbances to column hydraulics.

Feed flow control also assists in stabilizing upstream equipment. In many cases, upstream drum levels are controlled by adjusting the feed to the fractionator. In this configuration, the column acts as the downstream sink for those vessels, allowing the upstream inventory to remain stable.

The main objectives of feed flow control are to stabilize upstream separator levels, avoid abrupt feed surges, and maintain smooth vapor–liquid traffic within the column.

Column Pressure Control

Column pressure is a key variable in distillation because vapor–liquid equilibrium depends directly on pressure. In the control system, overhead pressure is measured by PT-701 and controlled by PC-701, which manipulates PV-701 connected to the fuel gas system.

If the pressure rises above the setpoint, the control valve opens and sends vapor to the fuel gas system, reducing column pressure. If the pressure falls, the valve closes to retain vapor in the column.

Maintaining a constant pressure protects the column from overpressure and stabilizes separation performance. Changes in pressure affect boiling points and relative volatility, which directly impacts product quality.

Reflux Drum (Accumulator) Level Control

The overhead condenser sends condensed liquid to the reflux drum, where it is collected before being returned as reflux or withdrawn as distillate product. The liquid level in the accumulator is measured by LT-702.

Accumulator level is controlled using a cascade control strategy. The level controller LC-702 adjusts the setpoint of the reflux flow controller FC-702, which regulates the reflux valve and pump system. By manipulating the reflux flow rate, the system maintains the desired liquid inventory in the accumulator.

This cascade configuration ensures stable operation of the reflux pump, maintains sufficient liquid for reflux, and prevents flooding or loss of liquid inventory in the condenser–reflux system.

Reflux Flow Control

Reflux flow is one of the most important manipulated variables in distillation. In the control loop, FT-702 measures reflux flow and FC-702 regulates the reflux valve to maintain the desired flowrate returning to the column.

Returning liquid to the top of the column improves mass transfer between vapor and liquid phases and helps prevent heavier components from entering the overhead product. However, excessive reflux increases energy consumption and column loading, while insufficient reflux leads to poor separation. Maintaining a stable reflux flow helps ensure consistent overhead product quality.

Top Temperature Control

Column top temperature is often used as an indirect indicator of overhead product composition. In the control scheme, TT-703 measures the temperature on a selected tray within the upper section of the column, typically the tray with the highest temperature change (ΔT) based on the column temperature profile, rather than the actual top tray, and TC-703 adjusts the distillate withdrawal valve TV-703.

If the temperature rises, it may indicate that heavier components are reaching the upper section of the column. The controller reduces distillate withdrawal, which increases the overhead receiver level, causing the reflux flow rate to increase, thereby restoring the desired operating condition. Temperature control therefore provides indirect regulation of overhead product quality.

Reboiler Duty / Steam Flow Control

The reboiler supplies the heat required to generate vapor within the column. Steam flow to the reboiler is measured by FT-704 and controlled by FC-704, which manipulates the steam control valve FV-704.

Increasing steam flow increases vapor generation and improves stripping of lighter components from the bottom section of the column. However, excessive reboiler duty may lead to flooding or excessive pressure. Stable steam flow control is therefore important to regulate column boil-up to maintain the desired separation between overhead and bottoms products.

COLUMN SUMP LEVEL CONTROL

The liquid inventory in the column sump must be maintained to ensure stable reboiler operation. The sump level is measured by LT-705 and controlled using a cascade level control strategy.

In this loop, the level controller adjusts the setpoint of the bottoms flow controller FC-705, which regulates the bottoms product valve FV-705. By manipulating the bottoms withdrawal rate, the control system maintains the desired liquid level in the column base.

This cascade control configuration prevents excessive liquid accumulation, avoids reboiler dry-out, and stabilizes bottoms product withdrawal.

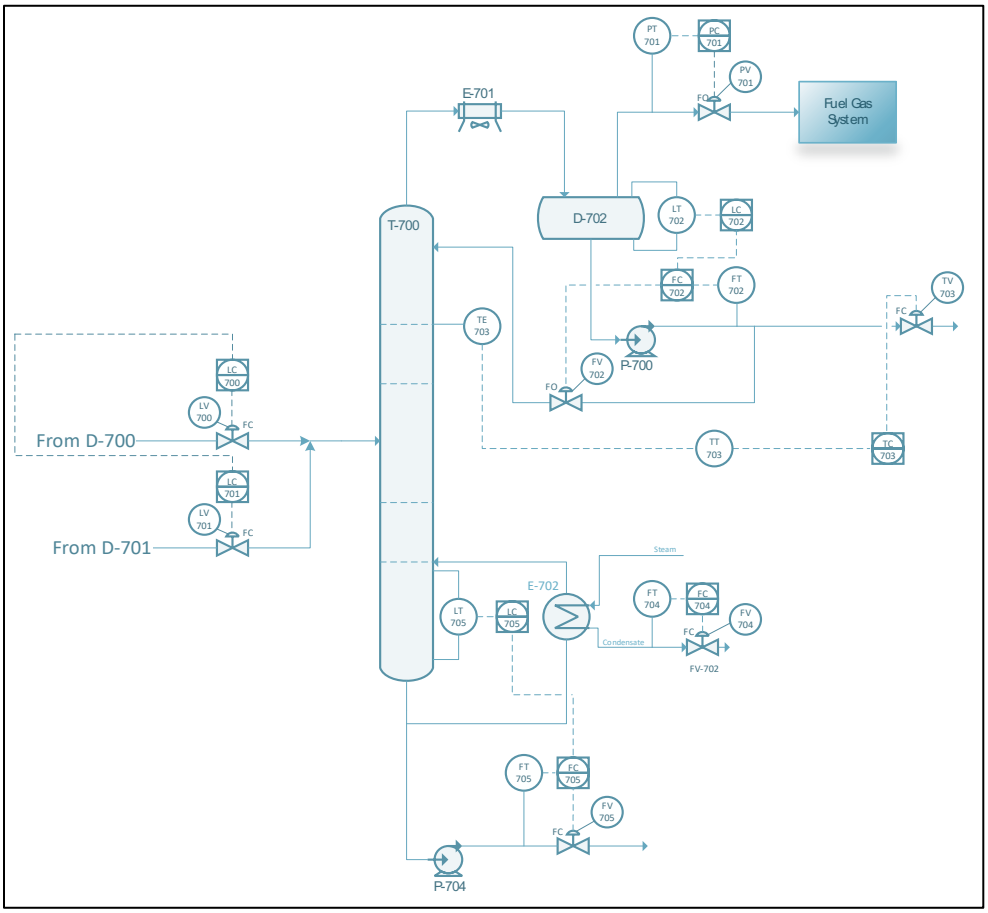


Figure 18- Process Control Loop on Fractionator Before HAZOP

Based on the recommendations from the HAZOP, the control schematic was revised to add the following instruments as shown below in Figure 19.

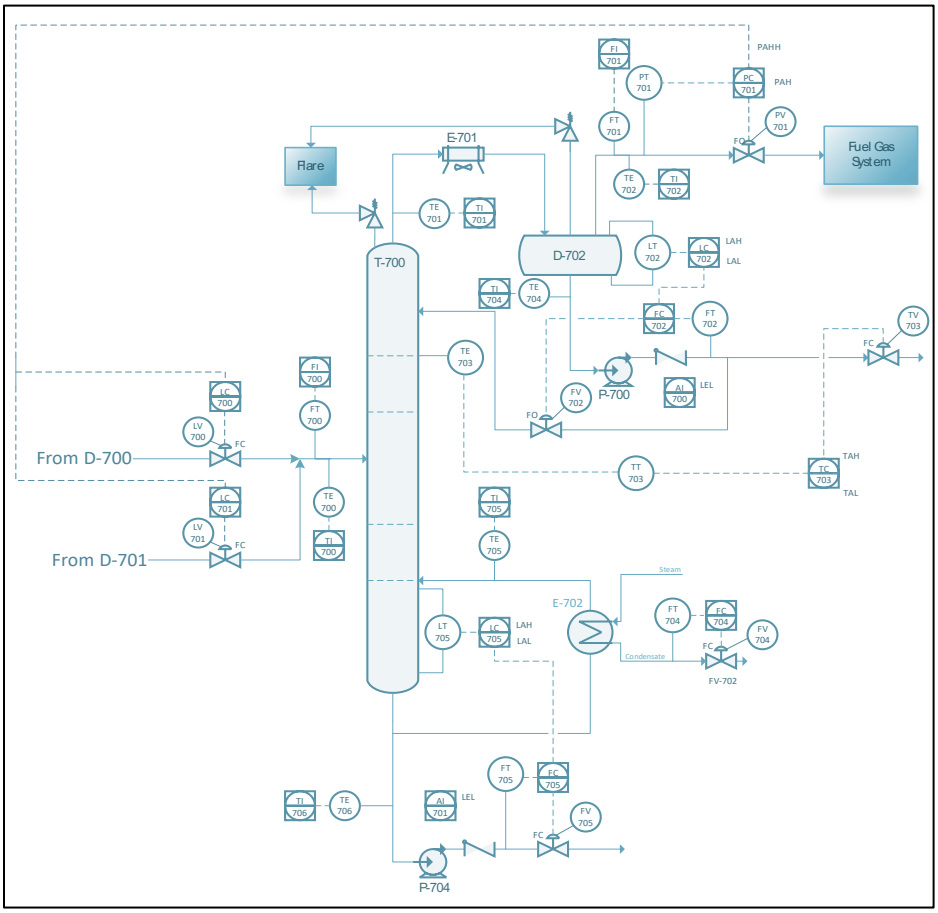


Figure 19- Process Control Loop on Fractionator After HAZOP

Process Economics Analysis

This section will detail the full economic breakdown on capital expenditure, including the ISBL, OSBL, WC, and Total Capital Cost, the costs of production including the VCOP and FCOP, the total revenue, economic evaluation metrics such as the NPV, DCFROR, and SPBMP, and a Sensitivity Analysis.

ISBL, OSBL, ECC, Contingency, WC and Total Capital Costs

Inside Battery Limits (ISBL)

The ISBL for the ETJ process facility was estimated using installed equipment costing obtained from Aspen Plus Process Economic Analyzer. Aspen reports capital costs in $USD on the Gulf Coast Basis in 2022 US dollars and thus must be corrected using a time factor and location factor to account for present-day pricing conditions on the East Coast.

The adjusted installed cost was calculated using:

$$\text{Adjusted Cost} = \text{Aspen Cost} \times \text{Time Factor} \times \text{Location Factor}$$

Using a time factor of 0.9865 taken from the Chemical Engineering Plant Cost Index (CEPCI) [24] and a location factor of 1.04 [25], the corrected ISBL was found to be:

$$ISBL = \$358.2 \text{ MM}$$

The ISBL includes all process equipment within the core ETJ production process, including reactors, distillation columns, compressors, heat exchangers, and associated piping and instrumentation. The remainder of the capital cost was estimated using standard chemical engineering heuristics based on fractions of ISBL. These include outside battery limits (OSBL), engineering and construction costs (ECC), contingency, and working capital (WC).

Outside Battery Limits (OSBL)

Outside battery limits include supporting infrastructure required for plant operation. These typically consists of:

- Utilities generation and distribution
- Cooling water systems
- Storage tanks
- Wastewater treatment
- Flare systems
- Plant piping and buildings

For conceptual design, OSBL is estimated as 40% of ISBL:

$$OSBL = 0.40 \times ISBL$$
$$OSBL = 0.40 \times 358.2 = \$143.3 \text{ MM}$$

Engineering and Construction Costs (ECC)

Engineering and construction costs include:

- Detailed plant design
- Project management
- Construction labor
- Contractor fees
- Commissioning and startup preparation

ECC is estimated as 10% of ISBL:

$$ECC = 0.10 \times ISBL$$
$$ECC = 0.10 \times 358.2 = \$35.8 \text{ MM}$$

Contingency

A contingency factor accounts for uncertainties associated with early-stage process design, such as:

- Equipment scaling uncertainty
- Construction cost fluctuations
- Design modifications

Contingency is estimated as 10% of ISBL:

$$Contingency = 0.10 \times ISBL$$
$$Contingency = 0.10 \times 358.2 = \$35.8 \text{ MM}$$

Working Capital (WC)

Working capital represents the funds required to cover operational costs before revenue is generated. This includes:

- Raw material inventory

- Catalyst inventory
- Initial operating expenses

Working capital is estimated as 15% of ISBL:

$$WC = 0.15 \times ISBL$$
$$WC = 0.15 \times 95.16 = \$53.7 \text{ MM}$$

TOTAL CAPITAL COST (TCC)

The total capital cost is the sum of the ISBL, OSBL, ECC, Contingency, and Working Capital costs. The TCC for this process is tabulated below:

Table 9 - Total Capital Costs Breakdown

Capital Cost Evaluation	
ISBL	$358.2MM
OSBL	$143.3MM
ECC	$35.8MM
Contingency	$35.8MM
WC	$53.7MM
TCC	**$626.8MM**

Variable Costs of Production (VCOP)

The VCOP encompasses the costs of utilities, raw materials, waste management, and other consumables. For this process, most of the VCOP is derived from the feedstock, specifically the massive amounts of ethanol required in the dehydration section. Ethanol feedstock was priced at $1.70/gal based on the average import price to the U.S. from Brazil reported from the United States Department of Agriculture [26]. Assuming a yearly production schedule of 8,000 hours, the ethanol expenditure is calculated in the Appendix to be $518MM per year, with prices based. Additional costs include hydrogen feedstock, catalysts, utilities (provided by Aspen Plus Economic Analyzer), and waste management for oxygenate-rich condensate and wastewater. Hydrogen was supplied by PlugPower and priced at $1.50/kg [27]. Waste management was calculated from the City of Atlanta Department of Watershed Management [28]. The total VCOP is tabulated below:

Table 10 - VCOP Breakdown

Type	Cost
Ethanol Feedstock	$518MM
Hydrogen Feedstock	$17MM
Utilities	$17MM
Catalysts	$5MM
Waste Management	$2MM
Total VCOP	**$560MM**

Fixed Costs of Production (FCOP)

The FCOP consists of costs that do not vary with the plant's production capacity. This cost is derived from paying operators to run shifts throughout the year, supervision, maintenance, and other overhead expenses. A full breakdown of the FCOP is tabulated below:

Table 11 - FCOP Breakdown

Fixed Cost of Production (FCOP)	
Number of Shifts	4
Operators per shift	5
Operator Salary ($/yr)	$70,000
Total Operator Compensation ($MM/yr)	$1.4MM
Supervision ($MM/yr)	$0.35MM
Direct Salary Overhead ($MM/yr)	$0.79MM
Maintenance ($MM/yr)	$18.7MM
Overhead Expenses (Plant, Taxes, Insurance, Land) (MM/yr)	$21MM
Total FCOP ($MM/yr)	**$42MM**

Revenue Analysis

Revenue for this ETJ process plant comes from the sale of hydrocarbon fuel products and renewable fuel incentives. The main product of the process is jet fuel, with smaller amounts of gasoline-range and diesel-range hydrocarbons recovered in the downstream fractionation section. These products can be blended with conventional fuels and sold through Georgia's existing fuel infrastructure.

Jet fuel is the largest revenue stream because it is the primary target product of the ETJ process. Based on current market pricing, jet fuel sales generate approximately $350 million per year [24]. Gasoline and diesel by-products contribute additional revenue of $41 million per year and $26.7 million per year, respectively. Together, direct fuel sales account for about $418 million annually.

The process also qualifies for renewable fuel incentives. Since ethanol-to-jet fuel is considered a renewable fuel pathway, it generates D3 Renewable Identification Number (RIN) credits under the Renewable Fuel Standard [29]. These credits significantly increase revenue, contributing approximately $745 million per year. In addition, the process qualifies for the 45Z clean fuel production tax credit, which adds roughly $245 million per year.

The full breakdown of calculations can be found in the Appendix. The total annual revenue from fuel sales and incentives is estimated to be nearly $1.6 billion per year, as summarized below:

Table 12 - Revenue Breakdown

Revenue	
Jet Fuel (SAF)	$350MM
Gasoline	$82MM
Diesel	$51MM
D3 RIN Credit	$745MM
45Z Tax Credit	$245MM
Total Revenue ($MM/yr)	**$1,565MM**

Simple Payback Period (SPBP) and Return on Investment (ROI)

The simple payback period is a common metric used in chemical engineering economic evaluations to estimate how long it will take for a project to recover its initial investment. In other words, it measures the time required for the plant's annual cash flow to repay the capital spent to build the facility. This metric is useful because it gives a quick sense of the financial risk associated with a project. A shorter payback period generally means the investment is recovered more quickly and the project carries less financial risk.

To calculate the simple payback period, the total capital investment is divided by the average annual cash flow generated by the facility:

$$SPBP = \frac{\text{Total Capital Investment}}{\text{Average Annual Cash Flow}}$$

Using the calculated capital cost of $626 million and an average annual cash flow in the first 3 years of plant life of $207 million, the payback period is:

$$SPBP = \frac{626}{207} = 3.01 \text{ years}$$

This result indicates that the facility would recover its initial investment in approximately 3 years, which is incredibly favorable for this large-scale chemical process and a good indicator of profitability.

The return on investment (ROI) is another key economic metric used to evaluate the profitability of a process. ROI compares the annual profit generated by the plant to the total capital required to build it. This metric helps determine how efficiently the invested capital is being used to generate profit. Higher ROI values indicate that the project produces a large return relative to the initial investment.

The ROI is calculated by dividing the net annual profit by the total capital investment, then multiplying by 100 to convert the result to a percentage:

$$ROI = \frac{\text{Net Annual Profit}}{\text{Total Capital Investment}} \times 100$$

Using a net annual profit in full production of $714 million and a total capital investment of $167 million, the return on investment is:

$$ROI = \frac{714}{626} \times 100 = 114\%$$

An ROI of 114% indicates that the project generates more than its capital investment in annual profit, which is another key indicator of the profitability of this process.

Net Present Value (NPV) and Discounted Cash Flow Rate of Return (DCFROR)

The net present value (NPV) calculation is utilized to evaluate the long-term profitability of the ETJ process while accounting for the time value of money. The NPV measures whether the discounted future cash flows generated by the project exceed the initial capital investment. A positive NPV indicates that the project creates economic value and is financially viable. The total capital cost (TCC) for the facility is approximately $166.5 million, and this investment is distributed over the first two years of the project. In year 1, the plant invests 25% of the total capital cost, while the remaining 75% is invested in year 2. Because the facility is still under construction during these years, no revenue is generated and the pre-tax cash flow is negative, representing the capital being invested into the project.

The plant begins operation in year 3, but only at 50% capacity during the startup period. As a result, the revenue and VCOP are all reduced by half. The pre-tax cash flow in year 3 is therefore calculated as:

$$\text{Pre-tax Cash Flow} = 0.5(\text{Revenue}) - 0.5(\text{VCOP}) - (\text{FCOP})$$

This results in approximately $462.4 million in pre-tax cash flow in the startup year. From year 4 onward, the facility operates at 100% capacity, and the pre-tax cash flow is calculated as the full revenue minus the full operating costs:

$$\text{Pre-tax Cash Flow} = \text{Revenue} - \text{VCOP} - \text{FCOP}$$

Under full operation, the plant generates a steady pre-tax cash flow of approximately $965 million per year.

Depreciation was included in the analysis to determine taxable income. The depreciation values were calculated using the MACRS-5 depreciation schedule, which is commonly used for chemical process equipment. The MACRS percentages were applied to the depreciable capital investment, defined as the total capital cost minus working capital, since working capital is not depreciable. Therefore, depreciation was determined by multiplying the MACRS-5 percentages by (TCC − WC) for each year in the depreciation schedule.

Taxable income was calculated by subtracting depreciation from pre-tax cash flow. A corporate tax rate of 26% was then applied to the taxable income from the previous year to determine the tax liability. The after-tax cash flow for each year is calculated as:

$$\text{Cash Flow} = \text{Pre-tax Cash Flow} - \text{Tax}$$

To account for the time value of money, each year's cash flow was multiplied by a discount factor to obtain the present value of that cash flow. The discount factor was calculated using a rate of 14%, which is inflated from the average 6-7% to account for the time value of money, inflation, and opportunity costs. The discount factor for each year was determined using:

$$\text{Discount Factor} = \frac{1}{(1+0.14)^{(\text{plant year})}}$$

The present value of the cash flow was then calculated as:

$$PV = (\text{Cash Flow}) \times (\text{Discount Factor})$$

Finally, the net present value was obtained by summing the present values of the cash flows over the entire project lifetime:

$$NPV = \sum PV$$

Using this method, the NPV is approximately $3.1 billion over the plant's lifetime. The Excel spreadsheet for this calculation can be located in the Appendix. Overall, the extremely large NPV indicates that this process is highly economically viable given the current capital and operating costs, market prices, and incentives.

To find the DCFROR, the interest rate of 14% is goal-seeked in Excel to yield an NPV of $0 at year 22 (or after 20 years of full operation), and the result is multiplied by 100 to yield a percentage. This value came out to be:

$$DCFROR = 85\%$$

This value represents the effective rate of return generated by the project, taking into account the time value of money. A DCFROR of 85% indicates that the project generates a return equivalent to earning 85% annually on the invested capital when future cash flows are discounted, which suggests that the facility produces very strong cash flows relative to the initial capital investment.

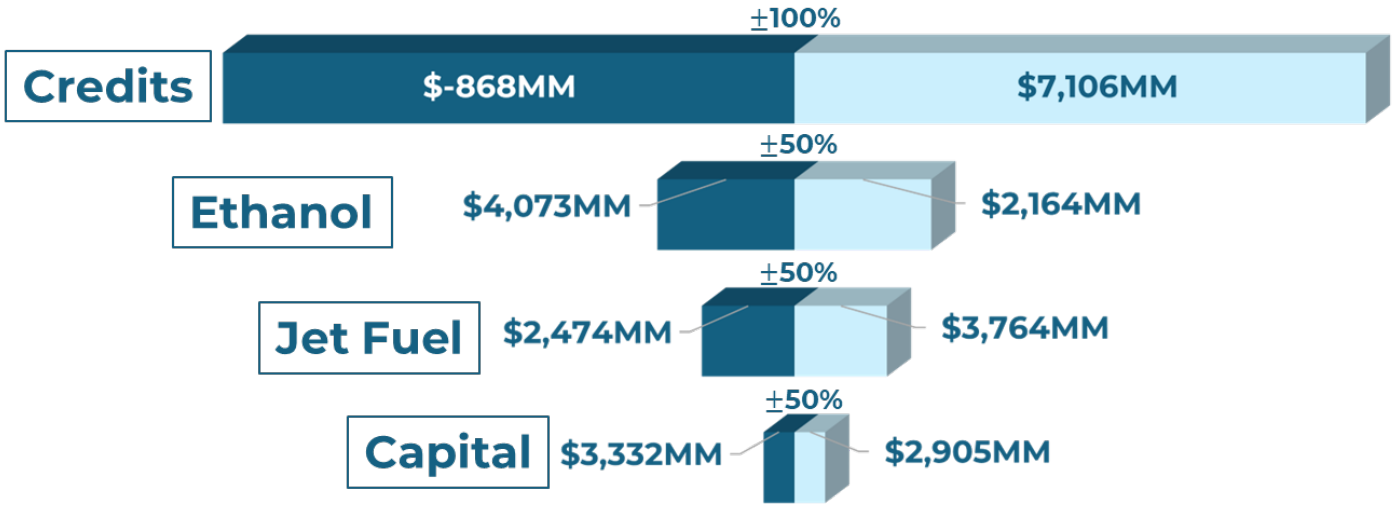
This result can also be compared to the assumed discount rate of 14%, which represents the minimum acceptable rate of return for the investment. Because the calculated DCFROR of 85% is significantly higher than the required 14% rate of return, the project is considered economically attractive under current conditions.

Sensitivity Analysis

A sensitivity analysis was performed to evaluate how changes in key economic parameters affect the net present value of the ETJ process. This type of analysis helps identify which variables have the greatest impact on project profitability and therefore represent the largest economic risks or uncertainties.

Four major economic parameters were analyzed: total capital costs, jet fuel sell price, ethanol feedstock buy price, and the total value of government incentives/credits. Each parameter was varied by ±50% of its base-case value, while all other variables were held constant. The credits were specifically varied by ±100% of their value to evaluate their volatility in this process's economics. For every case, the NPV was recalculated, and the results were plotted to visualize how sensitive the project economics are to each variable. The tornado-style diagram can be seen below:

Figure 20 - NPV Sensitivity Tornado Diagram



From the diagram, it is clear that credits produce the largest difference in NPV, indicating that the NPV is most sensitive to this factor. This result is expected because a major portion of the facility's revenue comes from tax credits such as D3 RINs and the 45Z tax credit. Furthermore, when the tax credits are varied to -100%, the process becomes unprofitable. This shows the project's dependence on government incentives and the volatility that may arise if government administrations shift away from renewable energy spending.

In contrast, total capital expenditure has a relatively minor effect on NPV. This is largely because the capital investments are relatively small compared to the facility's large revenue streams. As a result, even substantial changes in these values do not significantly affect the project's overall economic performance.

The ethanol buy price and jet fuel sell price moderately affect the NPV. This makes sense, as the ethanol feedstock is this process's largest expenditure and the jet fuel sell price, outside of government intervention, is the largest source of revenue.

Overall, the sensitivity analysis indicates that the economic viability of the ETJ facility is most strongly influenced by market and policy factors, particularly fuel prices and renewable fuel incentives. At the same time, the analysis shows that the project remains economically robust even when these variables decrease significantly, further demonstrating the strong financial potential of the ethanol-to-jet fuel process. The only instance of concern is when credits reach zero, which is a possibility based on the administrative agenda in the future.

Recommendations

Based on the technical, environmental, and economic analysis completed throughout this project, the ethanol-to-jet (ETJ) pathway should be strongly considered for future commercial development. The process demonstrated high jet fuel selectivity, strong greenhouse gas reduction potential, and excellent economic performance under the evaluated market conditions. Compared to competing SAF pathways such as HEFA, Fischer-Tropsch, and pyrolysis-based systems, the ETJ process offers a strong balance between scalability, feedstock flexibility, process maturity, and overall profitability. The chemistry involved in ethanol dehydration, oligomerization, and hydroprocessing is already well established in the petrochemical industry, reducing technical uncertainty relative to many emerging SAF technologies. Additionally, the process can produce approximately 80% jet-range hydrocarbons while remaining compatible with existing aviation fuel infrastructure and ASTM fuel specifications.

Economic analysis further supports pursuing the process. The project generated a highly positive net present value and a discounted cash flow rate of return that was significantly above industry benchmarks. Sensitivity analysis also showed that even under less favorable operating conditions, the process remained economically attractive due to the combination of strong SAF demand, government incentives, and renewable fuel credits. However, despite these strong results, several important concerns and uncertainties remain. One of the largest contributors to operating cost is the purchase of low-carbon ethanol feedstock and green hydrogen. Because the profitability of the process depends heavily on maintaining low-carbon-intensity fuel qualifications and incentive eligibility, fluctuations in feedstock pricing or policy changes could significantly impact long-term returns. This becomes especially apparent in the scenario where an administrative campaign champions against renewable energy.

For this reason, one of the strongest future recommendations is to produce low-carbon ethanol in-house rather than relying entirely on external suppliers such as LanzaTech. Ethanol could be produced internally through technologies such as industrial off-gas fermentation, biomass gasification followed by alcohol synthesis, or other waste-derived fermentation pathways. Integrating ethanol production directly into the facility would significantly reduce variable feedstock costs while also improving control over carbon intensity scoring and long-term supply stability. A similar recommendation applies to hydrogen production. Currently, hydrogen sourcing accounts for a major operating expense due to the high hydrogen demand in hydroprocessing. Installing an electrolyzer powered by renewable electricity would enable the facility to generate green hydrogen internally, reducing reliance on external hydrogen suppliers and improving process sustainability. While these additions would increase initial capital cost, they could substantially lower long-term operating expenses and improve overall economic resilience.

Additional process optimization opportunities also exist within the current ETJ design. One important improvement would involve greater heat integration between the

dehydration and oligomerization sections. Ethanol dehydration is strongly endothermic and requires substantial fired heater duty, while oligomerization is highly exothermic and requires extensive cooling to prevent catalyst damage and runaway reactions. Recovering heat from the oligomerization reactors and transferring it to the dehydration section could significantly reduce natural gas consumption in fired heaters, lowering both utility costs and plant emissions. This type of thermal integration would improve overall process efficiency and reduce operating costs without requiring major changes to the process chemistry.

Finally, future commercial development should evaluate increasing plant capacity beyond the current 10,000 barrels per day design basis. Expanding the facility to approximately 20,000 barrels per day could improve profitability through economies of scale. Although both capital expenditures and operating costs would increase, revenue would likely grow faster due to more efficient equipment utilization, lower per-unit production costs, and stronger overall process integration at a larger scale. Because the SAF market is expected to continue growing rapidly over the coming decades, scaling production capacity could position the facility more competitively within the future aviation fuel market.

Overall, the ethanol-to-jet process represents a technically feasible and economically promising pathway toward sustainable aviation fuel production. While challenges related to feedstock sourcing, hydrogen demand, utility consumption, and policy dependence still exist, the process shows strong potential for future commercialization. With continued optimization, additional integration of low-carbon feedstock production, and larger-scale implementation, the ETJ pathway could become one of the more competitive long-term solutions for decarbonizing the aviation industry.

Appendix

Design Basis

Our team's SAF production plant is planned for location in Wilkinson County, Georgia. This geographic region grants many strategic benefits. Firstly, LanzaTech, a producer of green ethanol, has provisions to prepare production in the same state. This offers our company the unique opportunity to source large volumes of low-carbon-intensity ethanol and possibly partner for business development in the area. Additionally, Wilkinson County sits right by Interstate 16, providing highway access to the Port of Savannah, one of the largest ports in the world, where international transactions can be conducted. This enables us to source additional low-carbon intensity ethanol from Brazil, specifically through sugarcane fermentation. The location can be seen below:



Figure 21 - State of Georgia Infrastructure

This flexibility of feedstock sourcing increases the robustness of our process, especially in uncertain times of government regulation. Having an in-state ethanol feedstock that's derived from woody biomass, agricultural waste, and industrial off-gas, and additionally foreign ethanol from sugarcane fermentation, ensures we can fulfill our design capacity. Moreover, being in Georgia grants us access to local and national jet fuel pipelines, as depicted in Figure 22:

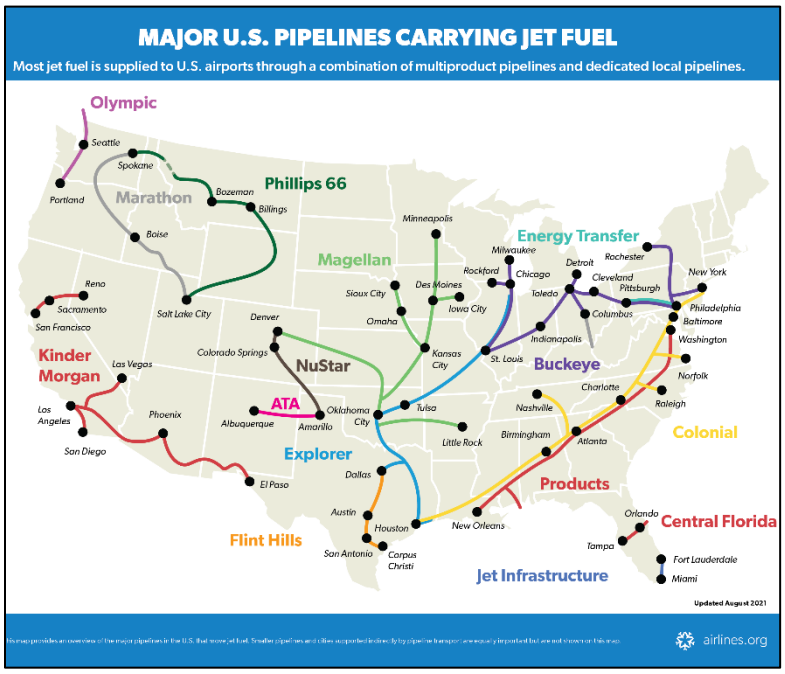


Figure 22 - American Jet Fuel Pipelines

These pipelines are not currently permitted to transport SAF, unless it's blended 50/50 with Jet Fuel A. However, many Jet Fuel A producers are in the surrounding area, and regulators are looking to approve 100% SAF for transport in pipelines by 2035. Thus, our location gives us the potential to sell our product to airports across the eastern and southeastern USA at the lowest cost via these jet fuel pipelines.

Our plan is to produce 10,000 barrels per day, for a total capacity of 3,650,000 barrels per year. This places us at the middle-to-high end of SAF production plants, with a manufacturing capacity large enough to make our business profitable and to cover the high capital investment required for our process. Additionally, 10,000 barrels per day is a good fit for the amount of low-carbon ethanol feedstock expected to be available from LanzaTech and Brazil.

Block Flow Diagram (BFD)

Our main feedstocks are ethanol in the form of E85 produced from low-carbon intensity feedstock and hydrogen gas as Green Hydrogen produce via electrolysis [30]. These feedstocks are readily available in bulk, making our process economically viable. The process produces 58.5 tons/hr of Jet Fuel (10,000 barrels/day) while requiring 62.8 tons/hr of ethanol and 36.9 tons/hr of Hydrogen. Our oligomerization process achieves near 100% conversion with recycling and a Jet Fuel selectivity of 80%. Additional feeds include fresh water and fuel gas, and the byproducts are fuel gas, flue gas, and wastewater, which are considered in the process economics.

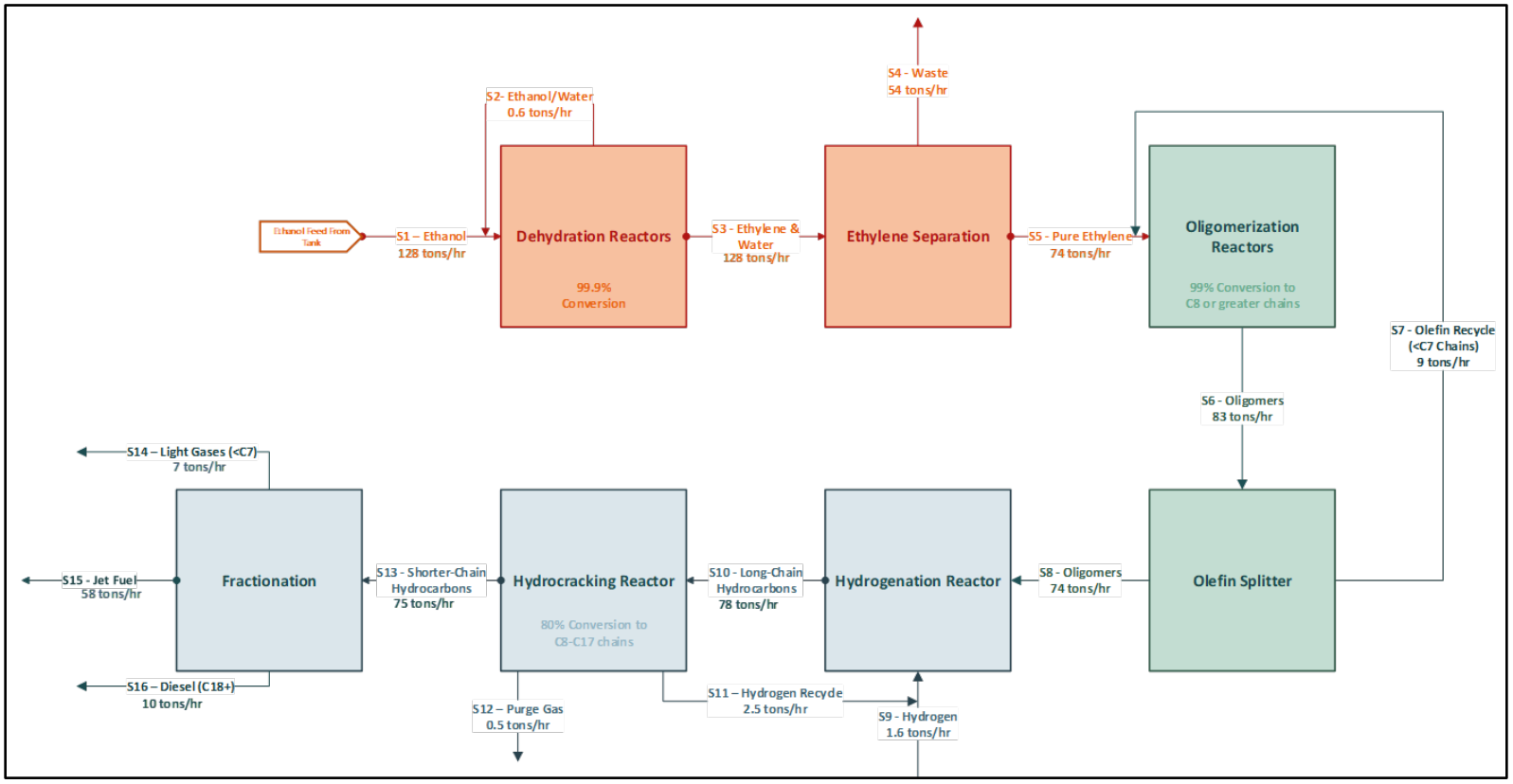


Figure 23- Ethanol to Jet Fuel BFD

Process Flow Diagrams

The first section of the ETJ process is Dehydration which can be split into three distinct sections: dehydration reaction, byproduct removal, and ethylene purification, as shown in the figures below:

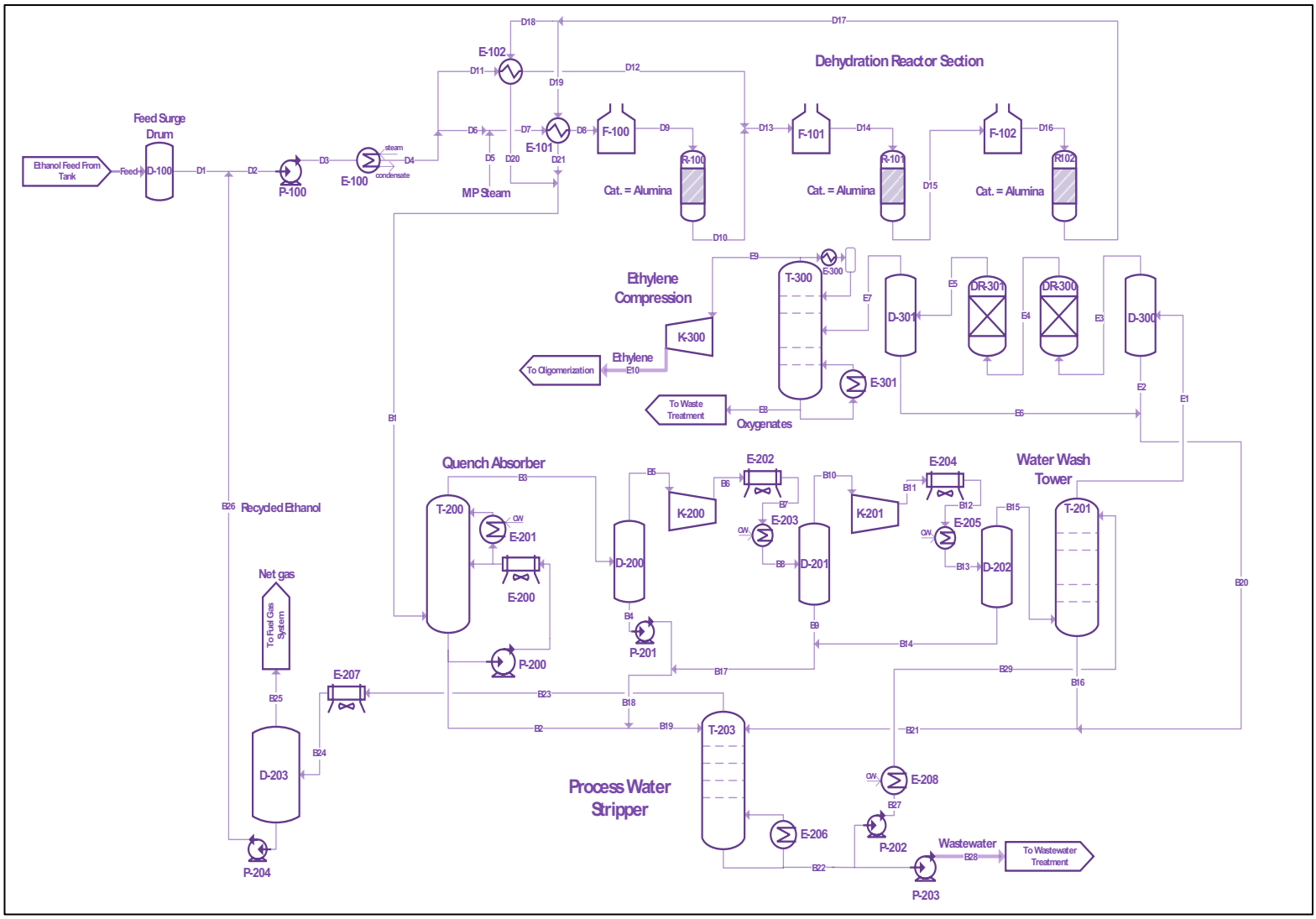


Figure 24- Dehydration of Ethanol to Ethylene PFD

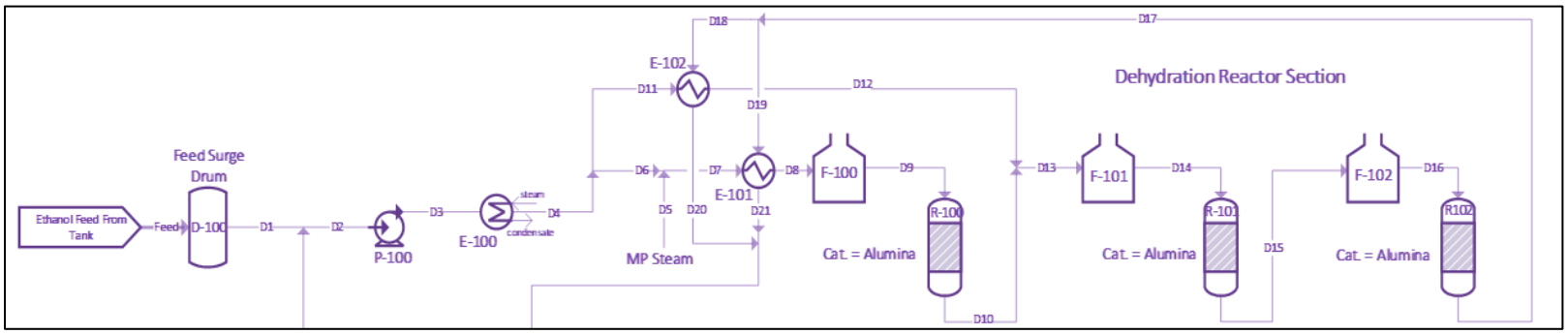


Figure 27- Dehydration Reactor Section

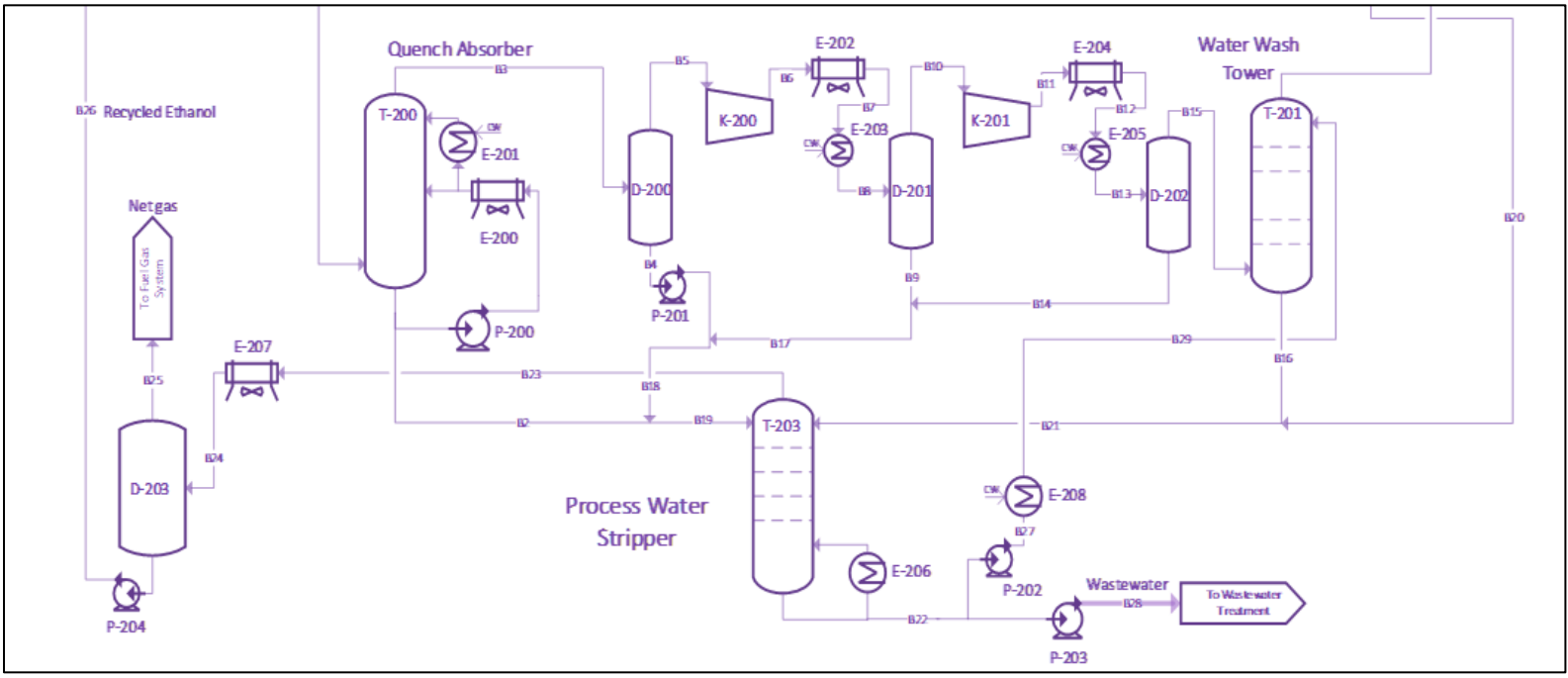


Figure 25- Byproduct Removal Section

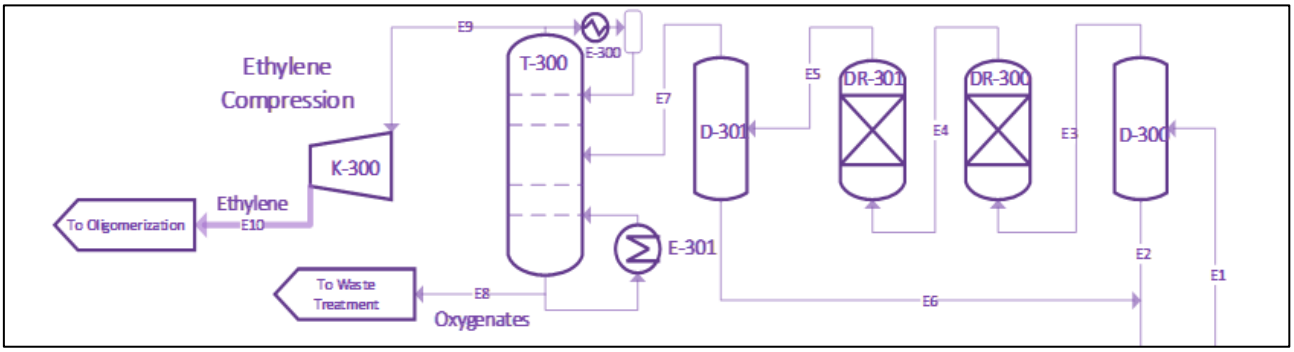


Figure 26- Ethylene Purification Section

The second section of the ETJ process is oligomerization which can be split into two distinct sections: oligomerization reaction and olefin separation, as shown in the figures below:

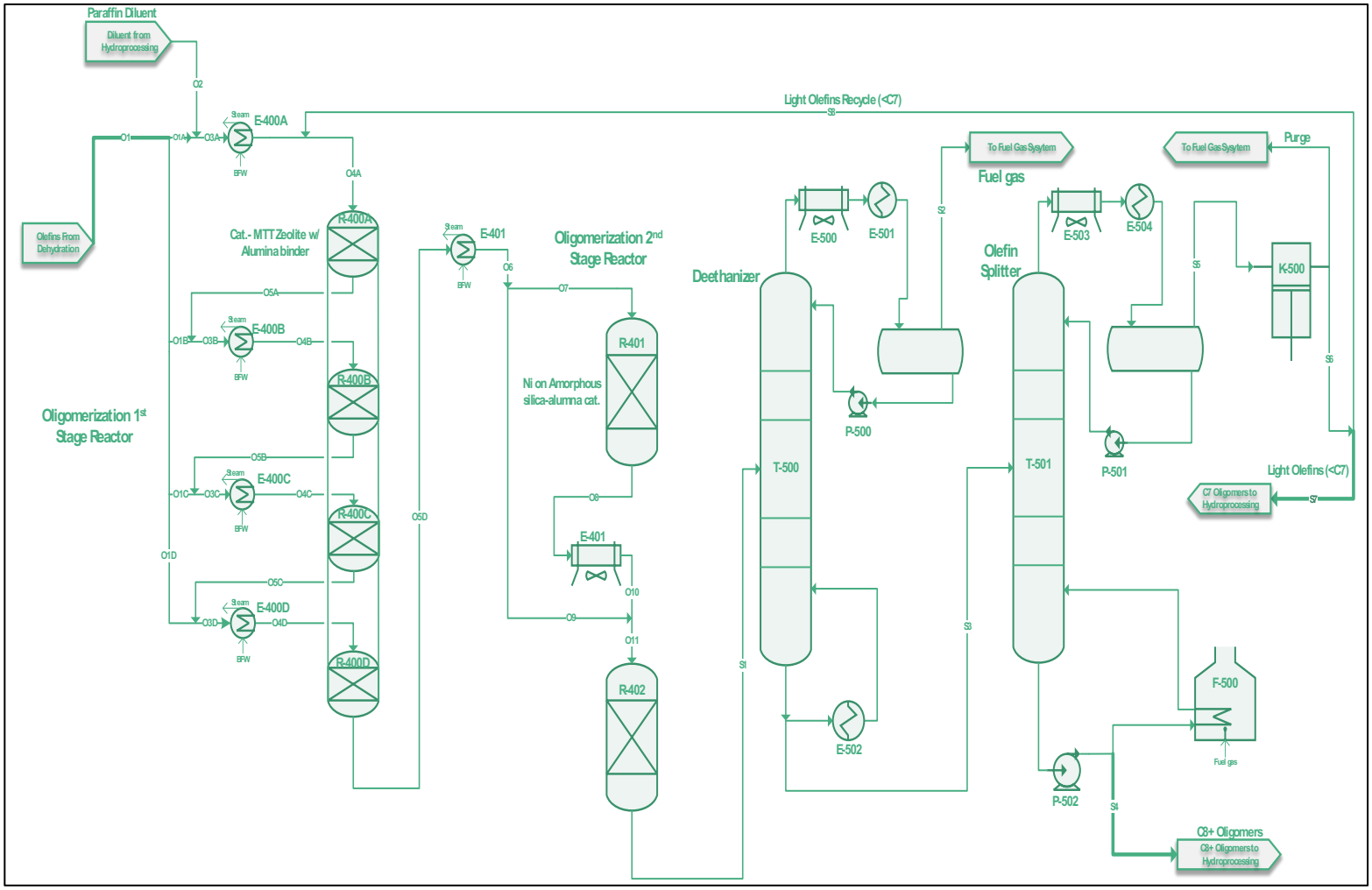


Figure 28- Oligomerization of Olefins to Oligomers PFD

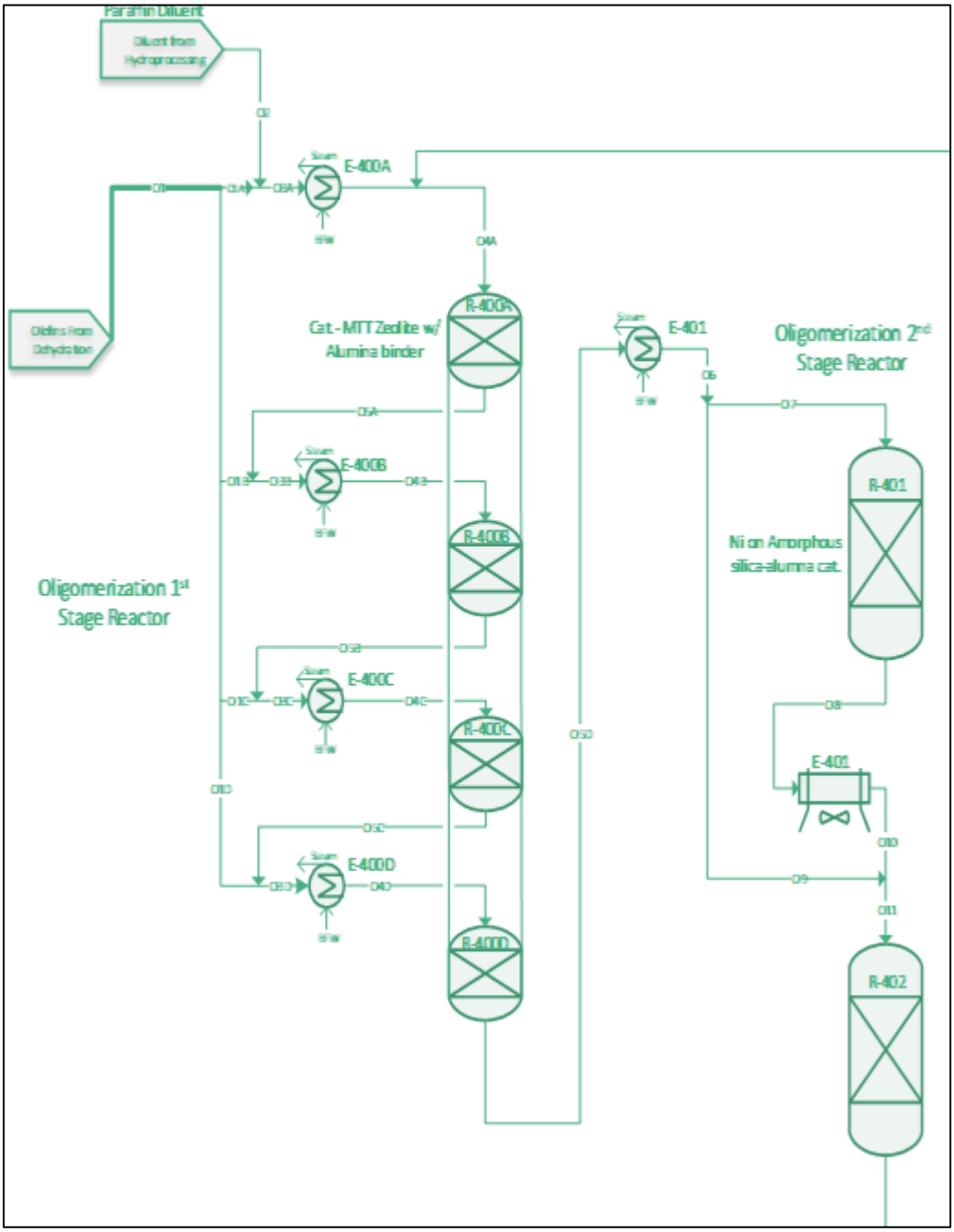


Figure 29- Oligomerization Reactor Section

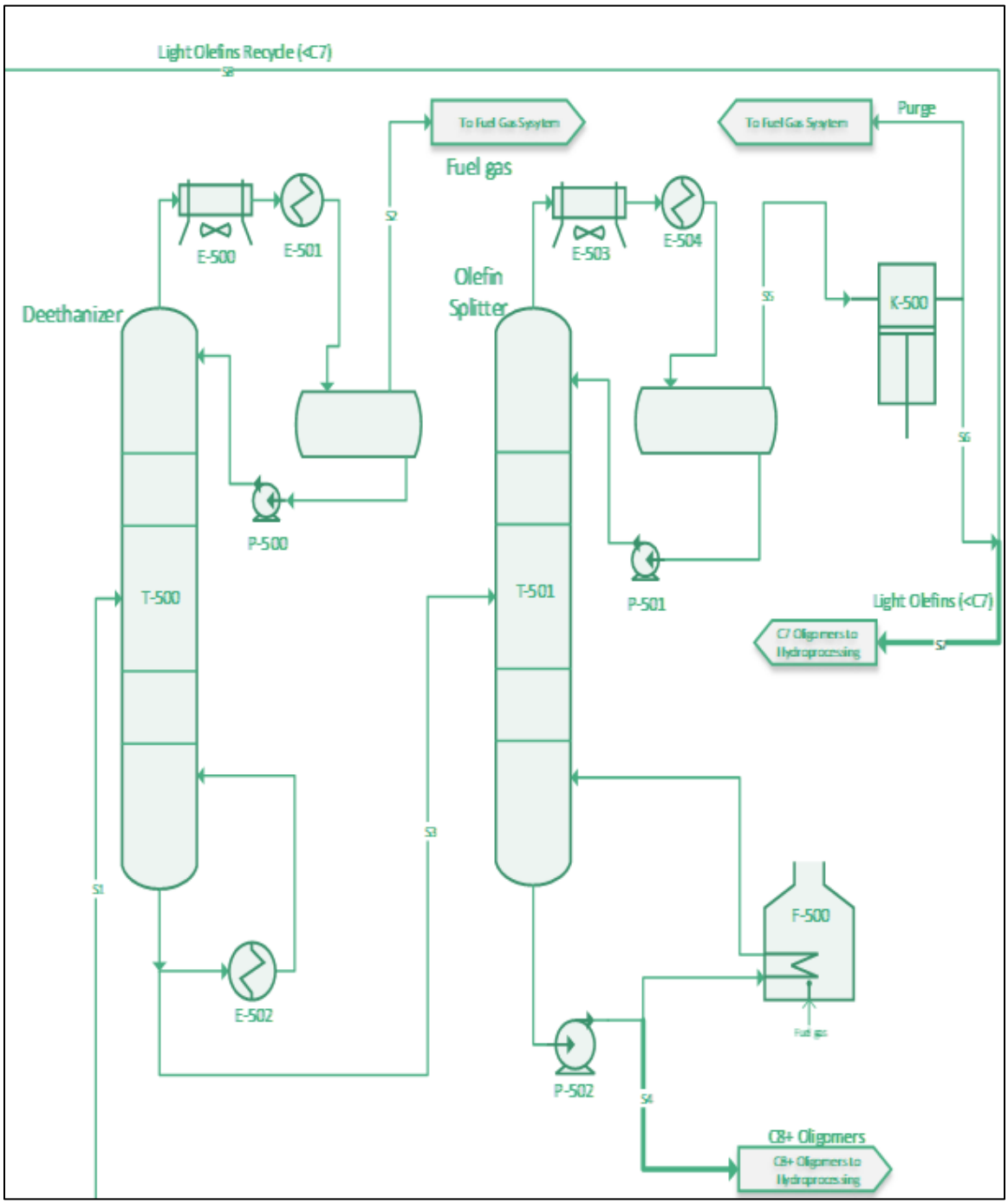


Figure 30- Olefin Separation Section

The third and final section of the ETJ process is hydroprocessing which can be split into two distinct sections: hyroprocessing reactions and Fractionation of products, as shown in the figures below:

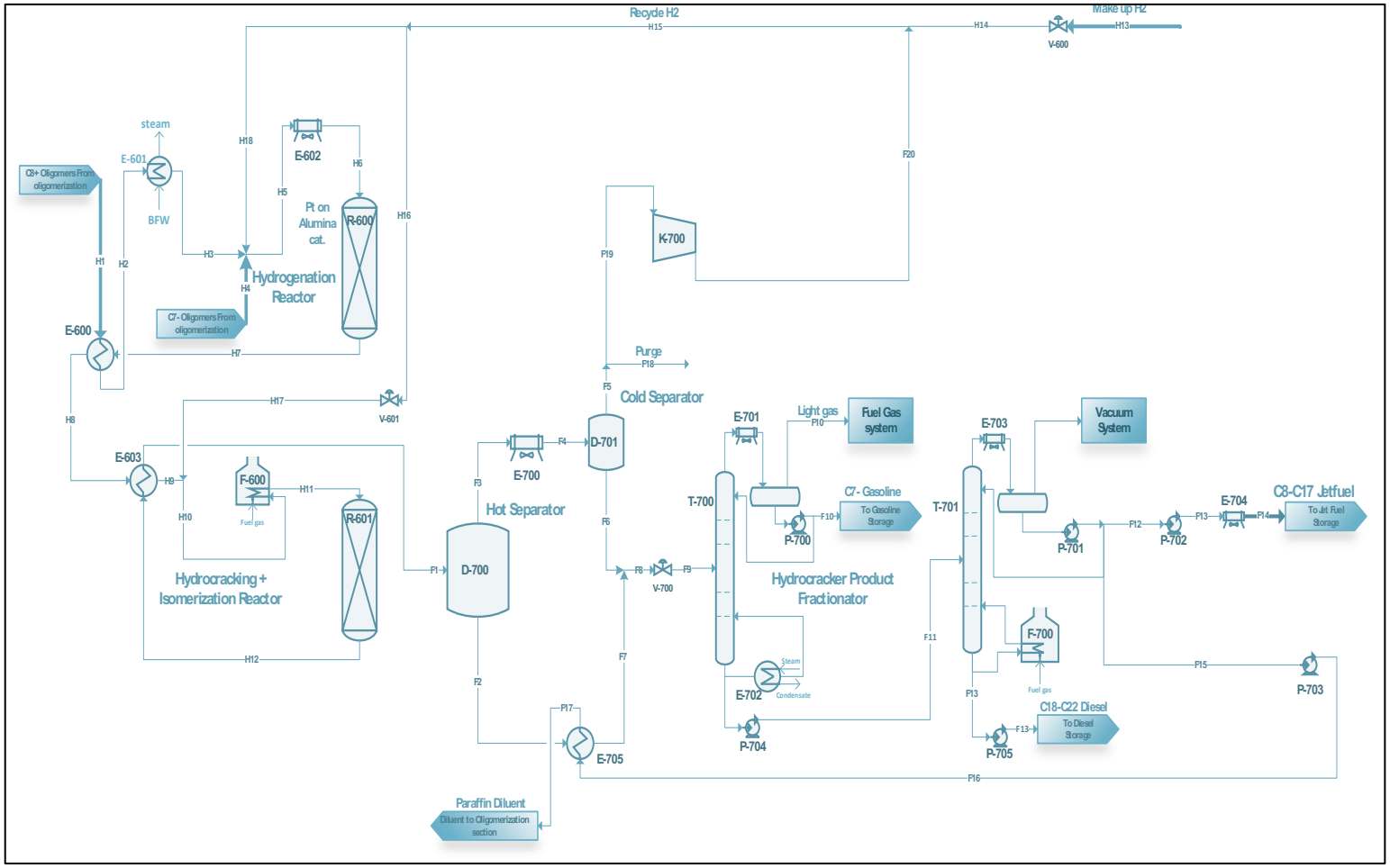


Figure 31- Hydroprocessing Oligomers to Jet Fuel PFD

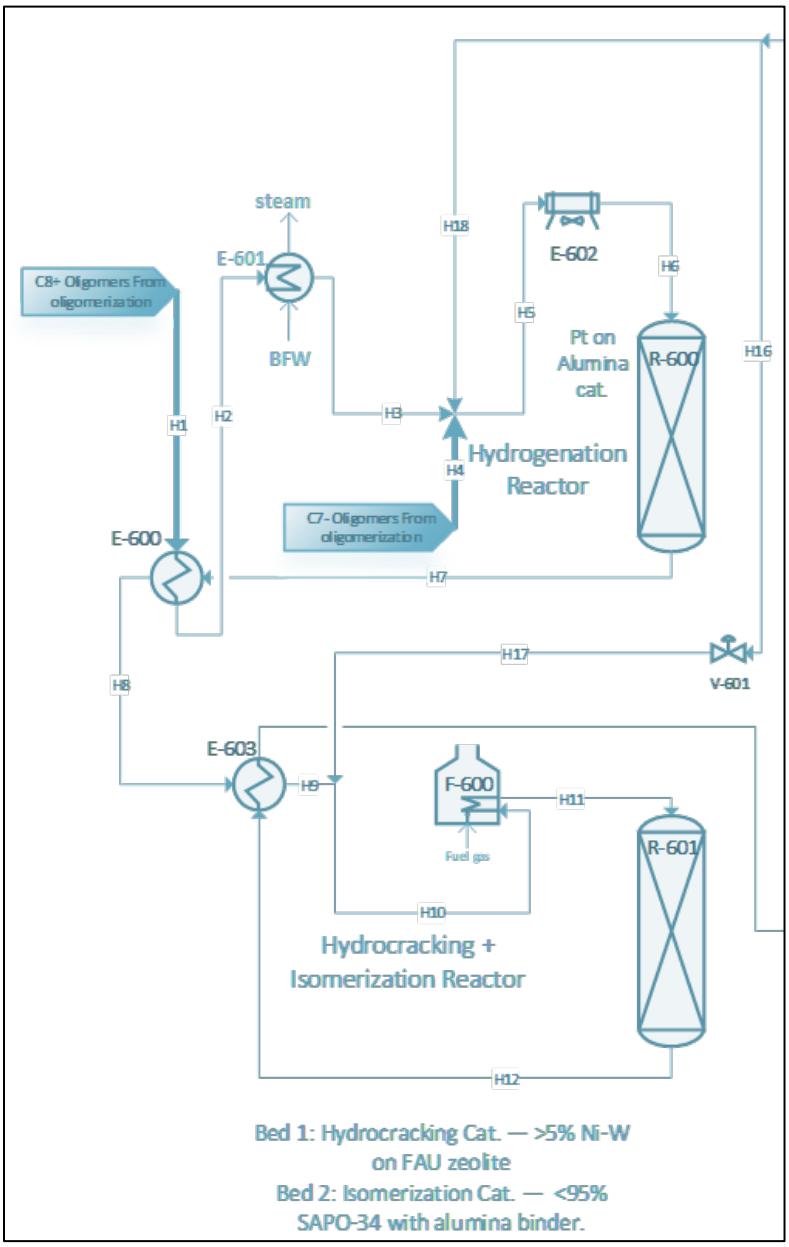


Figure 32- Hydroprocessing Reactions Section

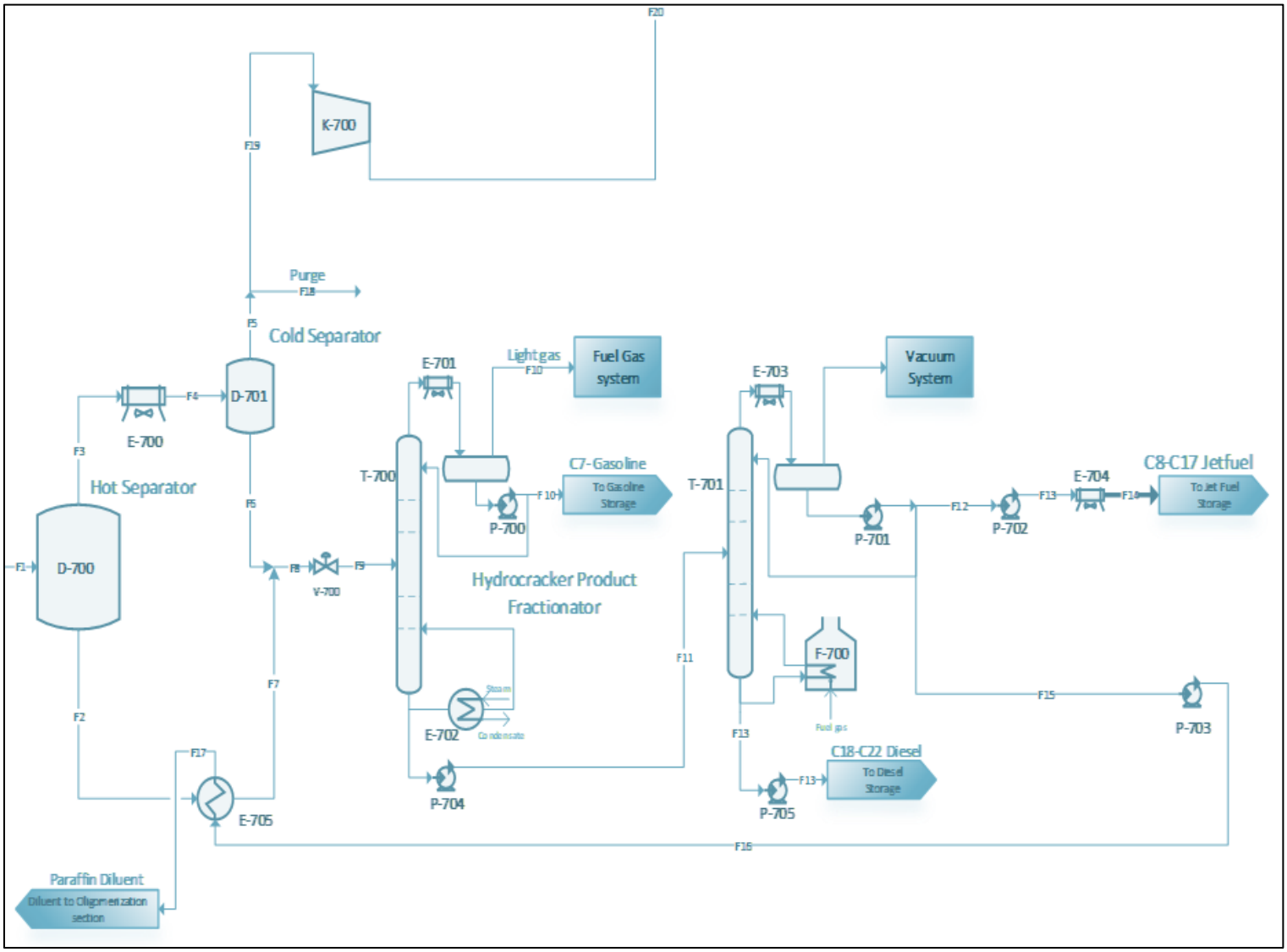


Figure 33- Fractionation of Products Section

Simulation PFD

The dehydration and purification section of the ethanol-to-jet (ETJ) process was simulated using Aspen Plus to evaluate mass balances, reaction performance, and separation efficiency under steady-state conditions. The simulation models the conversion of ethanol to purified ethylene prior to downstream processing.

Chemical reactions were modeled using RStoic reactor blocks, where reactions were defined using specified fractional conversions instead of detailed kinetics. This approach was selected because reliable catalyst kinetic data were not available. The primary reaction modeled was ethanol dehydration to ethylene and water, with multiple reactor stages used to represent the temperature drop associated with the endothermic reaction and to improve overall conversion.

The NRT and PR-BM thermodynamic packages was used to model vapor–liquid equilibrium and phase behavior for the polar components, hydrocarbon and light gas mixtures present in the system. Standard Aspen unit operation models were used for heat exchangers, compressors, separators, and dryers to simulate temperature, pressure, and product purity control throughout the process.

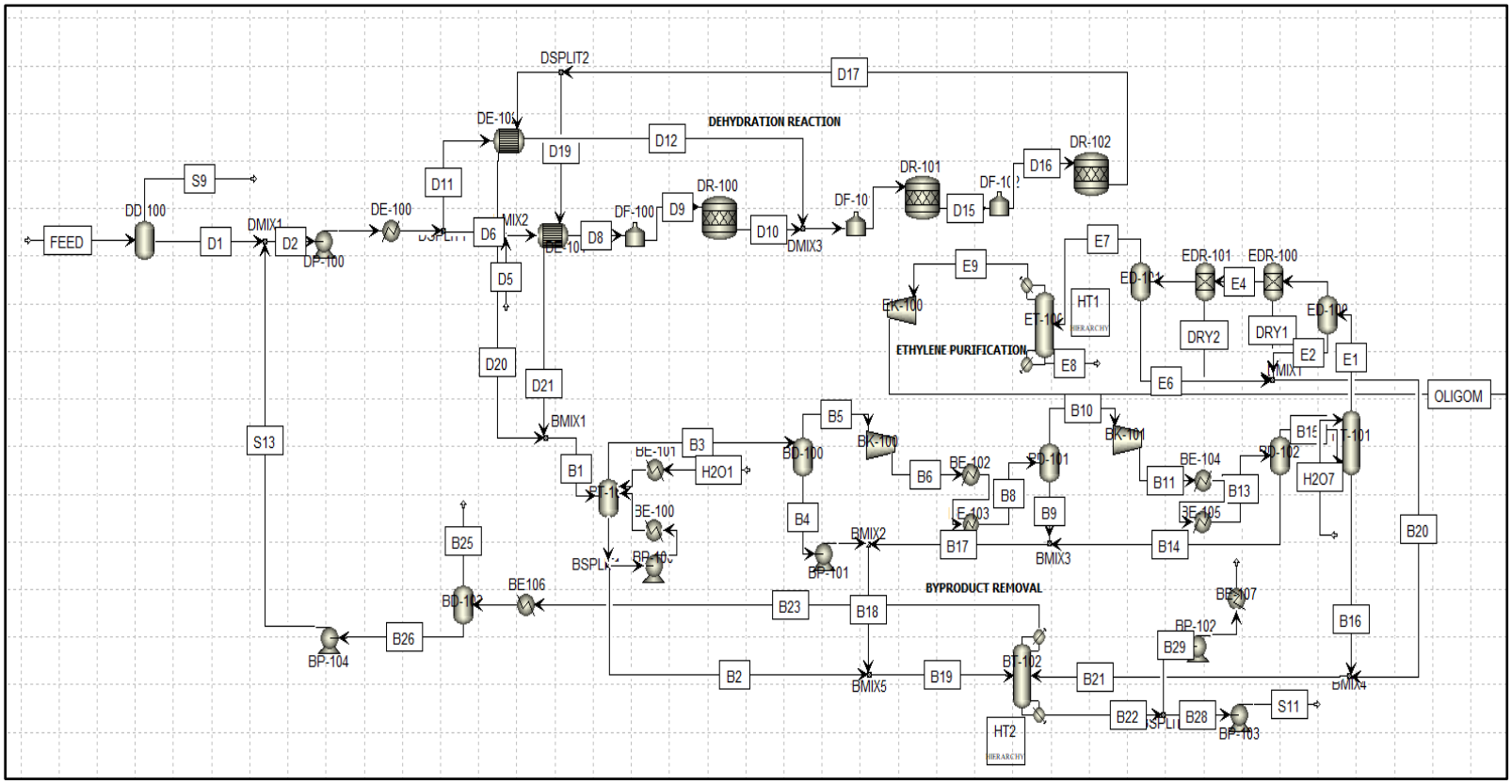


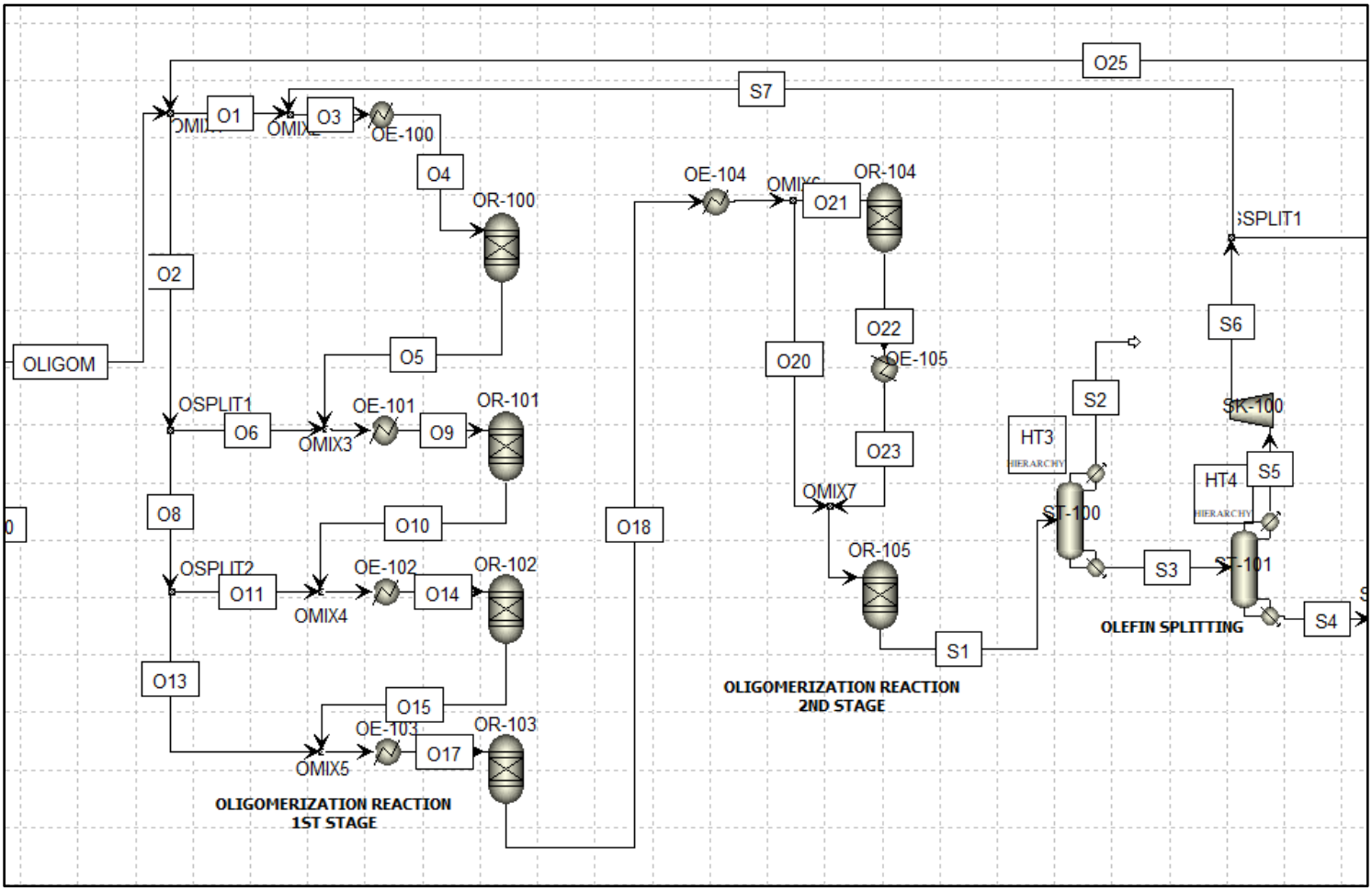
Figure 34- Dehydration of Ethanol to Ethylene Simulation PFD

The oligomerization section of the ethanol-to-jet (ETJ) process was simulated in Aspen Plus to model the conversion of ethylene into higher molecular weight olefins suitable for jet fuel production. This section includes multi-stage reactor systems, interstage cooling, and product separation to control the highly exothermic oligomerization reactions and achieve the desired hydrocarbon distribution.

The oligomerization reactors were modeled using RYield reactor blocks, where product distributions were specified based on yield data reported in LanzaJet and related patent literature. This approach was selected because detailed kinetic models for proprietary catalysts are not publicly available. The RYield reactors allow direct specification of product composition, enabling realistic representation of the formation of C8–C16 olefins required for downstream hydroprocessing.

The PR-BM thermodynamic package was used to model vapor–liquid equilibrium and phase behavior for the hydrocarbon mixtures present in the system. Standard Aspen unit operation models were used for heat exchangers, separators, and distillation units to simulate temperature control, reaction heat removal, and olefin recovery.

Figure 35- Oligomerization of Ethylene to Oligomers Simulation PFD



The hydroprocessing section of the ethanol-to-jet (ETJ) process was simulated in Aspen Plus to model the upgrading of olefin intermediates into finished jet fuel products through hydrogenation, hydrocracking, and isomerization reactions. This section includes reactor systems, heat exchangers, and fractionation units to achieve the required fuel specifications and product separation.

The hydrogenation reactor was modeled using an RStoic reactor block, where reactions were defined using specified fractional conversions to represent the saturation of olefins into paraffins in the presence of hydrogen. The hydrocracking and isomerization reactors were modeled using RYield reactor blocks, with product distributions defined based on literature and patent data to represent the formation of jet fuel range hydrocarbons and branched molecules that meet aviation fuel property requirements.

The Peng–Robinson thermodynamic package was used to model vapor–liquid equilibrium and phase behavior for the hydrocarbon and hydrogen mixtures present in the system. Standard Aspen unit operation models were used for heat exchangers, separators, and distillation columns to simulate reaction heat management and product fractionation into jet fuel, diesel, and light hydrocarbon streams.

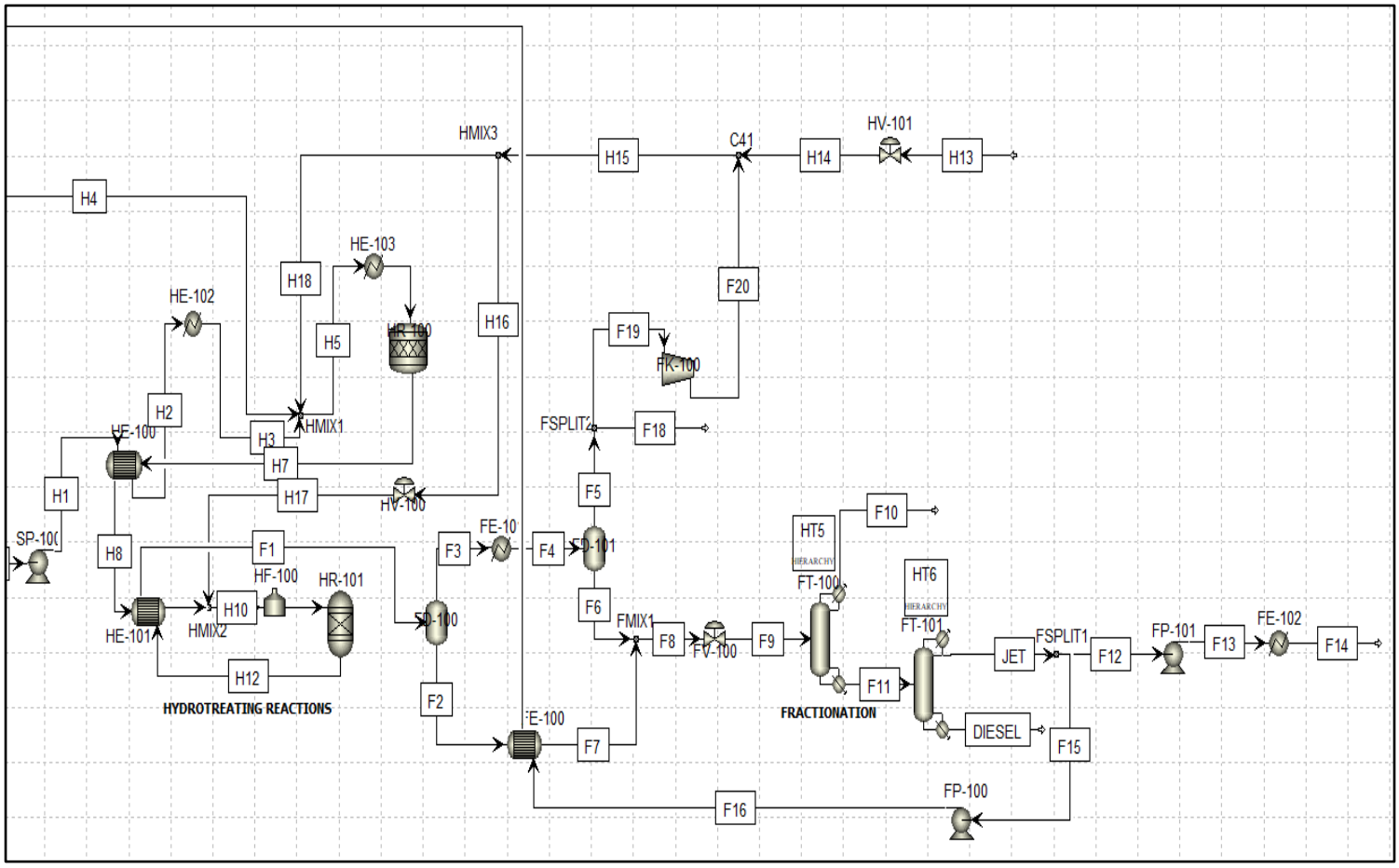


Figure 36- Hydroprocessing of Oligomers to Jet Fuel Simulation PFD

Plant Utilities

Table 13 - Total Plant-wide Utilities Expenditure

Name	Fluid	Rate	Rate Units	Cost/Year
Electricity	N/A	6,131	KW	**$2.9MM**
Cooling Water	Water	82	MMBTU/H	$0.24MM
Fuel Gas	Steam	283	MMBTU/H	**$10.1MM**
HP Steam	Steam	11	MMBTU/H	$0.24MM
LP Steam	Steam	4.6	MMBTU/H	$0.07MM
HP Steam	Steam	180	MMBTU/H	**$3.3MM**
Refrigerant	Propane	10	MMBTU/H	$0.23MM

Detailed Mass Balance

Table 14- Dehydration Reactor Section Mass Balance 1

Component	Units	D1	D2	D3	D4	D5	D6	D7
Temperature	F	77	77	78	77	347	278	319
Pressure	psia	40	39	106	40	123	101	101
Mass Flows	lb/hr	256,325	257,454	257,454	256,325	183,788	128,727	312,515
Ethanol	lb/hr	245,047	245,051	245,051	245,047	-	122,526	122,526
Water	lb/hr	11,278	12,291	12,291	11,278	183,788	6,145	189,934
Ethylene	lb/hr	-	92	92	-	-	46	46
Methane	lb/hr	-	0	0	-	-	0	0
Ethane	lb/hr	-	0	0	-	-	0	0
Propylene	lb/hr	-	1	1	-	-	1	1
Hydrogen	lb/hr	-	0	0	-	-	0	0
CO	lb/hr	-	0	0	-	-	0	0
CO2	lb/hr	-	0	0	-	-	0	0
Acetic Acid	lb/hr	-	0	0	-	-	0	0
2-Butanone	lb/hr	-	4	4	-	-	2	2
Acetaldehyde	lb/hr	-	15	15	-		8	8

Table 15- Dehydration Reactor Section Mass Balance 2

Component	Units	D8	D9	D10	D11	D12	D13	D14
Temperature	F	662	842	527	278	707	584	842
Pressure	psia	91	86	105	101	91	91	86
Mass Flows	lb/hr	312,515	312,515	312,515	128,727	128,727	441,242	441,242
Ethanol	lb/hr	122,526	122,526	1,225	122,526	122,526	123,751	123,751
Water	lb/hr	189,934	189,934	237,082	6,145	6,145	243,227	243,227
Ethylene	lb/hr	46	46	73,382	46	46	73,428	73,428
Methane	lb/hr	0	0	6	0	0	6	6
Ethane	lb/hr	0	0	61	0	0	61	61
Propylene	lb/hr	1	1	85	1	1	85	85
Hydrogen	lb/hr	0	0	28	0	0	28	28
CO	lb/hr	0	0	11	0	0	11	11
CO2	lb/hr	0	0	6	0	0	6	6
Acetic Acid	lb/hr	0	0	122	0	0	122	122
2-Butanone	lb/hr	2	2	18	2	2	19	19
Acetaldehyde	lb/hr	8	8	489	8	8	497	497

Table 16- Dehydration Reactor Section Mass Balance 3

Component	Units	D15	D16	D17	D18	D19	D20	D21
Temperature	F	586	860	837	837	837	599	381
Pressure	psia	95	90	85	85	85	75	75
Mass Flows	lb/hr	441,242	441,242	441,242	220,621	220,621	220,621	220,621
Ethanol	lb/hr	1,238	1,238	12	6	6	6	6
Water	lb/hr	290,846	290,846	291,323	145,661	145,661	145,661	145,661
Ethylene	lb/hr	147,496	147,496	148,237	74,119	74,119	74,119	74,119
Methane	lb/hr	12	12	12	6	6	6	6
Ethane	lb/hr	123	123	124	62	62	62	62
Propylene	lb/hr	171	171	171	86	86	86	86
Hydrogen	lb/hr	57	57	57	29	29	29	29
CO	lb/hr	22	22	22	11	11	11	11
CO2	lb/hr	13	13	13	7	7	7	7
Acetic Acid	lb/hr	246	246	247	123	123	123	123
2-Butanone	lb/hr	35	35	35	18	18	18	18
Acetaldehyde	lb/hr	983	983	988	494	494	494	494

Component	Units	B1	B2	B3	B4	B5	B6	B7
Temperature	F	492	209	240	240	240	374	149
Pressure	psia	75	73	73	73	73	175	174
Mass Flows	lb/hr	441,318	417,786	199,901	770	199,132	199,132	199,132
Ethanol	lb/hr	12	1	12	0	12	12	12
Water	lb/hr	291,373	417,730	50,012	749	49,263	49,263	49,263
Ethylene	lb/hr	148,263	31	148,231	14	148,217	148,217	148,217
Methane	lb/hr	13	0	12	0	12	12	12
Ethane	lb/hr	124	0	124	0	124	124	124
Propylene	lb/hr	171	0	171	0	171	171	171
Hydrogen	lb/hr	57	0	57	0	57	57	57
CO	lb/hr	22	0	22	0	22	22	22
CO2	lb/hr	13	0	13	0	13	13	13
Acetic Acid	lb/hr	247	22	225	6	219	219	219
2-Butanone	lb/hr	35	0	35	0	35	35	35
Acetaldehyde	lb/hr	988	3	985	0	985	985	985

Table 17- Byproduct Removal Section Mass Balance 1

Table 18- Byproduct Removal Section Mass Balance 2

Component	Units	B8	B9	B10	B11	B12	B13	B14
Temperature	F	95	116	116	211	149	95	133
Pressure	psia	169	169	169	421	420	415	415
Mass Flows	lb/hr	199,132	48,375	150,757	150,757	150,757	150,757	174
Ethanol	lb/hr	12	1	11	11	11	11	0
Water	lb/hr	49,263	48,361	902	902	902	902	174
Ethylene	lb/hr	148,217	2	148,215	148,215	148,215	148,215	0
Methane	lb/hr	12	0	12	12	12	12	0
Ethane	lb/hr	124	0	124	124	124	124	0
Propylene	lb/hr	171	0	171	171	171	171	0
Hydrogen	lb/hr	57	0	57	57	57	57	0
CO	lb/hr	22	0	22	22	22	22	0
CO2	lb/hr	13	0	13	13	13	13	0
Acetic Acid	lb/hr	219	10	209	209	209	209	0
2-Butanone	lb/hr	35	0	35	35	35	35	0
Acetaldehyde	lb/hr	985	0	985	985	985	985	0

Table 19- Byproduct Removal Section Mass Balance 3

Component	Units	B15	B16	B17	B18	B19	B20	B21
Temperature	F	133	133	78	81	196	91	96
Pressure	psia	415	406	169	73	73	406	406
Mass Flows	lb/hr	150,583	11,084	48,549	49,319	467,105	1,523	12,607
Ethanol	lb/hr	11	0	1	1	1	6	7
Water	lb/hr	728	11,079	48,535	49,284	467,014	672	11,751
Ethylene	lb/hr	148,215	1	2	17	48	571	572
Methane	lb/hr	12	0	0	0	0	0	0
Ethane	lb/hr	124	0	0	0	0	1	1
Propylene	lb/hr	171	0	0	0	0	4	4
Hydrogen	lb/hr	57	0	0	0	0	0	0
CO	lb/hr	22	0	0	0	0	0	0
CO2	lb/hr	13	0	0	0	0	0	0
Acetic Acid	lb/hr	209	3	11	17	38	186	190
2-Butanone	lb/hr	35	0	0	0	0	11	11
Acetaldehyde	lb/hr	985	0	0	1	3	72	72

Table 20- Byproduct Removal Section Mass Balance 4

Component	Units	B22	B23	B24	B25	B26	B27	B28	B29
Temperature	F	274	252	140	147	147	275	275	275
Pressure	psia	44	44	39	39	39	411	411	411
Mass Flows	lb/hr	477,948	1,765	1,765	636	1,129	477,948	466,924	11,023
Ethanol	lb/hr	2	6	6	2	5	2	2	0
Water	lb/hr	477,718	1,047	1,047	35	1,012	477,718	466,700	11,018
Ethylene	lb/hr	0	620	620	528	92	0	0	0
Methane	lb/hr	0	0	0	0	0	0	0	0
Ethane	lb/hr	0	1	1	1	0	0	0	0
Propylene	lb/hr	0	4	4	3	1	0	0	0
Hydrogen	lb/hr	0	0	0	0	0	0	0	0
CO	lb/hr	0	0	0	0	0	0	0	0
CO2	lb/hr	0	0	0	0	0	0	0	0
Acetic Acid	lb/hr	228	0	0	0	0	228	223	5
2-Butanone	lb/hr	0	11	11	8	4	0	0	0
Acetaldehyde	lb/hr	0	75	75	60	15	0	0	0

Table 21- Ethylene Purification Mass Balance

Component	Units	E1	E2	E3	E4	E5	E6	E7	E8	E9
Temperature	F	130	91	91	91	91	91	91	134	2
Pressure	psia	406	406	406	406	406	406	406	402	400
Mass Flows	lb/hr	150,522	1,425	149,096	148,999	148,999	98	148,999	1,307	147,692
Ethanol	lb/hr	11	6	5	5	5	-	5	5	0
Water	lb/hr	672	575	98	0	0	98	0	0	0
Ethylene	lb/hr	148,213	571	147,643	147,643	147,643	-	147,643	148	147,495
Methane	lb/hr	12	0	12	12	12	-	12	0	12
Ethane	lb/hr	124	1	123	123	123	-	123	30	93
Propylene	lb/hr	171	4	168	168	168	-	168	168	0
Hydrogen	lb/hr	57	0	57	57	57	-	57	0	57
CO	lb/hr	22	0	22	22	22	-	22	0	22
CO2	lb/hr	13	0	13	13	13	-	13	0	13
Acetic Acid	lb/hr	205	186	19	19	19	-	19	19	0
2-Butanone	lb/hr	35	11	24	24	24	-	24	24	0
Acetaldehyde	lb/hr	985	72	913	913	913	-	913	913	0

Table 22- Oligomerization Reactor Section Mass Balance 1

Component	Units	O1	O1A	O2	O3A	O4A	O5A
Temperature	F	112	198	410	265	248	320
Pressure	psia	1,015	1,005	1,005	1,005	1,000	1,010
Mass Flows	lb/hr	147,692	46,141	36,874	64,451	64,451	64,451
Ethylene	lb/hr	147,495	36,874	-	36,874	36,874	549
Methane	lb/hr	12	3	-	3	3	3
Ethane	lb/hr	93	23	-	24	24	24
Hydrogen	lb/hr	57	14	-	14	14	14
CO	lb/hr	22	5	-	5	5	5
CO2	lb/hr	13	3		3	3	3
Butene	lb/hr	-	-	-	4,097	4,097	39,678
Pentene	lb/hr	-	-	-	828	828	-
Hexene	lb/hr	-	-	-	12,402	12,402	13,045
Heptene	lb/hr	-	-	-	702	702	-
Octene	lb/hr	-	-	-	0	0	1,631
Parraffins	lb/hr			36,874	9,218	9,218	9,218

Table 23- Oligomerization Reactor Section Mass Balance 2

Component	Units	O1B	O3B	O4B	O5B	O1C	O3C
Temperature	F	198	283	248	320	198	294
Pressure	psia	1,005	1,005	1,000	1,005	1,005	1,005
Mass Flows	lb/hr	46,141	110,592	110,592	110,592	46,141	156,734
Ethylene	lb/hr	36,874	37,423	37,423	918	36,874	37,792
Methane	lb/hr	3	6	6	6	3	9
Ethane	lb/hr	23	47	47	47	23	70
Hydrogen	lb/hr	14	29	29	29	14	43
CO	lb/hr	5	11	11	11	5	16
CO2	lb/hr	3	6	6	6	3	9
Butene	lb/hr	-	39,678	39,678	66,327	-	66,327
Pentene	lb/hr	-	-	-	-	-	-
Hexene	lb/hr	-	13,045	13,045	21,806	-	21,806
Heptene	lb/hr	-	-	-	-	-	-
Octene	lb/hr	-	1,631	1,631	2,726	-	2,726
Paraffins	lb/hr	9,218	18,437	18,437	18,437	9,218	27,655

Table 24- Oligomerization Reactor Section Mass Balance 3

Component	Units	O4C	O5C	O1D	O3D	O4D	O5D
Temperature	F	248	320	198	295	248	320
Pressure	psia	1,000	1,000	1,005	1,000	995	995
Mass Flows	lb/hr	156,734	156,734	46,141	202,875	202,875	202,875
Ethylene	lb/hr	37,792	1,287	36,874	38,160	38,160	1,655
Methane	lb/hr	9	9	3	12	12	12
Ethane	lb/hr	70	70	23	93	93	93
Hydrogen	lb/hr	43	43	14	57	57	57
CO	lb/hr	16	16	5	22	22	22
CO2	lb/hr	9	9	3	13	13	13
Butene	lb/hr	66,327	92,976	-	92,976	92,976	119,624
Pentene	lb/hr	-	-	-	-	-	-
Hexene	lb/hr	21,806	30,567	-	30,567	30,567	39,329
Heptene	lb/hr	-	-	-	-	-	-
Octene	lb/hr	2,726	3,821	-	3,821	3,821	4,916
Parrafins	lb/hr	27,655	27,655	9,218	36,874	36,874	36,874

Table 25- Oligomerization Reactor Section Mass Balance 4

Component	Units	O6	O7	O8	O9	O10	O11
Temperature	F	248	248	320	248	248	248
Pressure	psia	990	990	990	990	985	985
Mass Flows	lb/hr	202,875	101,438	101,438	101,438	101,438	202,875
Ethylene	lb/hr	1,655	828	-	828	-	828
Methane	lb/hr	12	6	6	6	6	12
Ethane	lb/hr	93	47	47	47	47	93
Hydrogen	lb/hr	57	29	29	29	29	57
CO	lb/hr	22	11	11	11	11	22
CO2	lb/hr	13	6	6	6	6	13
Butene	lb/hr	119,624	59,812	4,138	59,812	4,138	63,950
Pentene	lb/hr	-	-	828	-	828	828
Hexene	lb/hr	39,329	19,664	12,414	19,664	12,414	32,079
Heptene	lb/hr	-	-	828	-	828	828
Octene	lb/hr	4,916	2,458	14,070	2,458	14,070	16,528
Nonene	lb/hr	-	-	828	-	18,208	828
Decene	lb/hr	-	-	18,208	-	828	18,208
Undecene	lb/hr	-	-	828	-	13,242	828
Dodecene	lb/hr	-	-	13,242	-	1,655	13,242
Tridecene	lb/hr	-	-	1,655	-	8,276	1,655
Tetradecene	lb/hr	-	-	8,276	-	5,793	8,276
Hexadecene	lb/hr	-	-	5,793	-	1,655	5,793
Octadecene	lb/hr	-	-	1,655	-		1,655

Component	Units	O6	O7	O8	O9	O10	O11
Parrafins	lb/hr	36,874	18,437	18,437	18,437	18,437	36,874

Table 26- Olefin Separation Section Mass Balance

Component	Units	S1	S2	S3	S4	S5	S6	S7	S8
Temperature	F	320	19	600	394	179	396	396	396
Pressure	psia	985	415	417	36	33	1,015	1,015	1,015
Mass Flows	lb/hr	202,875	279	202,597	165,977	36,619	36,619	18,310	18,310
Ethylene	lb/hr	-	-	-	-	-	-	-	-
Methane	lb/hr	12	12	0	0	0	0	0	0
Ethane	lb/hr	93	92	1	0	1	1	0	0
Hydrogen	lb/hr	57	57	0	0	0	0	0	0
CO	lb/hr	22	22	0	0	0	0	0	0
CO2	lb/hr	13	13	0	0	0	0	0	0
Butene	lb/hr	8,276	82	8,194	0	8,194	8,194	4,097	4,097
Pentene	lb/hr	1,655	0	1,655	0	1,655	1,655	828	828
Hexene	lb/hr	24,829	0	24,829	25	24,804	24,804	12,402	12,402
Heptene	lb/hr	1,655	0	1,655	251	1,405	1,405	702	702
Octene	lb/hr	28,139	0	28,139	28,138	1	1	0	0
Nonene	lb/hr	1,655	0	1,655	1,655	0	0	0	0
Decene	lb/hr	36,415	0	36,415	36,415	0	0	0	0
Undecene	lb/hr	1,655	0	1,655	1,655	0	0	0	0
Dodecene	lb/hr	26,484	0	26,484	26,484	0	0	0	0
Tridecene	lb/hr	3,310	0	3,310	3,310	0	0	0	0

Component	Units	S1	S2	S3	S4	S5	S6	S7	S8
Tetradecene	lb/hr	16,552	0	16,552	16,552	0	0	0	0
Hexadecene	lb/hr	11,587	0	11,587	11,587	0	0	0	0
Octadecene	lb/hr	3,310	0	3,310	3,310	0	0	0	0
Parrafins	lb/hr	36,874	-	36,874	36,874	-	-	-	-

Table 27 - Hydroprocessing Reactions Mass Balance 1

Component	Units	H1	H2	H3	H4	H5	H6
Temperature	F	402	536	410	396	375	356
Pressure	psia	1,015	1,005	1,000	1,015	1,000	995
Mass Flows	lb/hr	165,977	165,977	165,977	18,310	192,325	192,325
HYDROGEN	lb/hr	0	0	0	0	5,701	5,701
1-BUT-01	lb/hr	0	0	0	4,097	4,097	4,097
N-BUT-01	lb/hr	0	0	0	0	1,219	1,219
1-PEN-01	lb/hr	0	0	0	828	828	828
N-PEN-01	lb/hr	0	0	0	0	570	570
1-HEX-01	lb/hr	25	25	25	12,402	12,427	12,427
N-HEX-01	lb/hr	1	1	1	80	318	318
1-HEP-01	lb/hr	251	251	251	702	953	953
N-HEP-01	lb/hr	253	253	253	200	551	551
1-Oct-01	lb/hr	28,138	28,138	28,138	0	28,139	28,139
N-OCT-01	lb/hr	3,600	3,600	3,600	0	3,729	3,729
1-NON-01	lb/hr	1,655	1,655	1,655	0	1,655	1,655
N-NON-01	lb/hr	3,629	3,629	3,629	0	3,681	3,681

Component	Units	H1	H2	H3	H4	H5	H6
1-Dec-01	lb/hr	36,415	36,415	36,415	0	36,415	36,415
N-DEC-01	lb/hr	3,642	3,642	3,642	0	3,661	3,661
1-UND-01	lb/hr	1,655	1,655	1,655	0	1,655	1,655
N-UND-01	lb/hr	3,646	3,646	3,646	0	3,654	3,654
1-DOD-01	lb/hr	26,484	26,484	26,484	0	26,484	26,484
N-DOD-01	lb/hr	3,648	3,648	3,648	0	3,651	3,651
1-TRI-01	lb/hr	3,310	3,310	3,310	0	3,310	3,310
N-TRI-01	lb/hr	3,649	3,649	3,649	0	3,650	3,650
1-TET-01	lb/hr	16,552	16,552	16,552	0	16,552	16,552
N-TET-01	lb/hr	3,649	3,649	3,649	0	3,649	3,649
1-PEN-02	lb/hr	-	-	-	-	-	-
N-PEN-02	lb/hr	3,649	3,649	3,649	0	3,649	3,649
1-HEX-02	lb/hr	11,587	11,587	11,587	0	11,587	11,587
N-HEX-02	lb/hr	3,649	3,649	3,649	0	3,649	3,649
1-HEP-02	lb/hr	-	-	-	-	-	-
N-HEP-02	lb/hr	3,576	3,576	3,576	0	3,576	3,576
1-Oct-02	lb/hr	3,310	3,310	3,310	0	3,310	3,310
N-OCT-02	lb/hr	3	3	3	0	3	3
1-NON-02	lb/hr	-	-	-	-	-	-
N-NON-02	lb/hr	0	0	0	0	0	0
1-EIC-01	lb/hr	-	-	-	-	-	-
N-EIC-01	lb/hr	0	0	0	0	0	0

Table 28 - Hydroprocessing Reactions Mass Balance 2

Component	Units	H7	H8	H9	H10	H11	H12
Temperature	F	687	597	716	685	752	748
Pressure	psia	929	919	909	904	899	834
Mass Flows	lb/hr	192,325	192,325	192,325	195,769	195,769	195,769
HYDROGEN	lb/hr	3,484	3,484	3,484	5,927	5,927	5,605
1-BUT-01	lb/hr	4	4	4	4	4	-
N-BUT-01	lb/hr	5,459	5,459	5,459	5,981	5,981	4,754
1-PEN-01	lb/hr	1	1	1	1	1	-
N-PEN-01	lb/hr	1,420	1,420	1,420	1,664	1,664	4,754
1-HEX-01	lb/hr	12	12	12	12	12	-
N-HEX-01	lb/hr	13,030	13,030	13,030	13,132	13,132	4,754
1-HEP-01	lb/hr	1	1	1	1	1	-
N-HEP-01	lb/hr	1,523	1,523	1,523	1,565	1,565	4,754
1-Oct-01	lb/hr	28	28	28	28	28	-
N-OCT-01	lb/hr	32,345	32,345	32,345	32,401	32,401	15,213
1-NON-01	lb/hr	2	2	2	2	2	-
N-NON-01	lb/hr	5,361	5,361	5,361	5,384	5,384	15,213
1-Dec-01	lb/hr	36	36	36	36	36	-
N-DEC-01	lb/hr	40,563	40,563	40,563	40,571	40,571	15,213
1-UND-01	lb/hr	2	2	2	2	2	-
N-UND-01	lb/hr	5,329	5,329	5,329	5,332	5,332	15,213
1-DOD-01	lb/hr	26	26	26	26	26	-
N-DOD-01	lb/hr	30,425	30,425	30,425	30,426	30,426	15,213

Component	Units	H7	H8	H9	H10	H11	H12
1-TRI-01	lb/hr	3	3	3	3	3	-
N-TRI-01	lb/hr	6,993	6,993	6,993	6,994	6,994	15,213
1-TET-01	lb/hr	17	17	17	17	17	-
N-TET-01	lb/hr	20,355	20,355	20,355	20,355	20,355	15,213
1-PEN-02	lb/hr	-	-	-	-	-	-
N-PEN-02	lb/hr	3,649	3,649	3,649	3,649	3,649	15,213
1-HEX-02	lb/hr	12	12	12	12	12	-
N-HEX-02	lb/hr	15,328	15,328	15,328	15,328	15,328	15,213
1-HEP-02	lb/hr	-	-	-	-	-	-
N-HEP-02	lb/hr	3,576	3,576	3,576	3,576	3,576	15,213
1-Oct-02	lb/hr	3	3	3	3	3	-
N-OCT-02	lb/hr	3,336	3,336	3,336	3,336	3,336	6,339
1-NON-02	lb/hr	-	-	-	-	-	-
N-NON-02	lb/hr	0	0	0	0	0	6,339
1-EIC-01	lb/hr	-	-	-	-	-	-
N-EIC-01	lb/hr	0	0	0	0	0	6,339

Table 29 - Hydroprocessing Reactions Mass Balance 3

Component	Units	H13	H14	H15	H16	H17	H18
Temperature	F	257	257	233	233	233	233
Pressure	psia	2,176	1,015	1,015	1,015	904	1,015
Mass Flows	lb/hr	3,107	3,107	11,482	3,445	3,445	8,038
HYDROGEN	lb/hr	3,107	3,107	8,144	2,443	2,443	5,701

Component	Units	H13	H14	H15	H16	H17	H18
1-BUT-01	lb/hr	-	-	-	-	-	-
N-BUT-01	lb/hr	-	-	1,741	522	522	1,219
1-PEN-01	lb/hr	-	-	-	-	-	-
N-PEN-01	lb/hr	-	-	814	244	244	570
1-HEX-01	lb/hr	-	-	-	-	-	-
N-HEX-01	lb/hr	-	-	339	102	102	238
1-HEP-01	lb/hr	-	-	-	-	-	-
N-HEP-01	lb/hr	-	-	140	42	42	98
1-Oct-01	lb/hr	-	-	-	-	-	-
N-OCT-01	lb/hr	-	-	185	56	56	130
1-NON-01	lb/hr	-	-	-	-	-	-
N-NON-01	lb/hr	-	-	74	22	22	52
1-Dec-01	lb/hr	-	-	-	-	-	-
N-DEC-01	lb/hr	-	-	28	8	8	20
1-UND-01	lb/hr	-	-	-	-	-	-
N-UND-01	lb/hr	-	-	11	3	3	7
1-DOD-01	lb/hr	-	-	-	-	-	-
N-DOD-01	lb/hr	-	-	4	1	1	3
1-TRI-01	lb/hr	-	-	-	-	-	-
N-TRI-01	lb/hr	-	-	1	0	0	1
1-TET-01	lb/hr	-	-	-	-	-	-
N-TET-01	lb/hr	-	-	1	0	0	0
1-PEN-02	lb/hr	-	-	-	-	-	-

Component	Units	H13	H14	H15	H16	H17	H18
N-PEN-02	lb/hr	-	-	0	0	0	0
1-HEX-02	lb/hr	-	-	-	-	-	-
N-HEX-02	lb/hr	-	-	0	0	0	0
1-HEP-02	lb/hr	-	-	-	-	-	-
N-HEP-02	lb/hr	-	-	0	0	0	0
1-Oct-02	lb/hr	-	-	-	-	-	-
N-OCT-02	lb/hr	-	-	0	0	0	0
1-NON-02	lb/hr	-	-	-	-	-	-
N-NON-02	lb/hr	-	-	0	0	0	0
1-EIC-01	lb/hr	-	-	-	-	-	-
N-EIC-01	lb/hr	-	-	0	0	0	0

Table 30 - Fractionation Section Mass Balance 1

Component	Units	F1	F2	F3	F4	F5	F6	F7
Temperature	F	636	636	636	176	176	176	610
Pressure	psia	824	824	824	819	819	819	814
Mass Flows	lb/hr	195,769	91,693	104,076	104,076	9,306	94,770	91,693
HYDROGEN	lb/hr	5,605	3	5,602	5,602	5,596	5	3
1-BUT-01	lb/hr	-	-	-	-	-	-	-
N-BUT-01	lb/hr	4,754	126	4,628	4,628	1,935	2,693	126
1-PEN-01	lb/hr	-	-	-	-	-	-	-
N-PEN-01	lb/hr	4,754	210	4,545	4,545	904	3,640	210
1-HEX-01	lb/hr	-	-	-	-	-	-	-

Component	Units	F1	F2	F3	F4	F5	F6	F7
N-HEX-01	lb/hr	4,754	337	4,417	4,417	377	4,040	337
1-HEP-01	lb/hr	-	-	-	-	-	-	-
N-HEP-01	lb/hr	4,754	530	4,224	4,224	156	4,068	530
1-Oct-01	lb/hr	-	-	-	-	-	-	-
N-OCT-01	lb/hr	15,213	2,434	12,780	12,780	206	12,574	2,434
1-NON-01	lb/hr	-	-	-	-	-	-	-
N-NON-01	lb/hr	15,213	3,279	11,934	11,934	82	11,852	3,279
1-Dec-01	lb/hr	-	-	-	-	-	-	-
N-DEC-01	lb/hr	15,213	4,415	10,798	10,798	31	10,767	4,415
1-UND-01	lb/hr	-	-	-	-	-	-	-
N-UND-01	lb/hr	15,213	5,621	9,592	9,592	12	9,580	5,621
1-DOD-01	lb/hr	-	-	-	-	-	-	-
N-DOD-01	lb/hr	15,213	6,867	8,346	8,346	4	8,342	6,867
1-TRI-01	lb/hr	-	-	-	-	-	-	-
N-TRI-01	lb/hr	15,213	8,097	7,116	7,116	2	7,114	8,097
1-TET-01	lb/hr	-	-	-	-	-	-	-
N-TET-01	lb/hr	15,213	9,310	5,903	5,903	1	5,902	9,310
1-PEN-02	lb/hr	-	-	-	-	-	-	-
N-PEN-02	lb/hr	15,213	10,353	4,860	4,860	0	4,860	10,353
1-HEX-02	lb/hr	-	-	-	-	-	-	-
N-HEX-02	lb/hr	15,213	11,389	3,824	3,824	0	3,824	11,389
1-HEP-02	lb/hr	-	-	-	-	-	-	-
N-HEP-02	lb/hr	15,213	12,160	3,053	3,053	0	3,053	12,160

Component	Units	F1	F2	F3	F4	F5	F6	F7
1-Oct-02	lb/hr	-	-	-	-	-	-	-
N-OCT-02	lb/hr	6,339	5,321	1,018	1,018	0	1,018	5,321
1-NON-02	lb/hr	-	-	-	-	-	-	-
N-NON-02	lb/hr	6,339	5,536	802	802	0	802	5,536
1-EIC-01	lb/hr	-	-	-	-	-	-	-
N-EIC-01	lb/hr	6,339	5,705	633	633	0	633	5,705

Table 31 - Fractionation Section Mass Balance 2

Component	Units	F8	F9	F10	F11	F12	F13	F14
Temperature	F	404	391	161	377	321	321	113
Pressure	psia	814	44	19	22	12	25	20
Mass Flows	lb/hr	186,463	186,463	13,426	173,037	116,854	116,854	116,854
HYDROGEN	lb/hr	8	8	8	0	-	-	-
1-BUT-01	lb/hr	-	-	-	-	-	-	-
N-BUT-01	lb/hr	2,819	2,819	2,819	0	0	0	0
1-PEN-01	lb/hr	-	-	-	-	-	-	-
N-PEN-01	lb/hr	3,850	3,850	3,850	0	0	0	0
1-HEX-01	lb/hr	-	-	-	-	-	-	-
N-HEX-01	lb/hr	4,377	4,377	4,040	337	257	257	257
1-HEP-01	lb/hr	-	-	-	-	-	-	-
N-HEP-01	lb/hr	4,598	4,598	2,709	1,890	1,436	1,436	1,436
1-Oct-01	lb/hr	-	-	-	-	-	-	-
N-OCT-01	lb/hr	15,007	15,007	0	15,007	11,408	11,408	11,408

Component	Units	F8	F9	F10	F11	F12	F13	F14
1-NON-01	lb/hr	-	-	-	-	-	-	-
N-NON-01	lb/hr	15,131	15,131	0	15,131	11,501	11,501	11,501
1-Dec-01	lb/hr	-	-	-	-	-	-	-
N-DEC-01	lb/hr	15,182	15,182	0	15,182	11,540	11,540	11,540
1-UND-01	lb/hr	-	-	-	-	-	-	-
N-UND-01	lb/hr	15,201	15,201	0	15,201	11,555	11,555	11,555
1-DOD-01	lb/hr	-	-	-	-	-	-	-
N-DOD-01	lb/hr	15,209	15,209	0	15,209	11,561	11,561	11,561
1-TRI-01	lb/hr	-	-	-	-	-	-	-
N-TRI-01	lb/hr	15,212	15,212	0	15,212	11,563	11,563	11,563
1-TET-01	lb/hr	-	-	-	-	-	-	-
N-TET-01	lb/hr	15,213	15,213	0	15,213	11,564	11,564	11,564
1-PEN-02	lb/hr	-	-	-	-	-	-	-
N-PEN-02	lb/hr	15,213	15,213	0	15,213	11,564	11,564	11,564
1-HEX-02	lb/hr	-	-	-	-	-	-	-
N-HEX-02	lb/hr	15,213	15,213	0	15,213	11,564	11,564	11,564
1-HEP-02	lb/hr	-	-	-	-	-	-	-
N-HEP-02	lb/hr	15,213	15,213	0	15,213	11,333	11,333	11,333
1-Oct-02	lb/hr	-	-	-	-	-	-	-
N-OCT-02	lb/hr	6,339	6,339	0	6,339	8	8	8
1-NON-02	lb/hr	-	-	-	-	-	-	-
N-NON-02	lb/hr	6,339	6,339	0	6,339	0	0	0
1-EIC-01	lb/hr	-	-	-	-	-	-	-

Component	Units	F8	F9	F10	F11	F12	F13	F14
N-EIC-01	lb/hr	6,339	6,339	0	6,339	0	0	0

Table 32 - Fractionation Section Mass Balance 3

Component	Units	F15	F16	F17	F18	F19	F20
Temperature	F	321	330	330	176	176	220
Pressure	psia	12	1,015	1,015	819	819	1,015
Mass Flows	lb/hr	36,874	36,874	36,874	931	8,375	8,375
HYDROGEN	lb/hr	-	-	-	560	5,037	5,037
1-BUT-01	lb/hr	-	-	-	-	-	-
N-BUT-01	lb/hr	0	0	0	193	1,741	1,741
1-PEN-01	lb/hr	-	-	-	-	-	-
N-PEN-01	lb/hr	0	0	0	90	814	814
1-HEX-01	lb/hr	-	-	-	-	-	-
N-HEX-01	lb/hr	81	81	81	38	339	339
1-HEP-01	lb/hr	-	-	-	-	-	-
N-HEP-01	lb/hr	453	453	453	16	140	140
1-Oct-01	lb/hr	-	-	-	-	-	-
N-OCT-01	lb/hr	3,600	3,600	3,600	21	185	185
1-NON-01	lb/hr	-	-	-	-	-	-
N-NON-01	lb/hr	3,629	3,629	3,629	8	74	74
1-Dec-01	lb/hr	-	-	-	-	-	-
N-DEC-01	lb/hr	3,642	3,642	3,642	3	28	28
1-UND-01	lb/hr	-	-	-	-	-	-

Component	Units	F15	F16	F17	F18	F19	F20
N-UND-01	lb/hr	3,646	3,646	3,646	1	11	11
1-DOD-01	lb/hr	-	-	-	-	-	-
N-DOD-01	lb/hr	3,648	3,648	3,648	0	4	4
1-TRI-01	lb/hr	-	-	-	-	-	-
N-TRI-01	lb/hr	3,649	3,649	3,649	0	1	1
1-TET-01	lb/hr	-	-	-	-	-	-
N-TET-01	lb/hr	3,649	3,649	3,649	0	1	1
1-PEN-02	lb/hr	-	-	-	-	-	-
N-PEN-02	lb/hr	3,649	3,649	3,649	0	0	0
1-HEX-02	lb/hr	-	-	-	-	-	-
N-HEX-02	lb/hr	3,649	3,649	3,649	0	0	0
1-HEP-02	lb/hr	-	-	-	-	-	-
N-HEP-02	lb/hr	3,576	3,576	3,576	0	0	0
1-Oct-02	lb/hr	-	-	-	-	-	-
N-OCT-02	lb/hr	3	3	3	0	0	0
1-NON-02	lb/hr	-	-	-	-	-	-
N-NON-02	lb/hr	0	0	0	0	0	0
1-EIC-01	lb/hr	-	-	-	-	-	-
N-EIC-01	lb/hr	0	0	0	0	0	0

Equipment List & Sizing

Table 33 - Dehydration Reaction Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Pump	P-100	Stainless Steel 316	640 30	gal/min hp	0.1	MMBtu/hr	Electricity
Vaporizer	E-100	Stainless Steel 316	15 129	US Ton MMBtu/hr	2,480	lb/min	MP Steam
Shell and Tube Heat Exchanger	E-101	Stainless Steel 316	160	ft^2	11	MMBtu/hr	Process Stream
Shell and Tube Heat Exchanger	E-102	Stainless Steel 316	205	ft^2	12.4	MMBtu/hr	Process Stream
Fired Heater	F-100	Stainless Steel 316	114 30	US Ton MMBtu/hr	1,945	lb/min	Natural Gas
Fired Heater	F-101	Stainless Steel 316	189 62	US Ton MMBtu/hr	3,984	lb/min	Natural Gas
Fired Heater	F-102	Stainless Steel 316	199 65	US Ton MMBtu/hr	4,217	lb/min	Natural Gas
Surge Drum	D-100	Stainless Steel 316	4	US Ton	-	-	-
Packed Bed Reactor	R-100	Stainless Steel 316	29	US Ton	2,486	ft^3	Alumina Catalyst
Packed Bed Reactor	R-101	Stainless Steel 316	42	US Ton	2,511	ft^3	Alumina Catalyst
Packed Bed Reactor	R-102	Stainless Steel 316	33	US Ton	25.1	ft^3	Alumina Catalyst

Table 34 - Byproduct Removal Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Pump	P-200	Stainless Steel 316	8,158 16	gal/min hp	0.04	MMBtu/hr	Electricity
Centrifugal Pump	P-201	Stainless Steel 316	1.74 .115	gal/min hp	3E^-4	MMBtu/hr	Electricity
Centrifugal Pump	P-202	Stainless Steel 316	25 6.3	gal/min hp	0.02	MMBtu/hr	Electricity

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Pump	P-203	Stainless Steel 316	1,058 21	gal/min hp	0.05	MMBtu/hr	Electricity
Centrifugal Pump	P-204	Stainless Steel 316	2.5 .017	gal/min hp	4E^-5	MMBtu/hr	Electricity
Centrifugal Compressor	K-200	Stainless Steel 316	4,935	hp	13	MMBtu/hr	Electricity
Centrifugal Compressor	K-201	Stainless Steel 316	2,738	hp	7	MMBtu/hr	Electricity
Air Cooler	E-200	Stainless Steel 316	38,714	ft^2	-27	MMBtu/hr	Air
Air Cooler	E-202	Stainless Steel 316	55,098	ft^2	-68	MMBtu/hr	Air
Air Cooler	E-204	Stainless Steel 316	5,612	ft^2	-4	MMBtu/hr	Air
Air Cooler	E-207	Stainless Steel 316	1,364	ft^2	-1.12	MMBtu/hr	Air
Trim Cooler	E-201	Stainless Steel 316	2,655	ft^2	21 2,380	MMBtu/hr gal/min	Cooling Water
Trim Cooler	E-203	Stainless Steel 316	2,162	ft^2	7.4 820	MMBtu/hr gal/min	Cooling Water
Trim Cooler	E-205	Stainless Steel 316	1,092	ft^2	1.4 413	MMBtu/hr gal/min	Cooling Water
Trim Cooler	E-208	Stainless Steel 316	1,430	ft^2	1.92 214	MMBtu/hr gal/min	Cooling Water
Flash Drum	D-200	Stainless Steel 316	52.2	US Ton	-	-	-
Quench Absorber (Flash Drum)	T-200	Stainless Steel 316	52.2	US Ton	-	-	-
Knockout Drum	D-201	Stainless Steel 316	5.6	US Ton	-	-	-
Knockout Drum	D-202	Stainless Steel 316	3.55	US Ton	-	-	-
Knockout Drum	D-203	Stainless Steel 316	5.5	US Ton	-	-	-

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Knockout Drum	D-204	Stainless Steel 316	1.3	US Ton	-	-	-
Water Wash Tower (Trayed Column)	T-201	Stainless Steel 316	Diameter: 4.56 Weight: 13.45	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	6	stages	-	-	-
Condenser	-	-	-	-	-	-	-
Reboiler	-	-	-	-	-	-	-
Water Stripper (Trayed Column)	T-202	Stainless Steel 316	Diameter: 6.86 Weight: 11.7	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	9	stages	-	-	-
Shell & Tube Condenser	-	Stainless Steel 316	411	ft^2	10 1,035	MMBtu/hr gal/min	Cooling Water
Thermosiphon Reboiler	-	Stainless Steel 316	4,052	ft^2	52 1,013	MMBtu/hr lb/min	LP Steam

Table 35 - Ethylene Purification Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Compressor	K-100	Carbon Steel	2,299	hp	5.85	MMBtu/hr	Electricity
Knockout Drum	D-100	Stainless Steel 316	5.5	US Ton	-	-	-
Knockout Drum	D-101	Stainless Steel 316	4.7	US Ton	-	-	-
Adsorbent Drier	DR-100	Stainless Steel 316	-	-	366	ft^3	Adsorbent
Adsorbent Drier	DR-101	Stainless Steel 316	-	-	366	ft^3	Adsorbent
Oxygenate Stripper (Trayed Column)	T-100	Carbon Steel	Diameter: 7 Weight: 77.2	ft US Ton	-	-	-

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Sieve Trays	-	Carbon Steel	26	stages	-	-	-
Shell & Tube Condenser	-	Carbon Steel	12,347	ft^2	9.4 91,600	MMBtu/hr lb/min	Refrigerant
Thermosiphon Reboiler	-	Carbon Steel	235	ft^2	4.6 81	MMBtu/hr lb/min	LP Steam

Table 36 - Oligomerization Reaction Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Steam Generator	E-100	Carbon Steel w/ Stainless Internals	33.4	ft^2	0.79 88	MMBtu/hr gal/min	Process Water
Steam Generator	E-101	Carbon Steel w/ Stainless Internals	88	ft^2	3.47 386	MMBtu/hr gal/min	Process Water
Steam Generator	E-102	Carbon Steel w/ Stainless Internals	291	ft^2	7.71 860	MMBtu/hr gal/min	Process Water
Steam Generator	E-103	Carbon Steel w/ Stainless Internals	453	ft^2	12 1,342	MMBtu/hr gal/min	Process Water
Steam Generator	E-104	Carbon Steel w/ Stainless Internals	664	ft^2	18.7 2,087	MMBtu/hr gal/min	Process Water
Air Cooler	E-105	Carbon Steel	3,015	ft^2	5.05	MMBtu/hr	Air
Fixed Bed Reactor	R-100	Carbon Steel	17	US Ton	162.95	ft^3	MTT Zeolite on Alumina Catalyst
Fixed Bed Reactor	R-101	Carbon Steel	25.2	US Ton	165.37	ft^3	MTT Zeolite on Alumina Catalyst
Fixed Bed Reactor	R-102	Carbon Steel	32.3	US Ton	167.00	ft^3	MTT Zeolite on Alumina Catalyst
Fixed Bed Reactor	R-103	Carbon Steel	40.4	US Ton	168.63	ft^3	MTT Zeolite on Alumina Catalyst
Fixed Bed Reactor	R-104	Carbon Steel	21	US Ton	268.03	ft^3	Amorphous Alumina Catalyst
Fixed Bed Reactor	R-105	Carbon Steel	32.3	US Ton	286.32	ft^3	Amorphous Alumina Catalyst

Table 37 - Olefin Splitting Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Pump	P-100	Carbon Steel	561 377	gal/min hp	0.96	MMBtu/hr	Electricity
Centrifugal Compressor	K-100	Carbon Steel	782.4	hp	1.87	MMBtu/hr	Electricity
Deethanizer (Trayed Column)	T-100	Carbon Steel	Diameter: 12.3 Weight: 23	ft US Ton	-	-	-
Sieve Trays	-	Carbon Steel	25	stages	-	-	-
Shell & Tube Condenser	-	Carbon Steel	269.1	ft^2	0.67 6,495	MMBtu/hr lb/min	Refrigerant
Charged Heat Reboiler	-	Carbon Steel	156	US Ton	41.9 2,708	MMBtu/hr lb/min	Natural Gas
Olefin Splitter (Trayed Column)	T-101	Carbon Steel	Diameter: 11.4 Weight: 120.4	ft US Ton	-	-	-
Sieve Trays	-	Carbon Steel	33	stages	-	-	-
Shell & Tube Condenser	-	Carbon Steel	1,672	ft^2	38.3 35,587	MMBtu/hr gal/min	Cooling Water
Thermosiphon Reboiler	-	Carbon Steel	676.3	ft^2	5.85 132	MMBtu/hr lb/min	HP Steam

Table 38 - Hydroprocessing Reactions Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Shell and Tube Heat Exchanger	E-100	Stainless Steel 316	685	ft^2	18	MMBtu/hr	Process Stream
Shell and Tube Heat Exchanger	E-101	Stainless Steel 316	4,422	ft^2	23	MMBtu/hr	Process Stream
Air Cooler	E-103	Stainless Steel 316	1301	ft^2	3.13	MMBtu/hr	Air
Steam Generator	E-102	Carbon Steel w/ Stainless Internals	498	ft^2	16.7 1,861	MMBtu/hr gal/min	Process Water
Fired Heater	F-100	Stainless Steel 316	79 11.56	US Ton MMBtu/hr	339	kg/min	Natural Gas

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Control Valve	V-100	Stainless Steel 316	ΔP= 7.6	bar	-	-	-
Control Valve	V-101	Stainless Steel 316	ΔP= 80	bar	-	-	-
Hydrogenation Reactor	R-100	Stainless Steel 316	35.45	US Ton	1,187	ft^3	Platinum on Alumina Catalyst
Hydrocracking & Isomerization Reactor	R-101	Stainless Steel 316	14.8	US Ton	5,066	ft^3	Ni-W on FAU Zeolite Catalyst

Table 39 - Fractionation Equipment List

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Pump	P-100	Stainless Steel 316	117.5 80.9	gal/min hp	0.21	MMBtu/hr	Electricity
Centrifugal Pump	P-101	Stainless Steel 316	372.4 3.4	gal/min hp	0.01	MMBtu/hr	Electricity
Centrifugal Compressor	K-100	Stainless Steel 316	327.7	hp	0.83	MMBtu/hr	Electricity
Shell and Tube Heat Exchanger	E-100	Stainless Steel 316	55.3	ft^2	2.09	MMBtu/hr	Process Stream
Air Cooler	E-101	Stainless Steel 316	21,948	ft^2	48.5	MMBtu/hr	Air
Air Cooler	E-102	Stainless Steel 316	20,723	ft^2	14.7	MMBtu/hr	Air
Control Valve	V-100	Stainless Steel 316	ΔP= 53	bar	-	-	-
Hot Separator	D-100	Stainless Steel 316	18.2	US Ton	-	-	-
Cold Separator	D-101	Stainless Steel 316	13.8	US Ton	-	-	-
Naphtha Fractionator (Trayed Column)	T-100	Stainless Steel 316	Diameter: 6.04 Weight: 69.1	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	31	stages	-	-	-

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Condenser	-	Stainless Steel 316	13,264	ft^2	9.8	MMBtu/hr	Air
Thermosiphon Reboiler	-	Stainless Steel 316	508	ft^2	5.3 120	MMBtu/hr lb/min	HP Steam
Jet Fuel Fractionator (Trayed Column)	T-101	Stainless Steel 316	Diameter: 14.34 Weight: 300	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	60	stages	-	-	-
Condenser	-	Stainless Steel 316	18,191	ft^2	46	MMBtu/hr	Air
Charged Heat Reboiler	-	Stainless Steel 316	140	US Ton	44 2,836	MMBtu/hr lb/min	Natural Gas

Process Economics Data & Calculations

Table 40 - Detailed Equipment Costing from Aspen

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
BD-100-flash vessel	58400	266400	11200	36924
BD-101-flash vessel	36500	159000	5400	15274
BD-102-flash vessel	52000	171200	11000	21434
BD-103-flash vessel	22100	139100	2600	10646
BE-100	8674100	16147400	1579500	2624522
BE-101	81700	234000	13100	36580
BE-102	1374600	2027500	247200	363073
BE-103	65800	181700	11200	26711
BE-104	195000	400000	34800	68166
BE-105	46700	525600	7000	24635
BE106	44600	328500	7000	25158
BE-107	14800	468000	1200	7209
BK-100	-	198900	0	0
BK-101	1609800	78066900	30300	59690
BP-100	53000	735300	2900	45774
BP-101	6200	12371400	210	2447
BP-102	26500	592200	1500	16301
BP-103	18900	1755000	1100	13705
BP-104	5300	420300	170	2371
BT-100-flash vessel	382800	401400	104400	188891

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
BT-101-tower	137800	133200	26900	47183
BT-102-bottoms split	0	0	0	0
BT-102-cond	22800	14488200	2800	11362
BT-102-cond acc	18200	477000	2100	10664
BT-102-overhead split	0	55800	0	0
BT-102-reb	129500	238500	26800	58154
BT-102-reflux pump	6900	170100	250	3427
BT-102-tower	154700	47700	24800	59461
DD-100-flash vessel	48800	3445200	8500	29380
DE-100	169300	1240200	29600	65170
DE-101	110800	0	17700	54984
DE-102	40600	205200	5800	27107
DF-100	989800	163800	228300	283088
DF-101	1728800	0	378600	442924
DF-102	1810900	1165500	398800	464110
DP-100	19500	62100	1100	10152
DR-100	194300	1392300	58600	99417
DR-101	268100	439200	83700	134739
DR-102	511000	1523700	66500	113801
ED-100-flash vessel	52000	997200	11000	21434
ED-101-flash vessel	47700	365400	9300	19430

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
EDR-100	47700	8908200	9300	19430
EDR-101	47700	15559200	9300	19430
EK-100	1643400	16298100	27600	52053
ET-100-bottoms split	0	175500	0	0
ET-100-cond	1119700	1748700	204600	279073
ET-100-cond acc	61200	2412900	16000	30398
ET-100-overhead split	0	4599000	0	0
ET-100-reb	20600	468000	2500	10986
ET-100-reflux pump	11200	429300	660	6512
ET-100-tower	577300	429300	138300	172314
FD-100-flash vessel	133100	429300	36400	62270
FD-101-flash vessel	105000	14790600	27600	48129
FE-100	15000	0	1200	11986
FE-101	1103800	10077300	227200	302642
FE-102	456300	550800	86800	135357
FK-100	1537900	0	16800	32458
FP-100	175200	185400	8000	14565
FP-101	10000	100800	540	6966
FT-100-bottoms split	0	5195700	0	0
FT-100-cond	276600	1197900	52200	88380
FT-100-cond acc	28800	945000	3900	16099

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
FT-100-overhead split	0	135000	0	0
FT-100-reb	32400	9934200	5000	16521
FT-100-reflux pump	9500	4106700	600	5834
FT-100-tower	576000	13841100	120200	193857
FT-101-bottoms split	0	1576800	0	0
FT-101-cond	393800	90000	76600	119287
FT-101-cond acc	55800	0	9800	35581
FT-101-overhead split	0	2489400	0	0
FT-101-reb	1312700	259200	280700	316547
FT-101-reflux pump	19000	0	1100	12012
FT-101-tower	1710000	291600	423500	595827
HE-100	52700	85500	9000	36516
HE-101	337200	5184000	65600	118524
HE-102	35000	0	5400	21670
HE-103	80200	3544200	13800	36403
HF-100	705400	502200	157100	181842
HR-100	440800	0	70900	99530
HR-101	132500	11814300	29600	51855
OE-100	12400	171000	620	7255
OE-101	15600	15390000	1200	7885
OE-102	28100	474300	3800	13993

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
OE-103	33800	3034800	4900	15158
OE-104	43000	315000	6700	21983
OE-105	161600	721800	29600	60047
OR-100	120000	6348600	33100	55643
OR-101	169600	3967200	50400	80033
OR-102	205000	1192500	64500	96837
OR-103	248400	111600	80700	114118
OR-104	146600	140400	41300	63957
OR-105	205000	252900	64500	96837
SK-100	1436400	304200	17600	37236
SP-100	311500	387000	13500	31193
ST-100-bottoms split	0	1454400	0	0
ST-100-cond	22900	1080000	2800	12075
ST-100-cond acc	24600	1526400	3800	11066
ST-100-overhead split	0	1845000	0	0
ST-100-reb	1309300	2235600	311300	344021
ST-100-reflux pump	6000	1319400	220	2447
ST-100-tower	917300	1845000	208600	277014
ST-101-bottoms split	0	12927600	0	0
ST-101-cond	53600	2803500	9100	26693
ST-101-cond acc	45100	0	8100	30851

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
ST-101-overhead split	0	206100	0	0
ST-101-reb	40700	221400	7000	18655
ST-101-reflux pump	19700	0	1100	11352
ST-101-tower	718700	11783700	144500	233236

Table 41 - Catalyst Cost Calculations 1

	dehydration		oligo-merization 1		Drier		olefin hydro-genation		Hydro-cracking		olig 2	
bpd/SCFH	1,500		1,500		137,419		1,500		1,500		1,500	
m3/hr	142		19		-		33		137		16	
LHSV/GHSV	1		1		3,986		1		0.5		1	
catalyst	142	*m3*	19	*m3*	1,710	*lbs*	33	*m3*	273	*m3*	16	*m3*
density	680	*kg/m3*	680	*kg/m3*	49.6	*lb/ft3*	680	*kg/m3*	680	*kg/m3*	680	*kg/m3*
loading	96,693	*kgs*	12,785	*kgs*	34	*ft3*	22,406	*kgs*	185,823	*kgs*	10,674	*kgs*
	213,208	*lbs*	28,190	*lbs*	776	*kgs*	49,405	*lbs*	409,740	*lbs*	23,536	*lbs*
lifespan	3	*years*	3	*years*	3	*years*	3	*years*	3	*years*	3	*years*
consumption	71,069	*lbs/year*	9,397	*lbs/year*	259	*lbs/year*	16,468	*lbs/year*	136,580	*lbs/year*	7,845	*lbs/year*
Price	$10	*/lb*	$15	*/lb*	$15	*/lb*	$20	*/lb*	$25	*/lb*	$15	*/lb*
rate	**$710,694**	*/year*	**$140,952**	*/year*	**$3,879**	*/year*	**$329,367**	*/year*	**$3,414,502**	*/year*	**$117,682**	*/year*

Table 42 - Catalyst Cost Calculations 2

Platinum price	$/kg	$74,644	
Pt content	0.20%		
% of bed volume	5%	9,291.16	kgs
Pt quantity		18.58	kgs
Pt value	$	**$1,387,065**	
/year		$462,355	
Total Cost/yr	$5,179,431		

Table 43 - VCOP & Income Calculations

	Price	Amount (gal/year or kg/year)	value/year
Ethanol	1.70	304,776,000.00	$ 518,119,200.00
Hydrogen	1.5	11273360	$ 16,910,040.00
Jet Fuel	2.50	139,998,600.00	$ 349,996,500.00
D3 RIN	1.70	139,998,600.00	$ 575,954,240.40
45Z Tax Credit	1.75	139,998,600.00	$ 244,997,550.00
gasoline	1.00	41,105,190.72	$ 41,105,190.72
D3 RIN	1.70	41,105,190.72	$ 169,106,754.62
Diesel	1.20	22,246,345.43	$ 26,695,614.52
D3 RIN	1.70	22,246,345.43	$ 91,521,465.11
Process Water/Oxygenates	0.00554	394,975,389.42	$ 2,188,163.66
Fuel Gas	0.00255	6,693,677.25	$ 17,068.88

Table 44 - Total Utilities Calculation

Name	Fluid	Rate	Rate Units	Cost per Hour	Cost per year	Cost Units
Electricity		6130.861	KW	367.85166	$ 2,942,813.28	USD/H
AP-UTIL-CWDRC	Water	81945270	BTU/H	30.565586	$ 244,524.69	USD/H
AP-UTIL-FIRE	Steam	282696200	BTU/H	1266.478976	$ 10,131,831.81	USD/H
AP-UTIL-HPSTEAM	Steam	11155360	BTU/H	29.45015	$ 235,601.20	USD/H
AP-UTIL-LPSTEAM	Steam	4577057	BTU/H	9.154114	$ 73,232.91	USD/H
AP-UTIL-MPSTEAM	Steam	179717600	BTU/H	416.944832	$ 3,335,558.66	USD/H
AP-UTIL-REFRIGE2	Propane	10119320	BTU/H	29.244835	$ 233,958.68	USD/H
					Total Utilities	
					$ 17,197,521.22	

Table 44 - Net Present Value (NPV) Table

Years	Pre-Tax Cash Flow ($MM)	Depreciation	Taxable Income	Tax	Cash Flow	Discount Factor	Present Value of CF	NPV
1	(156,698,504.70)	-	-156698504.7	-	(156,698,504.70)	0.88	(137,454,828.68)	(137,454,828.68)
2	(470,095,514.09)	-	(470,095,514.09)	-	(470,095,514.09)	0.77	-361723233.4	-499178062.1
3	462,416,596.18	114,613,763.44	347,802,832.74	-	462,416,596.18	0.67	312,118,031.04	(187,060,031.02)
4	965,096,161.41	183,382,021.50	781,714,139.92	90,428,736.51	874,667,424.90	0.59	517,873,331.54	330,813,300.53
5	965,096,161.41	110,029,212.90	855,066,948.52	203,245,676.38	761,850,485.03	0.52	395,681,268.85	726,494,569.38
6	965,096,161.41	66,017,527.74	899,078,633.67	222,317,406.61	742,778,754.80	0.46	338,400,008.59	1,064,894,577.97
7	965,096,161.41	66,017,527.74	899,078,633.67	233,760,444.76	731,335,716.66	0.40	292,269,047.68	1,357,163,625.65
8	965,096,161.41	33,008,763.87	932,087,397.54	233,760,444.76	731,335,716.66	0.35	256,376,357.61	1,613,539,983.26
9	965,096,161.41	-	965,096,161.41	242,342,723.36	722,753,438.05	0.31	222,252,422.92	1,835,792,406.18
10	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.27	192,643,249.20	2,028,435,655.38
11	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.24	168,985,306.31	2,197,420,961.69
12	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.21	148,232,724.84	2,345,653,686.53
13	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.18	130,028,706.00	2,475,682,392.53
14	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.16	114,060,268.42	2,589,742,660.95
15	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.14	100,052,867.03	2,689,795,527.98
16	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.12	87,765,672.84	2,777,561,200.82
17	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.11	76,987,432.31	2,854,548,633.13
18	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.09	67,532,835.36	2,922,081,468.49
19	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.08	59,239,329.26	2,981,320,797.76
20	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.07	51,964,323.92	3,033,285,121.67
21	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.06	45,582,740.28	3,078,867,861.95
22	965,096,161.41	-	965,096,161.41	250,925,001.97	714,171,159.45	0.06	39,984,859.89	3,118,852,721.84

Table 46 - Fixed Capital Costs (FCOP) Calculations

FIXED OPERATING COSTS						
					$MM/yr	$/unit product
Labor						
	5 Operators per Shift Position					
Number of shift positions	4	70000	$/yr each		1.40	1400000.00
Supervision		25%	of Operating Labor		0.35	350000.00
Direct Ovhd.		45%	of Labor & Superv.		0.79	787500.00
Maintenance		3%	of ISBL Investment		18,726,960.89	18726960894352.90
Overhead Expense						
Plant Overhead		65%	of Labor & Maint.		12,172,526.23	12172526230704.40
Tax & Insurance		2%	of Fixed Investment		9,363,480.45	9363480447176.47
Interest on Debt Financing		0%	of Fixed Capital		0.00	0.00
		0%	of Working Capital		0.00	0.00
			Fixed Cost of Production (FCOP)		40,262,970.11	40,262,970,109,733.80

Hand Calculation

Hand calculations were performed to design and size the Olefin Splitter prior to rigorous simulation in Aspen Plus. The objective of these calculations was to estimate the minimum number of theoretical stages, the minimum reflux ratio, the operating number of equilibrium stages, the actual number of trays, and the physical dimensions of the column. The shortcut design methods used included the Fenske, Underwood, and Gilliland correlations, which are standard techniques for multicomponent distillation design.

The separation in the Olefin Splitter is governed by the behavior of the light key (LK) component and the heavy key (HK) component. For this system, the light key was selected as heptane/heptene, and the heavy key was selected as octene. The required mole fractions of the key components in the distillate and bottoms streams were obtained from the overall material balance and Aspen simulation results.

The compositions and relative volatility used in the calculations were:

$$x_{D,LK} = 0.0288$$
$$x_{D,HK} = 6.06 \times 10^{-6}$$
$$x_{B,LK} = 0.00268$$
$$x_{B,HK} = 0.2319$$
$$\alpha_{LK,HK} = 1.80$$

1. Minimum Number of Stages - Fenske Equation

The minimum number of theoretical stages was calculated using the Fenske equation, which assumes total reflux (no product withdrawal).

$$N_{min} = \frac{\ln\left[\left(\frac{x_{D,LK}}{x_{D,HK}}\right)\left(\frac{x_{B,HK}}{x_{B,LK}}\right)\right]}{\ln(\alpha_{LK,HK})}$$

Substituting the values:

$$N_{min} = \frac{\ln\left[\left(\frac{0.0288}{6.06 \times 10^{-6}}\right)\left(\frac{0.2319}{0.00268}\right)\right]}{\ln(1.80)}$$
$$N_{min} = 21.2 \text{ stages}$$

2. Minimum Reflux Ratio - Underwood Equation

The minimum reflux ratio was determined using the Underwood equation, which solves for the parameter θusing the feed composition and relative volatilities.

$$\sum\left[\frac{qz_i}{\alpha_i - \theta}\right] = 1$$

After solving for θ, the minimum reflux ratio was calculated from:

$$R_{min} = \sum\left(\frac{x_{D,i}\alpha_i}{\alpha_i - \theta}\right) - 1$$

Result:

$$R_{min} = 3.49$$

3. Operating Number of Stages - Gilliland Correlation

An operating reflux ratio greater than the minimum reflux ratio was selected:

$$R = 4.54$$

The Gilliland correlation relates reflux ratio and number of stages using:

$$X = \frac{R - R_{min}}{R + 1}$$
$$Y = \frac{N - N_{min}}{N + 1}$$

Substituting:

$$X = \frac{4.54 - 3.49}{4.54 + 1}$$
$$X = 0.19$$

From the Gilliland correlation:

$$Y = 0.24$$

Solving for the number of equilibrium stages:

$$0.24 = \frac{N - 21.2}{N + 1}$$
$$N_{eq} = 28.2 \text{ equilibrium stages}$$

4. Actual Number of Trays - Tray Efficiency

Because real trays do not operate at 100% efficiency, the theoretical number of stages was corrected using an assumed tray efficiency of:

$$E = 0.80$$
$$N_{actual} = \frac{N_{eq}}{E}$$
$$N_{actual} = \frac{28.2}{0.80}$$
$$N_{actual} = 35 \text{ trays}$$

5. Column Height Calculation

The overall column height was estimated using the number of trays, tray spacing, and mechanical allowances at the top and bottom of the column.

For this system:

- Tray spacing = 18 inches (1.5 ft) for pressure columns
- Top allowance = 12 ft
- Bottom allowance = 12.93 ft

$$H = (N_{trays} \times S) + H_{top} + H_{bot}$$
$$H = (35)(1.5) + 12 + 12.93$$
$$H = 77.43 \text{ ft}$$

6. Column Diameter - Souders–Brown Flooding Method

The column diameter was determined using the allowable vapor velocity based on flooding conditions. The column was designed to operate at 80% of flooding velocity to ensure stable operation and avoid entrainment [31].

The required cross-sectional area was first determined, and the column diameter was calculated from:

$$D = \sqrt{\frac{4A}{\pi}}$$

Result:

$$D = 11.96 \text{ ft}$$

7. Comparison with Aspen RadFrac Simulation

The hand calculation results were compared with the Aspen Plus RadFrac simulation to validate the preliminary design.

Table 45- Hand vs Simulation Calculations

Parameter	Hand Calculation	Aspen Simulation	Difference
Equilibrium stages	35	33	2
Physical trays	33	31	2
Column height	77.43 ft	80.38 ft	2.95 ft
Column diameter	11.96 ft	12.8 ft	0.84 ft

The hand calculations provide an initial design basis for the Olefin Splitter and demonstrate consistency with the Aspen RadFrac simulation results. The small differences between the shortcut calculations and the rigorous simulation are expected due to simplifying assumptions such as constant relative volatility and ideal tray efficiency. Overall, the close agreement confirms that the Olefin Splitter design is reasonable and that the Aspen simulation accurately represents the required separation performance.

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